Untitled8

February 4, 2019

1 Sunica Saki 193m/18 MB

Numela 1

2 ZADATAK 7

Na osnovu podataka iz fajla global_health_nutrition_data.csv za svaku evropsku zemlju formirati Gergorinove krugove za Vandermondovu matricu vremenske serije indikatora Population, total (kod je SP.POP.TOTL). Pored krugova, na istom grafikonu prikazati i karakteristine korene.

```
In [1]: import pandas as pd
        import numpy as np
        import scipy
        from scipy import linalg as la
        import matplotlib
        from matplotlib.patches import Circle
        from matplotlib.collections import PatchCollection
        import matplotlib.pyplot as plt
        import os
        import sys
        plt.style.use('ggplot')
        import warnings
        warnings.filterwarnings('ignore')

In [2]: path2file = r"C:\Users\zeks\Downloads\global_health_nutrition_data.csv"

In [3]: data = pd.read_csv(os.path.normpath(path2file)) #uitavanje podataka
        data = data[data['Indicator Code']== 'SP.POP.TOTL']

In [4]: european_code =['ALB','AND','AUT','BLR','BEL','BIH','BGR','HRV','CIP','CZE','DNK','EST

In [5]: european_countries=[]
        for i in list(range(data.shape[0])):
            european_countries.append(data['Country Code'].values[i] in european_code)
        data=data[european_countries] #filtriranje evropskih zemalja
        data
```

Out[5]:

	Country Name	Country Code	Indicator Name	Indicator Code	\
14757	Albania	ALB	Population, total	SP.POP.TOTL	
18207	Austria	AUT	Population, total	SP.POP.TOTL	
20277	Belarus	BLR	Population, total	SP.POP.TOTL	
20622	Belgium	BEL	Population, total	SP.POP.TOTL	
22692	Bosnia and Herzegovina	BIH	Population, total	SP.POP.TOTL	
24417	Bulgaria	BGR	Population, total	SP.POP.TOTL	
30972	Croatia	HRV	Population, total	SP.POP.TOTL	
32352	Czech Republic	CZE	Population, total	SP.POP.TOTL	
32697	Denmark	DNK	Population, total	SP.POP.TOTL	
35802	Estonia	EST	Population, total	SP.POP.TOTL	
36492	Faroe Islands	FRO	Population, total	SP.POP.TOTL	
37182	Finland	FIN	Population, total	SP.POP.TOTL	
37527	France	FRA	Population, total	SP.POP.TOTL	
39252	Germany	DEU	Population, total	SP.POP.TOTL	
39942	Gibraltar	GIB	Population, total	SP.POP.TOTL	
40287	Greece	GRC	Population, total	SP.POP.TOTL	
44082	Hungary	HUN	Population, total	SP.POP.TOTL	
44427	Iceland	ISL	Population, total	SP.POP.TOTL	
46152	Ireland	IRL	Population, total	SP.POP.TOTL	
47187	Italy	ITA	Population, total	SP.POP.TOTL	
51672	Latvia	LVA	Population, total	SP.POP.TOTL	
53397	Liechtenstein	LIE	Population, total	SP.POP.TOTL	
53742	Lithuania	LTU	Population, total	SP.POP.TOTL	
54087	Luxembourg	LUX	Population, total	SP.POP.TOTL	
54777	Macedonia, FYR	MKD	Population, total	SP.POP.TOTL	
56847	Malta	MLT	Population, total	SP.POP.TOTL	
58917	Moldova	MDA	Population, total	SP.POP.TOTL	
59262	Monaco	MCO	Population, total	SP.POP.TOTL	
59952	Montenegro	MNE	Population, total	SP.POP.TOTL	
62367	Netherlands	NLD	Population, total	SP.POP.TOTL	
64782	Norway	NOR	Population, total	SP.POP.TOTL	
67887	Poland	POL	Population, total	SP.POP.TOTL	
68232	Portugal	PRT	Population, total	SP.POP.TOTL	
69612	Russian Federation	RUS	Population, total	SP.POP.TOTL	
70647	San Marino	SMR	Population, total	SP.POP.TOTL	
72027	Serbia	SRB	Population, total	SP.POP.TOTL	
73752	Slovak Republic	SVK	Population, total	SP.POP.TOTL	
74097	Slovenia	SVN	Population, total	SP.POP.TOTL	
75822	Spain	ESP	Population, total	SP.POP.TOTL	
78927	Sweden	SWE	Population, total	SP.POP.TOTL	
79272	Switzerland	CHE	Population, total	SP.POP.TOTL	
84447	Ukraine	UKR	Population, total	SP.POP.TOTL	
85137	United Kingdom	GBR	Population, total	SP.POP.TOTL	

	1960	1961	1962	1963	1964	\
14757	1608800.0	1659800.0	1711319.0	1762621.0	1814135.0	
18207	7047539.0	7086299.0	7129864.0	7175811.0	7223801.0	

20277	8198000.0	8271216.0	8351928.0	8437232.0	8524224.0
20622	9153489.0	9183948.0	9220578.0	9289770.0	9378113.0
22692	3214520.0	3277096.0	3341809.0	3406466.0	3468083.0
24417	7867374.0	7943118.0	8012946.0	8078145.0	8144340.0
30972	4140000.0	4171672.0	4202104.0	4231408.0	4259680.0
32352	9602006.0	9586651.0	9624660.0	9670685.0	9727804.0
32697	4579603.0	4611687.0	4647727.0	4684483.0	4722072.0
35802	1211537.0	1225077.0	1241623.0	1258857.0	1277086.0
36492	34266.0	34730.0	35153.0	35550.0	35946.0
37182	4429634.0	4461005.0	4491443.0	4523309.0	4548543.0
37527	46814237.0	47444751.0	48119649.0	48803680.0	49449403.0
39252	72814900.0	73377632.0	74025784.0	74714353.0	75318337.0
39942	21529.0	21797.0	22133.0	22510.0	22894.0
40287	8331725.0	8398050.0	8448233.0	8479625.0	8510429.0
44082	9983967.0	10029321.0	10061734.0	10087947.0	10119835.0
44427	175574.0	179029.0	182378.0	185653.0	188983.0
46152	2828600.0	2824400.0	2836050.0	2852650.0	2866550.0
47187	50199700.0	50536350.0	50879450.0	51252000.0	51675350.0
51672	2120979.0	2152681.0	2181586.0	2210919.0	2240623.0
53397	16504.0	16894.0	17298.0	17722.0	18170.0
53742	2778550.0	2823550.0	2863350.0	2898950.0	2935200.0
54087	313970.0	316845.0	320750.0	324100.0	327750.0
54777	1488664.0	1507653.0	1527105.0	1547445.0	1569143.0
56847	326550.0	325250.0	323900.0	322550.0	321250.0
58917	2544000.0	2605000.0	2664000.0	2721000.0	2774000.0
59262	22454.0	22812.0	23042.0	23169.0	23237.0
59952	480579.0	491140.0	502558.0	513409.0	521753.0
62367	11486631.0	11638712.0	11805689.0	11965966.0	12127120.0
64782	3581239.0	3609800.0	3638918.0	3666537.0	3694339.0
67887	29637450.0	29964000.0	30308500.0	30712000.0	31139450.0
68232	8857716.0	8929316.0	8993985.0	9030355.0	9035365.0
69612	119897000.0	121236000.0	122591000.0	123960000.0	125345000.0
70647	15393.0	15787.0	16197.0	16616.0	17033.0
72027	NaN	NaN	NaN	NaN	NaN
73752	4068095.0	4191667.0	4238188.0	4282017.0	4327341.0
74097	1584720.0	1594131.0	1603649.0	1616971.0	1632114.0
75822	30455000.0	30739250.0	31023366.0	31296651.0	31609195.0
78927	7484656.0	7519998.0	7561588.0	7604328.0	7661354.0
79272	5327827.0	5434294.0	5573815.0	5694247.0	5789228.0
84447	42662150.0	43203635.0	43749469.0	44285897.0	44794325.0
85137	52400000.0	52800000.0	53250000.0	53650000.0	54000000.0

	1965	...	2007	2008	2009 \
14757	1864791.0	...	2970017.0	2947314.0	2927519.0
18207	7270889.0	...	8295487.0	8321496.0	8343323.0
20277	8610000.0	...	9560000.0	9528000.0	9507000.0
20622	9463667.0	...	10625700.0	10709973.0	10796493.0
22692	3524596.0	...	3840418.0	3839749.0	3837732.0

24417	8204168.0	...	7545338.0	7492561.0	7444443.0
30972	4287000.0	...	4436000.0	4434508.0	4429078.0
32352	9779358.0	...	10298828.0	10384603.0	10443936.0
32697	4759012.0	...	5461438.0	5493621.0	5523095.0
35802	1294566.0	...	1340680.0	1337090.0	1334515.0
36492	36363.0	...	48599.0	48629.0	48613.0
37182	4563732.0	...	5288720.0	5313399.0	5338871.0
37527	50023774.0	...	64016229.0	64374990.0	64707044.0
39252	75963695.0	...	82266372.0	82110097.0	81902307.0
39942	23256.0	...	29770.0	30089.0	30407.0
40287	8550333.0	...	11048473.0	11077841.0	11107017.0
44082	10147935.0	...	10055780.0	10038188.0	10022650.0
44427	192286.0	...	311566.0	317414.0	318499.0
46152	2877300.0	...	4398942.0	4489544.0	4535375.0
47187	52112350.0	...	58438310.0	58826731.0	59095365.0
51672	2265919.0	...	2200325.0	2177322.0	2141669.0
53397	18647.0	...	35431.0	35721.0	36004.0
53742	2971450.0	...	3231294.0	3198231.0	3162916.0
54087	331500.0	...	479993.0	488650.0	497783.0
54777	1592434.0	...	2051427.0	2055266.0	2058920.0
56847	318800.0	...	406724.0	409379.0	412477.0
58917	2825000.0	...	3576910.0	3570108.0	3565604.0
59262	23283.0	...	35046.0	35731.0	36351.0
59952	526327.0	...	615875.0	616969.0	618294.0
62367	12294732.0	...	16381696.0	16445593.0	16530388.0
64782	3723168.0	...	4709153.0	4768212.0	4828726.0
67887	31444950.0	...	38120560.0	38125759.0	38151603.0
68232	8998595.0	...	10542964.0	10558177.0	10568247.0
69612	126745000.0	...	142805088.0	142742350.0	142785342.0
70647	17440.0	...	29874.0	30160.0	30431.0
72027	NaN	...	7381579.0	7350222.0	7320807.0
73752	4370983.0	...	5374622.0	5379233.0	5386406.0
74097	1649160.0	...	2018122.0	2021316.0	2039669.0
75822	31954292.0	...	45226803.0	45954106.0	46362946.0
78927	7733853.0	...	9148092.0	9219637.0	9298515.0
79272	5856472.0	...	7551117.0	7647675.0	7743831.0
84447	45261939.0	...	46509350.0	46258200.0	46053300.0
85137	54348050.0	...	61322463.0	61806995.0	62276270.0

	2010	2011	2012	2013	2014 \
14757	2913021.0	2904780.0	2900247.0	2896652.0	2893654.0
18207	8363404.0	8391643.0	8429991.0	8479375.0	8541575.0
20277	9490000.0	9473000.0	9464000.0	9466000.0	9483000.0
20622	10895586.0	11047744.0	11128246.0	11182817.0	11231213.0
22692	3835258.0	3832310.0	3828419.0	3823533.0	3817554.0
24417	7395599.0	7348328.0	7305888.0	7265115.0	7223938.0
30972	4417781.0	4280622.0	4267558.0	4255689.0	4238389.0
32352	10474410.0	10496088.0	10510785.0	10514272.0	10525347.0

32697	5547683.0	5570572.0	5591572.0	5614932.0	5643475.0
35802	1331475.0	1327439.0	1322696.0	1317997.0	1314545.0
36492	48567.0	48492.0	48393.0	48292.0	48221.0
37182	5363352.0	5388272.0	5413971.0	5438972.0	5461512.0
37527	65027512.0	65342776.0	65659790.0	65972097.0	66495940.0
39252	81776930.0	81797673.0	80425823.0	82132753.0	80982500.0
39942	30732.0	31067.0	31402.0	31720.0	31997.0
40287	11121341.0	11104899.0	11045011.0	10965211.0	10892413.0
44082	10000023.0	9971727.0	9920362.0	9893082.0	9866468.0
44427	318041.0	319014.0	320716.0	323764.0	327386.0
46152	4560155.0	4576794.0	4586897.0	4598294.0	4617225.0
47187	59277417.0	59379449.0	59539717.0	60233948.0	60789140.0
51672	2097555.0	2059709.0	2034319.0	2012647.0	1993782.0
53397	36276.0	36537.0	36791.0	37040.0	37286.0
53742	3097282.0	3028115.0	2987773.0	2957689.0	2932367.0
54087	506953.0	518347.0	530946.0	543360.0	556319.0
54777	2062443.0	2065888.0	2069270.0	2072543.0	2075625.0
56847	414508.0	416268.0	419455.0	423374.0	427364.0
58917	3562045.0	3559986.0	3559519.0	3558566.0	3556397.0
59262	36845.0	37189.0	37404.0	37528.0	37623.0
59952	619428.0	620079.0	620601.0	621207.0	621810.0
62367	16615394.0	16693074.0	16754962.0	16804432.0	16865008.0
64782	4889252.0	4953088.0	5018573.0	5079623.0	5137232.0
67887	38042794.0	38063255.0	38063164.0	38040196.0	38011735.0
68232	10573100.0	10557560.0	10514844.0	10457295.0	10401062.0
69612	142849449.0	142960868.0	143201676.0	143506911.0	143819569.0
70647	30690.0	30938.0	31172.0	31391.0	31595.0
72027	7291436.0	7234099.0	7199077.0	7164132.0	7130576.0
73752	5391428.0	5398384.0	5407579.0	5413393.0	5418649.0
74097	2048583.0	2052843.0	2057159.0	2059953.0	2061980.0
75822	46576897.0	46742697.0	46773055.0	46620045.0	46480882.0
78927	9378126.0	9449213.0	9519374.0	9600379.0	9696110.0
79272	7824909.0	7912398.0	7996861.0	8089346.0	8188649.0
84447	45870700.0	45706100.0	45593300.0	45489600.0	45362900.0
85137	62766365.0	63258918.0	63700300.0	64128226.0	64613160.0

	2015	Unnamed: 60
14757	2889167.0	NaN
18207	8611088.0	NaN
20277	9513000.0	NaN
20622	11285721.0	NaN
22692	3810416.0	NaN
24417	7177991.0	NaN
30972	4224404.0	NaN
32352	10551219.0	NaN
32697	5676002.0	NaN
35802	1311998.0	NaN
36492	48199.0	NaN

```
37182    5482013.0          NaN
37527   66808385.0          NaN
39252   81413145.0          NaN
39942      32217.0          NaN
40287   10823732.0          NaN
44082    9844686.0          NaN
44427     330823.0          NaN
46152    4640703.0          NaN
47187   60802085.0          NaN
51672    1978440.0          NaN
53397      37531.0          NaN
53742    2910199.0          NaN
54087     569676.0          NaN
54777    2078453.0          NaN
56847     431333.0          NaN
58917    3554150.0          NaN
59262      37731.0          NaN
59952     622388.0          NaN
62367   16936520.0          NaN
64782    5195921.0          NaN
67887   37999494.0          NaN
68232   10348648.0          NaN
69612  144096812.0          NaN
70647      31781.0          NaN
72027    7098247.0          NaN
73752    5424050.0          NaN
74097    2063768.0          NaN
75822   46418269.0          NaN
78927    9798871.0          NaN
79272    8286976.0          NaN
84447   45198200.0          NaN
85137   65138232.0          NaN

[43 rows x 61 columns]
```

In [6]:
```
data=data.drop(columns=['Country Name','Country Code','Indicator Name','Indicator Code
data
```

Out[6]:
```
              1960         1961         1962         1963         1964  \
14757    1608800.0    1659800.0    1711319.0    1762621.0    1814135.0
18207    7047539.0    7086299.0    7129864.0    7175811.0    7223801.0
20277    8198000.0    8271216.0    8351928.0    8437232.0    8524224.0
20622    9153489.0    9183948.0    9220578.0    9289770.0    9378113.0
22692    3214520.0    3277096.0    3341809.0    3406466.0    3468083.0
24417    7867374.0    7943118.0    8012946.0    8078145.0    8144340.0
30972    4140000.0    4171672.0    4202104.0    4231408.0    4259680.0
32352    9602006.0    9586651.0    9624660.0    9670685.0    9727804.0
32697    4579603.0    4611687.0    4647727.0    4684483.0    4722072.0
```

35802	1211537.0	1225077.0	1241623.0	1258857.0	1277086.0
36492	34266.0	34730.0	35153.0	35550.0	35946.0
37182	4429634.0	4461005.0	4491443.0	4523309.0	4548543.0
37527	46814237.0	47444751.0	48119649.0	48803680.0	49449403.0
39252	72814900.0	73377632.0	74025784.0	74714353.0	75318337.0
39942	21529.0	21797.0	22133.0	22510.0	22894.0
40287	8331725.0	8398050.0	8448233.0	8479625.0	8510429.0
44082	9983967.0	10029321.0	10061734.0	10087947.0	10119835.0
44427	175574.0	179029.0	182378.0	185653.0	188983.0
46152	2828600.0	2824400.0	2836050.0	2852650.0	2866550.0
47187	50199700.0	50536350.0	50879450.0	51252000.0	51675350.0
51672	2120979.0	2152681.0	2181586.0	2210919.0	2240623.0
53397	16504.0	16894.0	17298.0	17722.0	18170.0
53742	2778550.0	2823550.0	2863350.0	2898950.0	2935200.0
54087	313970.0	316845.0	320750.0	324100.0	327750.0
54777	1488664.0	1507653.0	1527105.0	1547445.0	1569143.0
56847	326550.0	325250.0	323900.0	322550.0	321250.0
58917	2544000.0	2605000.0	2664000.0	2721000.0	2774000.0
59262	22454.0	22812.0	23042.0	23169.0	23237.0
59952	480579.0	491140.0	502558.0	513409.0	521753.0
62367	11486631.0	11638712.0	11805689.0	11965966.0	12127120.0
64782	3581239.0	3609800.0	3638918.0	3666537.0	3694339.0
67887	29637450.0	29964000.0	30308500.0	30712000.0	31139450.0
68232	8857716.0	8929316.0	8993985.0	9030355.0	9035365.0
69612	119897000.0	121236000.0	122591000.0	123960000.0	125345000.0
70647	15393.0	15787.0	16197.0	16616.0	17033.0
72027	NaN	NaN	NaN	NaN	NaN
73752	4068095.0	4191667.0	4238188.0	4282017.0	4327341.0
74097	1584720.0	1594131.0	1603649.0	1616971.0	1632114.0
75822	30455000.0	30739250.0	31023366.0	31296651.0	31609195.0
78927	7484656.0	7519998.0	7561588.0	7604328.0	7661354.0
79272	5327827.0	5434294.0	5573815.0	5694247.0	5789228.0
84447	42662150.0	43203635.0	43749469.0	44285897.0	44794325.0
85137	52400000.0	52800000.0	53250000.0	53650000.0	54000000.0

	1965	1966	1967	1968	1969 \
14757	1864791.0	1914573.0	1965598.0	2022272.0	2081695.0
18207	7270889.0	7322066.0	7376998.0	7415403.0	7441055.0
20277	8610000.0	8696496.0	8785648.0	8874552.0	8960304.0
20622	9463667.0	9527807.0	9580991.0	9618756.0	9646032.0
22692	3524596.0	3574972.0	3619997.0	3661642.0	3702834.0
24417	8204168.0	8258057.0	8310226.0	8369603.0	8434172.0
30972	4287000.0	4313000.0	4339000.0	4364000.0	4387000.0
32352	9779358.0	9821040.0	9852899.0	9876346.0	9896580.0
32697	4759012.0	4797381.0	4835354.0	4864883.0	4891860.0
35802	1294566.0	1308597.0	1318946.0	1331214.0	1345249.0
36492	36363.0	36804.0	37258.0	37718.0	38164.0
37182	4563732.0	4580869.0	4605744.0	4626469.0	4623785.0

37527	50023774.0	50508717.0	50915456.0	51276054.0	51638260.0
39252	75963695.0	76600311.0	76951336.0	77294314.0	77909682.0
39942	23256.0	23590.0	23901.0	24185.0	24438.0
40287	8550333.0	8613651.0	8684088.0	8740765.0	8772764.0
44082	10147935.0	10178653.0	10216604.0	10255815.0	10298723.0
44427	192286.0	195570.0	198751.0	201488.0	203369.0
46152	2877300.0	2888800.0	2902450.0	2915550.0	2932650.0
47187	52112350.0	52519000.0	52900500.0	53235750.0	53537950.0
51672	2265919.0	2283217.0	2301220.0	2323619.0	2343173.0
53397	18647.0	19156.0	19691.0	20235.0	20765.0
53742	2971450.0	3008050.0	3044400.0	3078850.0	3107321.0
54087	331500.0	333895.0	334995.0	335850.0	337500.0
54777	1592434.0	1617797.0	1644944.0	1672401.0	1698140.0
56847	318800.0	315200.0	311550.0	307900.0	304300.0
58917	2825000.0	2873000.0	2918000.0	2960000.0	3002000.0
59262	23283.0	23304.0	23297.0	23296.0	23348.0
59952	526327.0	526419.0	522796.0	517481.0	513340.0
62367	12294732.0	12456251.0	12598201.0	12729721.0	12877984.0
64782	3723168.0	3753012.0	3784539.0	3816486.0	3847707.0
67887	31444950.0	31681000.0	31987155.0	32294655.0	32548300.0
68232	8998595.0	8930990.0	8874520.0	8836650.0	8757705.0
69612	126745000.0	127468000.0	128196000.0	128928000.0	129664000.0
70647	17440.0	17839.0	18228.0	18587.0	18895.0
72027	NaN	NaN	NaN	NaN	NaN
73752	4370983.0	4411666.0	4449367.0	4483915.0	4518607.0
74097	1649160.0	1669905.0	1689528.0	1704546.0	1713874.0
75822	31954292.0	32283194.0	32682947.0	33113134.0	33441054.0
78927	7733853.0	7807797.0	7867931.0	7912273.0	7968072.0
79272	5856472.0	5918002.0	5991785.0	6067714.0	6136387.0
84447	45261939.0	45682314.0	46060456.0	46409007.0	46746668.0
85137	54348050.0	54648500.0	54943600.0	55211700.0	55441750.0

	...	2006	2007	2008	2009 \
14757	...	2992547.0	2970017.0	2947314.0	2927519.0
18207	...	8268641.0	8295487.0	8321496.0	8343323.0
20277	...	9604000.0	9560000.0	9528000.0	9507000.0
20622	...	10547958.0	10625700.0	10709973.0	10796493.0
22692	...	3838504.0	3840418.0	3839749.0	3837732.0
24417	...	7699020.0	7545338.0	7492561.0	7444443.0
30972	...	4440000.0	4436000.0	4434508.0	4429078.0
32352	...	10238905.0	10298828.0	10384603.0	10443936.0
32697	...	5437272.0	5461438.0	5493621.0	5523095.0
35802	...	1346810.0	1340680.0	1337090.0	1334515.0
36492	...	48505.0	48599.0	48629.0	48613.0
37182	...	5266268.0	5288720.0	5313399.0	5338871.0
37527	...	63621376.0	64016229.0	64374990.0	64707044.0
39252	...	82376451.0	82266372.0	82110097.0	81902307.0
39942	...	29437.0	29770.0	30089.0	30407.0

40287	...	11020362.0	11048473.0	11077841.0	11107017.0
44082	...	10071370.0	10055780.0	10038188.0	10022650.0
44427	...	303782.0	311566.0	317414.0	318499.0
46152	...	4273591.0	4398942.0	4489544.0	4535375.0
47187	...	58143979.0	58438310.0	58826731.0	59095365.0
51672	...	2218357.0	2200325.0	2177322.0	2141669.0
53397	...	35141.0	35431.0	35721.0	36004.0
53742	...	3269909.0	3231294.0	3198231.0	3162916.0
54087	...	472637.0	479993.0	488650.0	497783.0
54777	...	2047330.0	2051427.0	2055266.0	2058920.0
56847	...	405308.0	406724.0	409379.0	412477.0
58917	...	3585209.0	3576910.0	3570108.0	3565604.0
59262	...	34384.0	35046.0	35731.0	36351.0
59952	...	615025.0	615875.0	616969.0	618294.0
62367	...	16346101.0	16381696.0	16445593.0	16530388.0
64782	...	4660677.0	4709153.0	4768212.0	4828726.0
67887	...	38141267.0	38120560.0	38125759.0	38151603.0
68232	...	10522288.0	10542964.0	10558177.0	10568247.0
69612	...	143049528.0	142805088.0	142742350.0	142785342.0
70647	...	29568.0	29874.0	30160.0	30431.0
72027	...	7411569.0	7381579.0	7350222.0	7320807.0
73752	...	5373054.0	5374622.0	5379233.0	5386406.0
74097	...	2006868.0	2018122.0	2021316.0	2039669.0
75822	...	44397319.0	45226803.0	45954106.0	46362946.0
78927	...	9080505.0	9148092.0	9219637.0	9298515.0
79272	...	7483934.0	7551117.0	7647675.0	7743831.0
84447	...	46787750.0	46509350.0	46258200.0	46053300.0
85137	...	60846820.0	61322463.0	61806995.0	62276270.0

	2010	2011	2012	2013	2014 \
14757	2913021.0	2904780.0	2900247.0	2896652.0	2893654.0
18207	8363404.0	8391643.0	8429991.0	8479375.0	8541575.0
20277	9490000.0	9473000.0	9464000.0	9466000.0	9483000.0
20622	10895586.0	11047744.0	11128246.0	11182817.0	11231213.0
22692	3835258.0	3832310.0	3828419.0	3823533.0	3817554.0
24417	7395599.0	7348328.0	7305888.0	7265115.0	7223938.0
30972	4417781.0	4280622.0	4267558.0	4255689.0	4238389.0
32352	10474410.0	10496088.0	10510785.0	10514272.0	10525347.0
32697	5547683.0	5570572.0	5591572.0	5614932.0	5643475.0
35802	1331475.0	1327439.0	1322696.0	1317997.0	1314545.0
36492	48567.0	48492.0	48393.0	48292.0	48221.0
37182	5363352.0	5388272.0	5413971.0	5438972.0	5461512.0
37527	65027512.0	65342776.0	65659790.0	65972097.0	66495940.0
39252	81776930.0	81797673.0	80425823.0	82132753.0	80982500.0
39942	30732.0	31067.0	31402.0	31720.0	31997.0
40287	11121341.0	11104899.0	11045011.0	10965211.0	10892413.0
44082	10000023.0	9971727.0	9920362.0	9893082.0	9866468.0
44427	318041.0	319014.0	320716.0	323764.0	327386.0

46152	4560155.0	4576794.0	4586897.0	4598294.0	4617225.0
47187	59277417.0	59379449.0	59539717.0	60233948.0	60789140.0
51672	2097555.0	2059709.0	2034319.0	2012647.0	1993782.0
53397	36276.0	36537.0	36791.0	37040.0	37286.0
53742	3097282.0	3028115.0	2987773.0	2957689.0	2932367.0
54087	506953.0	518347.0	530946.0	543360.0	556319.0
54777	2062443.0	2065888.0	2069270.0	2072543.0	2075625.0
56847	414508.0	416268.0	419455.0	423374.0	427364.0
58917	3562045.0	3559986.0	3559519.0	3558566.0	3556397.0
59262	36845.0	37189.0	37404.0	37528.0	37623.0
59952	619428.0	620079.0	620601.0	621207.0	621810.0
62367	16615394.0	16693074.0	16754962.0	16804432.0	16865008.0
64782	4889252.0	4953088.0	5018573.0	5079623.0	5137232.0
67887	38042794.0	38063255.0	38063164.0	38040196.0	38011735.0
68232	10573100.0	10557560.0	10514844.0	10457295.0	10401062.0
69612	142849449.0	142960868.0	143201676.0	143506911.0	143819569.0
70647	30690.0	30938.0	31172.0	31391.0	31595.0
72027	7291436.0	7234099.0	7199077.0	7164132.0	7130576.0
73752	5391428.0	5398384.0	5407579.0	5413393.0	5418649.0
74097	2048583.0	2052843.0	2057159.0	2059953.0	2061980.0
75822	46576897.0	46742697.0	46773055.0	46620045.0	46480882.0
78927	9378126.0	9449213.0	9519374.0	9600379.0	9696110.0
79272	7824909.0	7912398.0	7996861.0	8089346.0	8188649.0
84447	45870700.0	45706100.0	45593300.0	45489600.0	45362900.0
85137	62766365.0	63258918.0	63700300.0	64128226.0	64613160.0

	2015
14757	2889167.0
18207	8611088.0
20277	9513000.0
20622	11285721.0
22692	3810416.0
24417	7177991.0
30972	4224404.0
32352	10551219.0
32697	5676002.0
35802	1311998.0
36492	48199.0
37182	5482013.0
37527	66808385.0
39252	81413145.0
39942	32217.0
40287	10823732.0
44082	9844686.0
44427	330823.0
46152	4640703.0
47187	60802085.0
51672	1978440.0

```
53397      37531.0
53742    2910199.0
54087     569676.0
54777    2078453.0
56847     431333.0
58917    3554150.0
59262      37731.0
59952     622388.0
62367   16936520.0
64782    5195921.0
67887   37999494.0
68232   10348648.0
69612  144096812.0
70647      31781.0
72027    7098247.0
73752    5424050.0
74097    2063768.0
75822   46418269.0
78927    9798871.0
79272    8286976.0
84447   45198200.0
85137   65138232.0

[43 rows x 56 columns]
```

In [7]:
```
data=data.fillna(method ='ffill') #pupunjavamo NaN i inf vrednosti
data
```

Out[7]:
```
               1960         1961         1962         1963         1964  \
14757    1608800.0    1659800.0    1711319.0    1762621.0    1814135.0
18207    7047539.0    7086299.0    7129864.0    7175811.0    7223801.0
20277    8198000.0    8271216.0    8351928.0    8437232.0    8524224.0
20622    9153489.0    9183948.0    9220578.0    9289770.0    9378113.0
22692    3214520.0    3277096.0    3341809.0    3406466.0    3468083.0
24417    7867374.0    7943118.0    8012946.0    8078145.0    8144340.0
30972    4140000.0    4171672.0    4202104.0    4231408.0    4259680.0
32352    9602006.0    9586651.0    9624660.0    9670685.0    9727804.0
32697    4579603.0    4611687.0    4647727.0    4684483.0    4722072.0
35802    1211537.0    1225077.0    1241623.0    1258857.0    1277086.0
36492      34266.0      34730.0      35153.0      35550.0      35946.0
37182    4429634.0    4461005.0    4491443.0    4523309.0    4548543.0
37527   46814237.0   47444751.0   48119649.0   48803680.0   49449403.0
39252   72814900.0   73377632.0   74025784.0   74714353.0   75318337.0
39942      21529.0      21797.0      22133.0      22510.0      22894.0
40287    8331725.0    8398050.0    8448233.0    8479625.0    8510429.0
44082    9983967.0   10029321.0   10061734.0   10087947.0   10119835.0
44427     175574.0     179029.0     182378.0     185653.0     188983.0
46152    2828600.0    2824400.0    2836050.0    2852650.0    2866550.0
```

47187	50199700.0	50536350.0	50879450.0	51252000.0	51675350.0
51672	2120979.0	2152681.0	2181586.0	2210919.0	2240623.0
53397	16504.0	16894.0	17298.0	17722.0	18170.0
53742	2778550.0	2823550.0	2863350.0	2898950.0	2935200.0
54087	313970.0	316845.0	320750.0	324100.0	327750.0
54777	1488664.0	1507653.0	1527105.0	1547445.0	1569143.0
56847	326550.0	325250.0	323900.0	322550.0	321250.0
58917	2544000.0	2605000.0	2664000.0	2721000.0	2774000.0
59262	22454.0	22812.0	23042.0	23169.0	23237.0
59952	480579.0	491140.0	502558.0	513409.0	521753.0
62367	11486631.0	11638712.0	11805689.0	11965966.0	12127120.0
64782	3581239.0	3609800.0	3638918.0	3666537.0	3694339.0
67887	29637450.0	29964000.0	30308500.0	30712000.0	31139450.0
68232	8857716.0	8929316.0	8993985.0	9030355.0	9035365.0
69612	119897000.0	121236000.0	122591000.0	123960000.0	125345000.0
70647	15393.0	15787.0	16197.0	16616.0	17033.0
72027	15393.0	15787.0	16197.0	16616.0	17033.0
73752	4068095.0	4191667.0	4238188.0	4282017.0	4327341.0
74097	1584720.0	1594131.0	1603649.0	1616971.0	1632114.0
75822	30455000.0	30739250.0	31023366.0	31296651.0	31609195.0
78927	7484656.0	7519998.0	7561588.0	7604328.0	7661354.0
79272	5327827.0	5434294.0	5573815.0	5694247.0	5789228.0
84447	42662150.0	43203635.0	43749469.0	44285897.0	44794325.0
85137	52400000.0	52800000.0	53250000.0	53650000.0	54000000.0

	1965	1966	1967	1968	1969 \
14757	1864791.0	1914573.0	1965598.0	2022272.0	2081695.0
18207	7270889.0	7322066.0	7376998.0	7415403.0	7441055.0
20277	8610000.0	8696496.0	8785648.0	8874552.0	8960304.0
20622	9463667.0	9527807.0	9580991.0	9618756.0	9646032.0
22692	3524596.0	3574972.0	3619997.0	3661642.0	3702834.0
24417	8204168.0	8258057.0	8310226.0	8369603.0	8434172.0
30972	4287000.0	4313000.0	4339000.0	4364000.0	4387000.0
32352	9779358.0	9821040.0	9852899.0	9876346.0	9896580.0
32697	4759012.0	4797381.0	4835354.0	4864883.0	4891860.0
35802	1294566.0	1308597.0	1318946.0	1331214.0	1345249.0
36492	36363.0	36804.0	37258.0	37718.0	38164.0
37182	4563732.0	4580869.0	4605744.0	4626469.0	4623785.0
37527	50023774.0	50508717.0	50915456.0	51276054.0	51638260.0
39252	75963695.0	76600311.0	76951336.0	77294314.0	77909682.0
39942	23256.0	23590.0	23901.0	24185.0	24438.0
40287	8550333.0	8613651.0	8684088.0	8740765.0	8772764.0
44082	10147935.0	10178653.0	10216604.0	10255815.0	10298723.0
44427	192286.0	195570.0	198751.0	201488.0	203369.0
46152	2877300.0	2888800.0	2902450.0	2915550.0	2932650.0
47187	52112350.0	52519000.0	52900500.0	53235750.0	53537950.0
51672	2265919.0	2283217.0	2301220.0	2323619.0	2343173.0
53397	18647.0	19156.0	19691.0	20235.0	20765.0

53742	2971450.0	3008050.0	3044400.0	3078850.0	3107321.0
54087	331500.0	333895.0	334995.0	335850.0	337500.0
54777	1592434.0	1617797.0	1644944.0	1672401.0	1698140.0
56847	318800.0	315200.0	311550.0	307900.0	304300.0
58917	2825000.0	2873000.0	2918000.0	2960000.0	3002000.0
59262	23283.0	23304.0	23297.0	23296.0	23348.0
59952	526327.0	526419.0	522796.0	517481.0	513340.0
62367	12294732.0	12456251.0	12598201.0	12729721.0	12877984.0
64782	3723168.0	3753012.0	3784539.0	3816486.0	3847707.0
67887	31444950.0	31681000.0	31987155.0	32294655.0	32548300.0
68232	8998595.0	8930990.0	8874520.0	8836650.0	8757705.0
69612	126745000.0	127468000.0	128196000.0	128928000.0	129664000.0
70647	17440.0	17839.0	18228.0	18587.0	18895.0
72027	17440.0	17839.0	18228.0	18587.0	18895.0
73752	4370983.0	4411666.0	4449367.0	4483915.0	4518607.0
74097	1649160.0	1669905.0	1689528.0	1704546.0	1713874.0
75822	31954292.0	32283194.0	32682947.0	33113134.0	33441054.0
78927	7733853.0	7807797.0	7867931.0	7912273.0	7968072.0
79272	5856472.0	5918002.0	5991785.0	6067714.0	6136387.0
84447	45261939.0	45682314.0	46060456.0	46409007.0	46746668.0
85137	54348050.0	54648500.0	54943600.0	55211700.0	55441750.0

	...	2006	2007	2008	2009 \
14757	...	2992547.0	2970017.0	2947314.0	2927519.0
18207	...	8268641.0	8295487.0	8321496.0	8343323.0
20277	...	9604000.0	9560000.0	9528000.0	9507000.0
20622	...	10547958.0	10625700.0	10709973.0	10796493.0
22692	...	3838504.0	3840418.0	3839749.0	3837732.0
24417	...	7699020.0	7545338.0	7492561.0	7444443.0
30972	...	4440000.0	4436000.0	4434508.0	4429078.0
32352	...	10238905.0	10298828.0	10384603.0	10443936.0
32697	...	5437272.0	5461438.0	5493621.0	5523095.0
35802	...	1346810.0	1340680.0	1337090.0	1334515.0
36492	...	48505.0	48599.0	48629.0	48613.0
37182	...	5266268.0	5288720.0	5313399.0	5338871.0
37527	...	63621376.0	64016229.0	64374990.0	64707044.0
39252	...	82376451.0	82266372.0	82110097.0	81902307.0
39942	...	29437.0	29770.0	30089.0	30407.0
40287	...	11020362.0	11048473.0	11077841.0	11107017.0
44082	...	10071370.0	10055780.0	10038188.0	10022650.0
44427	...	303782.0	311566.0	317414.0	318499.0
46152	...	4273591.0	4398942.0	4489544.0	4535375.0
47187	...	58143979.0	58438310.0	58826731.0	59095365.0
51672	...	2218357.0	2200325.0	2177322.0	2141669.0
53397	...	35141.0	35431.0	35721.0	36004.0
53742	...	3269909.0	3231294.0	3198231.0	3162916.0
54087	...	472637.0	479993.0	488650.0	497783.0
54777	...	2047330.0	2051427.0	2055266.0	2058920.0

56847	...	405308.0	406724.0	409379.0	412477.0
58917	...	3585209.0	3576910.0	3570108.0	3565604.0
59262	...	34384.0	35046.0	35731.0	36351.0
59952	...	615025.0	615875.0	616969.0	618294.0
62367	...	16346101.0	16381696.0	16445593.0	16530388.0
64782	...	4660677.0	4709153.0	4768212.0	4828726.0
67887	...	38141267.0	38120560.0	38125759.0	38151603.0
68232	...	10522288.0	10542964.0	10558177.0	10568247.0
69612	...	143049528.0	142805088.0	142742350.0	142785342.0
70647	...	29568.0	29874.0	30160.0	30431.0
72027	...	7411569.0	7381579.0	7350222.0	7320807.0
73752	...	5373054.0	5374622.0	5379233.0	5386406.0
74097	...	2006868.0	2018122.0	2021316.0	2039669.0
75822	...	44397319.0	45226803.0	45954106.0	46362946.0
78927	...	9080505.0	9148092.0	9219637.0	9298515.0
79272	...	7483934.0	7551117.0	7647675.0	7743831.0
84447	...	46787750.0	46509350.0	46258200.0	46053300.0
85137	...	60846820.0	61322463.0	61806995.0	62276270.0

	2010	2011	2012	2013	2014 \
14757	2913021.0	2904780.0	2900247.0	2896652.0	2893654.0
18207	8363404.0	8391643.0	8429991.0	8479375.0	8541575.0
20277	9490000.0	9473000.0	9464000.0	9466000.0	9483000.0
20622	10895586.0	11047744.0	11128246.0	11182817.0	11231213.0
22692	3835258.0	3832310.0	3828419.0	3823533.0	3817554.0
24417	7395599.0	7348328.0	7305888.0	7265115.0	7223938.0
30972	4417781.0	4280622.0	4267558.0	4255689.0	4238389.0
32352	10474410.0	10496088.0	10510785.0	10514272.0	10525347.0
32697	5547683.0	5570572.0	5591572.0	5614932.0	5643475.0
35802	1331475.0	1327439.0	1322696.0	1317997.0	1314545.0
36492	48567.0	48492.0	48393.0	48292.0	48221.0
37182	5363352.0	5388272.0	5413971.0	5438972.0	5461512.0
37527	65027512.0	65342776.0	65659790.0	65972097.0	66495940.0
39252	81776930.0	81797673.0	80425823.0	82132753.0	80982500.0
39942	30732.0	31067.0	31402.0	31720.0	31997.0
40287	11121341.0	11104899.0	11045011.0	10965211.0	10892413.0
44082	10000023.0	9971727.0	9920362.0	9893082.0	9866468.0
44427	318041.0	319014.0	320716.0	323764.0	327386.0
46152	4560155.0	4576794.0	4586897.0	4598294.0	4617225.0
47187	59277417.0	59379449.0	59539717.0	60233948.0	60789140.0
51672	2097555.0	2059709.0	2034319.0	2012647.0	1993782.0
53397	36276.0	36537.0	36791.0	37040.0	37286.0
53742	3097282.0	3028115.0	2987773.0	2957689.0	2932367.0
54087	506953.0	518347.0	530946.0	543360.0	556319.0
54777	2062443.0	2065888.0	2069270.0	2072543.0	2075625.0
56847	414508.0	416268.0	419455.0	423374.0	427364.0
58917	3562045.0	3559986.0	3559519.0	3558566.0	3556397.0
59262	36845.0	37189.0	37404.0	37528.0	37623.0

59952	619428.0	620079.0	620601.0	621207.0	621810.0
62367	16615394.0	16693074.0	16754962.0	16804432.0	16865008.0
64782	4889252.0	4953088.0	5018573.0	5079623.0	5137232.0
67887	38042794.0	38063255.0	38063164.0	38040196.0	38011735.0
68232	10573100.0	10557560.0	10514844.0	10457295.0	10401062.0
69612	142849449.0	142960868.0	143201676.0	143506911.0	143819569.0
70647	30690.0	30938.0	31172.0	31391.0	31595.0
72027	7291436.0	7234099.0	7199077.0	7164132.0	7130576.0
73752	5391428.0	5398384.0	5407579.0	5413393.0	5418649.0
74097	2048583.0	2052843.0	2057159.0	2059953.0	2061980.0
75822	46576897.0	46742697.0	46773055.0	46620045.0	46480882.0
78927	9378126.0	9449213.0	9519374.0	9600379.0	9696110.0
79272	7824909.0	7912398.0	7996861.0	8089346.0	8188649.0
84447	45870700.0	45706100.0	45593300.0	45489600.0	45362900.0
85137	62766365.0	63258918.0	63700300.0	64128226.0	64613160.0

	2015
14757	2889167.0
18207	8611088.0
20277	9513000.0
20622	11285721.0
22692	3810416.0
24417	7177991.0
30972	4224404.0
32352	10551219.0
32697	5676002.0
35802	1311998.0
36492	48199.0
37182	5482013.0
37527	66808385.0
39252	81413145.0
39942	32217.0
40287	10823732.0
44082	9844686.0
44427	330823.0
46152	4640703.0
47187	60802085.0
51672	1978440.0
53397	37531.0
53742	2910199.0
54087	569676.0
54777	2078453.0
56847	431333.0
58917	3554150.0
59262	37731.0
59952	622388.0
62367	16936520.0
64782	5195921.0

```
67887   37999494.0
68232   10348648.0
69612  144096812.0
70647      31781.0
72027    7098247.0
73752    5424050.0
74097    2063768.0
75822   46418269.0
78927    9798871.0
79272    8286976.0
84447   45198200.0
85137   65138232.0

[43 rows x 56 columns]
```

In [8]:
```
data =data.reset_index()
data=data.drop(columns=['index']) #reindeksiramo vrste
data
```

Out[8]:

	1960	1961	1962	1963	1964	\
0	1608800.0	1659800.0	1711319.0	1762621.0	1814135.0	
1	7047539.0	7086299.0	7129864.0	7175811.0	7223801.0	
2	8198000.0	8271216.0	8351928.0	8437232.0	8524224.0	
3	9153489.0	9183948.0	9220578.0	9289770.0	9378113.0	
4	3214520.0	3277096.0	3341809.0	3406466.0	3468083.0	
5	7867374.0	7943118.0	8012946.0	8078145.0	8144340.0	
6	4140000.0	4171672.0	4202104.0	4231408.0	4259680.0	
7	9602006.0	9586651.0	9624660.0	9670685.0	9727804.0	
8	4579603.0	4611687.0	4647727.0	4684483.0	4722072.0	
9	1211537.0	1225077.0	1241623.0	1258857.0	1277086.0	
10	34266.0	34730.0	35153.0	35550.0	35946.0	
11	4429634.0	4461005.0	4491443.0	4523309.0	4548543.0	
12	46814237.0	47444751.0	48119649.0	48803680.0	49449403.0	
13	72814900.0	73377632.0	74025784.0	74714353.0	75318337.0	
14	21529.0	21797.0	22133.0	22510.0	22894.0	
15	8331725.0	8398050.0	8448233.0	8479625.0	8510429.0	
16	9983967.0	10029321.0	10061734.0	10087947.0	10119835.0	
17	175574.0	179029.0	182378.0	185653.0	188983.0	
18	2828600.0	2824400.0	2836050.0	2852650.0	2866550.0	
19	50199700.0	50536350.0	50879450.0	51252000.0	51675350.0	
20	2120979.0	2152681.0	2181586.0	2210919.0	2240623.0	
21	16504.0	16894.0	17298.0	17722.0	18170.0	
22	2778550.0	2823550.0	2863350.0	2898950.0	2935200.0	
23	313970.0	316845.0	320750.0	324100.0	327750.0	
24	1488664.0	1507653.0	1527105.0	1547445.0	1569143.0	
25	326550.0	325250.0	323900.0	322550.0	321250.0	
26	2544000.0	2605000.0	2664000.0	2721000.0	2774000.0	
27	22454.0	22812.0	23042.0	23169.0	23237.0	

28	480579.0	491140.0	502558.0	513409.0	521753.0
29	11486631.0	11638712.0	11805689.0	11965966.0	12127120.0
30	3581239.0	3609800.0	3638918.0	3666537.0	3694339.0
31	29637450.0	29964000.0	30308500.0	30712000.0	31139450.0
32	8857716.0	8929316.0	8993985.0	9030355.0	9035365.0
33	119897000.0	121236000.0	122591000.0	123960000.0	125345000.0
34	15393.0	15787.0	16197.0	16616.0	17033.0
35	15393.0	15787.0	16197.0	16616.0	17033.0
36	4068095.0	4191667.0	4238188.0	4282017.0	4327341.0
37	1584720.0	1594131.0	1603649.0	1616971.0	1632114.0
38	30455000.0	30739250.0	31023366.0	31296651.0	31609195.0
39	7484656.0	7519998.0	7561588.0	7604328.0	7661354.0
40	5327827.0	5434294.0	5573815.0	5694247.0	5789228.0
41	42662150.0	43203635.0	43749469.0	44285897.0	44794325.0
42	52400000.0	52800000.0	53250000.0	53650000.0	54000000.0

	1965	1966	1967	1968	1969 \
0	1864791.0	1914573.0	1965598.0	2022272.0	2081695.0
1	7270889.0	7322066.0	7376998.0	7415403.0	7441055.0
2	8610000.0	8696496.0	8785648.0	8874552.0	8960304.0
3	9463667.0	9527807.0	9580991.0	9618756.0	9646032.0
4	3524596.0	3574972.0	3619997.0	3661642.0	3702834.0
5	8204168.0	8258057.0	8310226.0	8369603.0	8434172.0
6	4287000.0	4313000.0	4339000.0	4364000.0	4387000.0
7	9779358.0	9821040.0	9852899.0	9876346.0	9896580.0
8	4759012.0	4797381.0	4835354.0	4864883.0	4891860.0
9	1294566.0	1308597.0	1318946.0	1331214.0	1345249.0
10	36363.0	36804.0	37258.0	37718.0	38164.0
11	4563732.0	4580869.0	4605744.0	4626469.0	4623785.0
12	50023774.0	50508717.0	50915456.0	51276054.0	51638260.0
13	75963695.0	76600311.0	76951336.0	77294314.0	77909682.0
14	23256.0	23590.0	23901.0	24185.0	24438.0
15	8550333.0	8613651.0	8684088.0	8740765.0	8772764.0
16	10147935.0	10178653.0	10216604.0	10255815.0	10298723.0
17	192286.0	195570.0	198751.0	201488.0	203369.0
18	2877300.0	2888800.0	2902450.0	2915550.0	2932650.0
19	52112350.0	52519000.0	52900500.0	53235750.0	53537950.0
20	2265919.0	2283217.0	2301220.0	2323619.0	2343173.0
21	18647.0	19156.0	19691.0	20235.0	20765.0
22	2971450.0	3008050.0	3044400.0	3078850.0	3107321.0
23	331500.0	333895.0	334995.0	335850.0	337500.0
24	1592434.0	1617797.0	1644944.0	1672401.0	1698140.0
25	318800.0	315200.0	311550.0	307900.0	304300.0
26	2825000.0	2873000.0	2918000.0	2960000.0	3002000.0
27	23283.0	23304.0	23297.0	23296.0	23348.0
28	526327.0	526419.0	522796.0	517481.0	513340.0
29	12294732.0	12456251.0	12598201.0	12729721.0	12877984.0
30	3723168.0	3753012.0	3784539.0	3816486.0	3847707.0

31	31444950.0	31681000.0	31987155.0	32294655.0	32548300.0
32	8998595.0	8930990.0	8874520.0	8836650.0	8757705.0
33	126745000.0	127468000.0	128196000.0	128928000.0	129664000.0
34	17440.0	17839.0	18228.0	18587.0	18895.0
35	17440.0	17839.0	18228.0	18587.0	18895.0
36	4370983.0	4411666.0	4449367.0	4483915.0	4518607.0
37	1649160.0	1669905.0	1689528.0	1704546.0	1713874.0
38	31954292.0	32283194.0	32682947.0	33113134.0	33441054.0
39	7733853.0	7807797.0	7867931.0	7912273.0	7968072.0
40	5856472.0	5918002.0	5991785.0	6067714.0	6136387.0
41	45261939.0	45682314.0	46060456.0	46409007.0	46746668.0
42	54348050.0	54648500.0	54943600.0	55211700.0	55441750.0

	...	2006	2007	2008	2009	\
0	...	2992547.0	2970017.0	2947314.0	2927519.0	
1	...	8268641.0	8295487.0	8321496.0	8343323.0	
2	...	9604000.0	9560000.0	9528000.0	9507000.0	
3	...	10547958.0	10625700.0	10709973.0	10796493.0	
4	...	3838504.0	3840418.0	3839749.0	3837732.0	
5	...	7699020.0	7545338.0	7492561.0	7444443.0	
6	...	4440000.0	4436000.0	4434508.0	4429078.0	
7	...	10238905.0	10298828.0	10384603.0	10443936.0	
8	...	5437272.0	5461438.0	5493621.0	5523095.0	
9	...	1346810.0	1340680.0	1337090.0	1334515.0	
10	...	48505.0	48599.0	48629.0	48613.0	
11	...	5266268.0	5288720.0	5313399.0	5338871.0	
12	...	63621376.0	64016229.0	64374990.0	64707044.0	
13	...	82376451.0	82266372.0	82110097.0	81902307.0	
14	...	29437.0	29770.0	30089.0	30407.0	
15	...	11020362.0	11048473.0	11077841.0	11107017.0	
16	...	10071370.0	10055780.0	10038188.0	10022650.0	
17	...	303782.0	311566.0	317414.0	318499.0	
18	...	4273591.0	4398942.0	4489544.0	4535375.0	
19	...	58143979.0	58438310.0	58826731.0	59095365.0	
20	...	2218357.0	2200325.0	2177322.0	2141669.0	
21	...	35141.0	35431.0	35721.0	36004.0	
22	...	3269909.0	3231294.0	3198231.0	3162916.0	
23	...	472637.0	479993.0	488650.0	497783.0	
24	...	2047330.0	2051427.0	2055266.0	2058920.0	
25	...	405308.0	406724.0	409379.0	412477.0	
26	...	3585209.0	3576910.0	3570108.0	3565604.0	
27	...	34384.0	35046.0	35731.0	36351.0	
28	...	615025.0	615875.0	616969.0	618294.0	
29	...	16346101.0	16381696.0	16445593.0	16530388.0	
30	...	4660677.0	4709153.0	4768212.0	4828726.0	
31	...	38141267.0	38120560.0	38125759.0	38151603.0	
32	...	10522288.0	10542964.0	10558177.0	10568247.0	
33	...	143049528.0	142805088.0	142742350.0	142785342.0	

34	...	29568.0	29874.0	30160.0	30431.0
35	...	7411569.0	7381579.0	7350222.0	7320807.0
36	...	5373054.0	5374622.0	5379233.0	5386406.0
37	...	2006868.0	2018122.0	2021316.0	2039669.0
38	...	44397319.0	45226803.0	45954106.0	46362946.0
39	...	9080505.0	9148092.0	9219637.0	9298515.0
40	...	7483934.0	7551117.0	7647675.0	7743831.0
41	...	46787750.0	46509350.0	46258200.0	46053300.0
42	...	60846820.0	61322463.0	61806995.0	62276270.0

	2010	2011	2012	2013	2014 \
0	2913021.0	2904780.0	2900247.0	2896652.0	2893654.0
1	8363404.0	8391643.0	8429991.0	8479375.0	8541575.0
2	9490000.0	9473000.0	9464000.0	9466000.0	9483000.0
3	10895586.0	11047744.0	11128246.0	11182817.0	11231213.0
4	3835258.0	3832310.0	3828419.0	3823533.0	3817554.0
5	7395599.0	7348328.0	7305888.0	7265115.0	7223938.0
6	4417781.0	4280622.0	4267558.0	4255689.0	4238389.0
7	10474410.0	10496088.0	10510785.0	10514272.0	10525347.0
8	5547683.0	5570572.0	5591572.0	5614932.0	5643475.0
9	1331475.0	1327439.0	1322696.0	1317997.0	1314545.0
10	48567.0	48492.0	48393.0	48292.0	48221.0
11	5363352.0	5388272.0	5413971.0	5438972.0	5461512.0
12	65027512.0	65342776.0	65659790.0	65972097.0	66495940.0
13	81776930.0	81797673.0	80425823.0	82132753.0	80982500.0
14	30732.0	31067.0	31402.0	31720.0	31997.0
15	11121341.0	11104899.0	11045011.0	10965211.0	10892413.0
16	10000023.0	9971727.0	9920362.0	9893082.0	9866468.0
17	318041.0	319014.0	320716.0	323764.0	327386.0
18	4560155.0	4576794.0	4586897.0	4598294.0	4617225.0
19	59277417.0	59379449.0	59539717.0	60233948.0	60789140.0
20	2097555.0	2059709.0	2034319.0	2012647.0	1993782.0
21	36276.0	36537.0	36791.0	37040.0	37286.0
22	3097282.0	3028115.0	2987773.0	2957689.0	2932367.0
23	506953.0	518347.0	530946.0	543360.0	556319.0
24	2062443.0	2065888.0	2069270.0	2072543.0	2075625.0
25	414508.0	416268.0	419455.0	423374.0	427364.0
26	3562045.0	3559986.0	3559519.0	3558566.0	3556397.0
27	36845.0	37189.0	37404.0	37528.0	37623.0
28	619428.0	620079.0	620601.0	621207.0	621810.0
29	16615394.0	16693074.0	16754962.0	16804432.0	16865008.0
30	4889252.0	4953088.0	5018573.0	5079623.0	5137232.0
31	38042794.0	38063255.0	38063164.0	38040196.0	38011735.0
32	10573100.0	10557560.0	10514844.0	10457295.0	10401062.0
33	142849449.0	142960868.0	143201676.0	143506911.0	143819569.0
34	30690.0	30938.0	31172.0	31391.0	31595.0
35	7291436.0	7234099.0	7199077.0	7164132.0	7130576.0
36	5391428.0	5398384.0	5407579.0	5413393.0	5418649.0

37	2048583.0	2052843.0	2057159.0	2059953.0	2061980.0
38	46576897.0	46742697.0	46773055.0	46620045.0	46480882.0
39	9378126.0	9449213.0	9519374.0	9600379.0	9696110.0
40	7824909.0	7912398.0	7996861.0	8089346.0	8188649.0
41	45870700.0	45706100.0	45593300.0	45489600.0	45362900.0
42	62766365.0	63258918.0	63700300.0	64128226.0	64613160.0

	2015
0	2889167.0
1	8611088.0
2	9513000.0
3	11285721.0
4	3810416.0
5	7177991.0
6	4224404.0
7	10551219.0
8	5676002.0
9	1311998.0
10	48199.0
11	5482013.0
12	66808385.0
13	81413145.0
14	32217.0
15	10823732.0
16	9844686.0
17	330823.0
18	4640703.0
19	60802085.0
20	1978440.0
21	37531.0
22	2910199.0
23	569676.0
24	2078453.0
25	431333.0
26	3554150.0
27	37731.0
28	622388.0
29	16936520.0
30	5195921.0
31	37999494.0
32	10348648.0
33	144096812.0
34	31781.0
35	7098247.0
36	5424050.0
37	2063768.0
38	46418269.0
39	9798871.0

```
40    8286976.0
41   45198200.0
42   65138232.0

[43 rows x 56 columns]
```

Nai podaci su sada spremni za dalju analizu. Prvi na cilj je da definiemo funkciju koja e iscrtavati Gerschgorin-ove krugove za zadatu matricu, kao i karakteristine korene na istom grafiku. Pored toga, ispisaemo i vrednosti karakteristinih korena date matrice u okviru svakog grafika, sve u cilju bolje reprezentacije istraivanja. Formirajmo sada prethodno navedenu funkciju.

```
In [9]: def myGerschgorin(A):

            n = len(A)
            eval,evec = la.eig(A)
            print(eval) #ispisuje vrednosti karakteristinih korena za matricu A

            patches = []

            for i in range(n):
                xi = np.real(A[i,i])
                yi = np.imag(A[i,i]) #ove dve take e biti koordinate centra Gerschgorin-ovog k
                ri = np.sum(np.abs(A[i,:])) - np.abs(A[i,i]) #poluprenik Gerschgorin-ovog krug

                circle = Circle((xi, yi), ri)
                patches.append(circle) #iscrtava krugove

            fig, ax = plt.subplots()

            p = PatchCollection(patches, cmap=matplotlib.cm.jet, alpha=0.1)
            ax.add_collection(p)
            plt.axis('equal')

            for xi, yi in zip(np.real(eval), np.imag(eval)):
                plt.plot(xi, yi,'o') #iscrtava karakteristine korene od A

            plt.show()
```

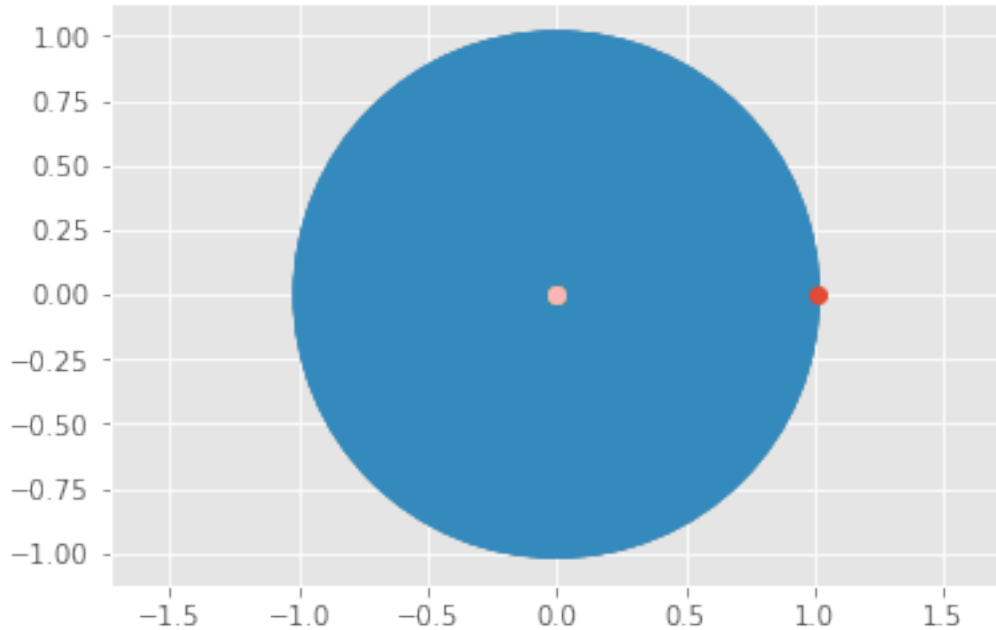
U nastavku pravimo Vondermonde-ove matrice od vrsta koje ine na data. Drugim reima, formiramo Vondermonde-ove matrice za svaku evropsku zemlju(posmatramo globalno zdravlje u Evropi, tj. njenih zemalja, po godinama). Ovo emo uiniti pomou for petlje koju emo istovremeno iskoristiti i za iscrtavanje Gerschgorin-ovih krugova i karakteristinih krugova za Vondermonde-ovu matricu svake evropske zemlje. Kako karakteristine korene traimo kod kvadratnih matrica i kako imamo 56 godina(vektor vrsta je dimenzije 56), zakljuujemo da e vrednosti u poslednjim kolonama Vondermonde-ove matrice biti jako velike(raunar ne moe da ih skladiti). Stoga emo, pre svega, pronai maksimalnu vrednosti u data-e, a potom njenom recipronom vrednou skalirati sve vrednosti data-e.

```
In [10]: m=np.max(data)
         M=np.max(m) #max of data
         M
```

```
Out[10]: 148689000.0
```

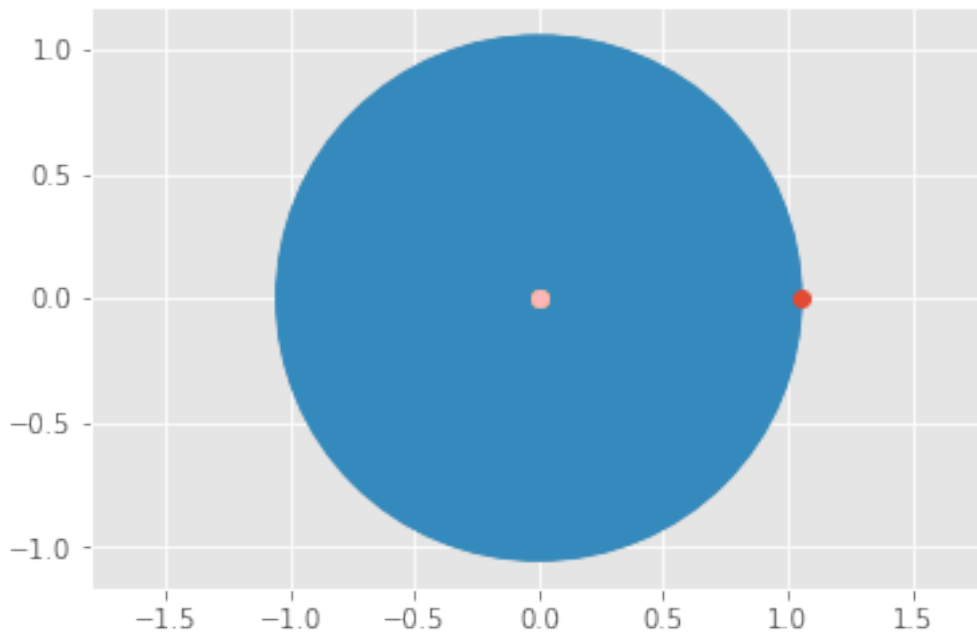
```
In [11]: for i in list(range(data.shape[0])):
             vector=data.iloc[i].values/M
             V=np.vander(vector,len(vector),increasing=True)
             myGerschgorin(V)
```

```
[ 1.01094209e+000+0.00000000e+000j  3.54710824e-004+0.00000000e+000j
  2.52790624e-007+0.00000000e+000j  2.67977322e-010+0.00000000e+000j
  3.79927698e-013+0.00000000e+000j  6.51868881e-016+0.00000000e+000j
  1.35459623e-018+0.00000000e+000j  3.73318988e-021+0.00000000e+000j
  1.27943579e-023+0.00000000e+000j  4.65406291e-026+0.00000000e+000j
  2.28145946e-028+0.00000000e+000j  2.21518976e-029+0.00000000e+000j
 -1.63200937e-031+0.00000000e+000j  1.72313561e-032+0.00000000e+000j
 -7.07612319e-034+0.00000000e+000j -2.32461414e-035+0.00000000e+000j
 -5.33166154e-037+0.00000000e+000j  1.37003077e-037+0.00000000e+000j
 -1.53329593e-039+0.00000000e+000j  5.12571628e-041+0.00000000e+000j
  3.25252177e-043+0.00000000e+000j -4.47705867e-045+0.00000000e+000j
  2.19086108e-045+0.00000000e+000j  2.00569607e-047+0.00000000e+000j
  1.63621235e-048+0.00000000e+000j -3.02425763e-050+0.00000000e+000j
  5.94896786e-052+0.00000000e+000j  5.07862492e-054+0.00000000e+000j
  2.41595502e-055+0.00000000e+000j -3.47394140e-057+0.00000000e+000j
 -3.15248888e-058+0.00000000e+000j -1.18320926e-060+0.00000000e+000j
  1.15796096e-062+0.00000000e+000j -1.93302588e-066+0.00000000e+000j
  5.51376972e-068+0.00000000e+000j  7.03344992e-072+4.24688620e-071j
  7.03344992e-072-4.24688620e-071j  2.56376187e-073+0.00000000e+000j
 -4.84477308e-075+0.00000000e+000j -7.46481653e-078+0.00000000e+000j
 -5.48801170e-080+0.00000000e+000j -9.49710765e-082+0.00000000e+000j
 -2.29946726e-083+0.00000000e+000j  1.38224500e-085+0.00000000e+000j
  6.06864117e-088+0.00000000e+000j  8.21176446e-089+0.00000000e+000j
 -1.49097740e-090+0.00000000e+000j  2.29762435e-092+0.00000000e+000j
 -1.14203673e-094+0.00000000e+000j -1.01214286e-096+0.00000000e+000j
 -2.84844572e-098+0.00000000e+000j -2.42946327e-100+0.00000000e+000j
  4.71849673e-103+4.86997467e-103j  4.71849673e-103-4.86997467e-103j
 -7.93217518e-106+0.00000000e+000j  1.19038745e-107+0.00000000e+000j]
```




```
[ 1.04977057e+00+0.00000000e+00j  3.05360017e-04+0.00000000e+00j
  2.10974264e-07+0.00000000e+00j  2.32316118e-10+0.00000000e+00j
  3.50326183e-13+0.00000000e+00j  6.86488633e-16+0.00000000e+00j
  1.30882921e-18+0.00000000e+00j  2.78194186e-21+0.00000000e+00j
  9.70732935e-24+2.38399337e-23j  9.70732935e-24-2.38399337e-23j
 -1.80449225e-24+0.00000000e+00j -3.18189625e-27+1.70197049e-26j
 -3.18189625e-27-1.70197049e-26j  9.05701107e-28+0.00000000e+00j
  1.70084758e-29+0.00000000e+00j -3.35162230e-30+0.00000000e+00j
 -6.32769788e-34+6.85779888e-33j -6.32769788e-34-6.85779888e-33j
 -1.23888031e-35+0.00000000e+00j -7.32887329e-37+0.00000000e+00j
  8.95078846e-38+0.00000000e+00j  6.18192778e-39+0.00000000e+00j
 -7.91476081e-40+0.00000000e+00j  1.44236862e-41+0.00000000e+00j
  1.20195432e-42+0.00000000e+00j -1.94681841e-44+0.00000000e+00j
  1.23254065e-45+0.00000000e+00j -8.07810923e-47+0.00000000e+00j
 -4.34481931e-47+0.00000000e+00j  1.52655508e-48+0.00000000e+00j
  2.64724327e-49+6.54795068e-49j  2.64724327e-49-6.54795068e-49j
 -6.19587936e-50+0.00000000e+00j  2.64089977e-52+4.21910734e-52j
  2.64089977e-52-4.21910734e-52j  5.10032911e-55+0.00000000e+00j
 -2.78437045e-56+0.00000000e+00j -1.17010181e-56+0.00000000e+00j
  1.74054065e-58+0.00000000e+00j  2.85312856e-60+0.00000000e+00j
 -5.79939714e-61+0.00000000e+00j  1.54526187e-62+0.00000000e+00j
 -9.28009516e-63+0.00000000e+00j  2.86348963e-63+0.00000000e+00j
 -5.37428449e-64+0.00000000e+00j -1.12670436e-65+0.00000000e+00j
  1.04372075e-66+0.00000000e+00j -1.43456082e-67+0.00000000e+00j
  2.37337604e-69+0.00000000e+00j -1.41292547e-70+0.00000000e+00j
  9.01639791e-73+0.00000000e+00j -2.80281001e-73+0.00000000e+00j
```

```
  9.15598500e-74+0.00000000e+00j  5.95890003e-75+0.00000000e+00j
 -2.18239076e-75+0.00000000e+00j  1.73976676e-76+0.00000000e+00j]
```




```
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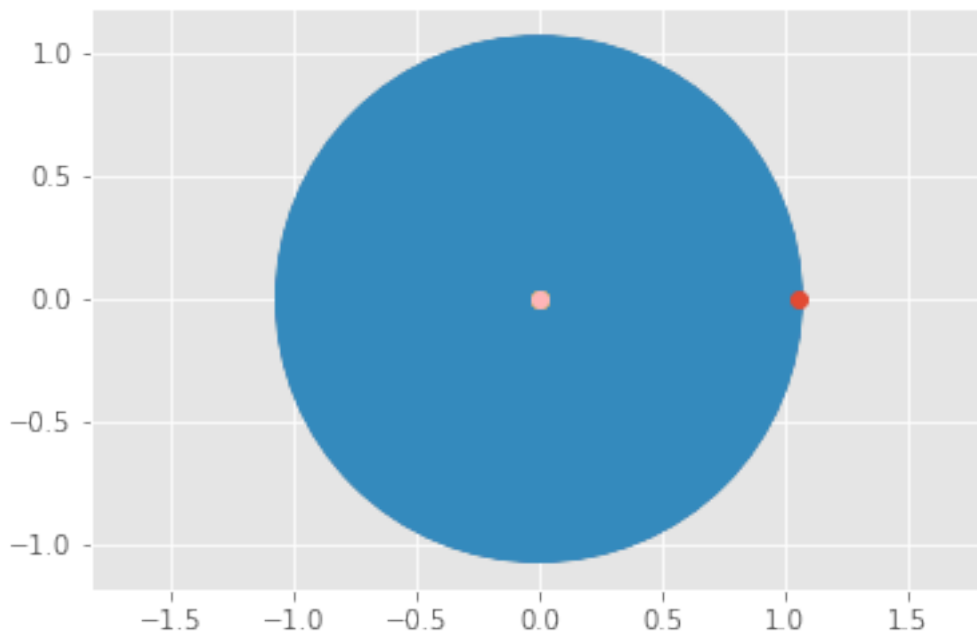
  7.65058450e-07+0.00000000e+00j  1.56028864e-09+0.00000000e+00j
  4.22566980e-12+0.00000000e+00j  1.44747339e-14+0.00000000e+00j
  5.82444916e-17+0.00000000e+00j  2.54708315e-19+0.00000000e+00j
  9.24791540e-22+0.00000000e+00j -1.41537859e-22+0.00000000e+00j
  6.78679472e-25+9.60838647e-25j  6.78679472e-25-9.60838647e-25j
  5.06857275e-27+2.80528528e-26j  5.06857275e-27-2.80528528e-26j
 -2.66297697e-28+1.50979306e-28j -2.66297697e-28-1.50979306e-28j
 -3.95330775e-29+0.00000000e+00j  4.52112791e-30+0.00000000e+00j
 -3.37830821e-30+0.00000000e+00j  8.69176656e-34+1.43125199e-33j
  8.69176656e-34-1.43125199e-33j -1.01917418e-34+0.00000000e+00j
  1.46920266e-35+0.00000000e+00j -3.37050857e-36+0.00000000e+00j
 -1.14147311e-37+0.00000000e+00j  2.97461716e-39+1.96249827e-39j
  2.97461716e-39-1.96249827e-39j -1.78693782e-39+0.00000000e+00j
 -7.27507504e-42+5.18520985e-42j -7.27507504e-42-5.18520985e-42j
 -2.29243417e-45+0.00000000e+00j  4.43733995e-47+0.00000000e+00j
  6.81866723e-48+0.00000000e+00j -9.17837006e-49+0.00000000e+00j
 -1.21043738e-50+0.00000000e+00j  1.05010660e-51+0.00000000e+00j
 -6.92830633e-53+0.00000000e+00j  3.62513896e-54+0.00000000e+00j
 -3.42895698e-55+0.00000000e+00j -2.45357405e-58+9.55217691e-58j
 -2.45357405e-58-9.55217691e-58j  2.04987866e-59+0.00000000e+00j
  1.15291901e-60+0.00000000e+00j -9.20030773e-62+0.00000000e+00j
```

```
-1.33644258e-63+0.00000000e+00j  1.00845029e-64+0.00000000e+00j
-1.81127609e-66+0.00000000e+00j  2.59492833e-67+0.00000000e+00j
-1.86433563e-69+4.95635641e-69j -1.86433563e-69-4.95635641e-69j
 7.35054046e-71+0.00000000e+00j -6.57050982e-72+0.00000000e+00j
-4.79996159e-74+0.00000000e+00j -1.11457010e-75+0.00000000e+00j
 4.76114092e-77+0.00000000e+00j -3.29372275e-77+0.00000000e+00j]
```




```
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  2.30874434e-07+0.00000000e+00j  5.45056608e-10+0.00000000e+00j
  1.56536032e-12+0.00000000e+00j  4.31061288e-15+0.00000000e+00j
  9.87101327e-18+0.00000000e+00j  1.36026162e-20+0.00000000e+00j
 -2.98725343e-23+0.00000000e+00j -3.36356572e-24+0.00000000e+00j
  8.85460374e-26+0.00000000e+00j -1.76695699e-26+0.00000000e+00j
  4.51821947e-27+0.00000000e+00j  9.34288569e-29+0.00000000e+00j
  3.11818092e-30+0.00000000e+00j  1.01550851e-31+7.68381632e-32j
  1.01550851e-31-7.68381632e-32j -4.13562199e-32+0.00000000e+00j
 -1.70206804e-34+0.00000000e+00j -1.83732499e-35+0.00000000e+00j
 -1.99567217e-36+0.00000000e+00j  1.27403467e-38+0.00000000e+00j
 -6.40422835e-41+3.11304707e-40j -6.40422835e-41-3.11304707e-40j
  6.58544625e-42+0.00000000e+00j  2.35345261e-43+0.00000000e+00j
 -1.44251375e-43+1.57162103e-43j -1.44251375e-43-1.57162103e-43j
  6.52823719e-45+3.29321110e-45j  6.52823719e-45-3.29321110e-45j
  6.39750193e-47+1.44225491e-46j  6.39750193e-47-1.44225491e-46j
  1.59475159e-47+0.00000000e+00j -1.62248857e-48+0.00000000e+00j
  1.09419373e-49+0.00000000e+00j  3.05211131e-52+1.83445677e-52j
```

```
 3.05211131e-52-1.83445677e-52j -1.61192846e-53+0.00000000e+00j
 2.31062183e-54+0.00000000e+00j  4.03262226e-56+0.00000000e+00j
 4.00256890e-57+0.00000000e+00j -3.28660284e-58+0.00000000e+00j
-9.53380094e-59+0.00000000e+00j -1.07607698e-59+0.00000000e+00j
 1.91142682e-60+0.00000000e+00j -2.46792564e-61+5.15737365e-62j
-2.46792564e-61-5.15737365e-62j  2.16737842e-61+0.00000000e+00j
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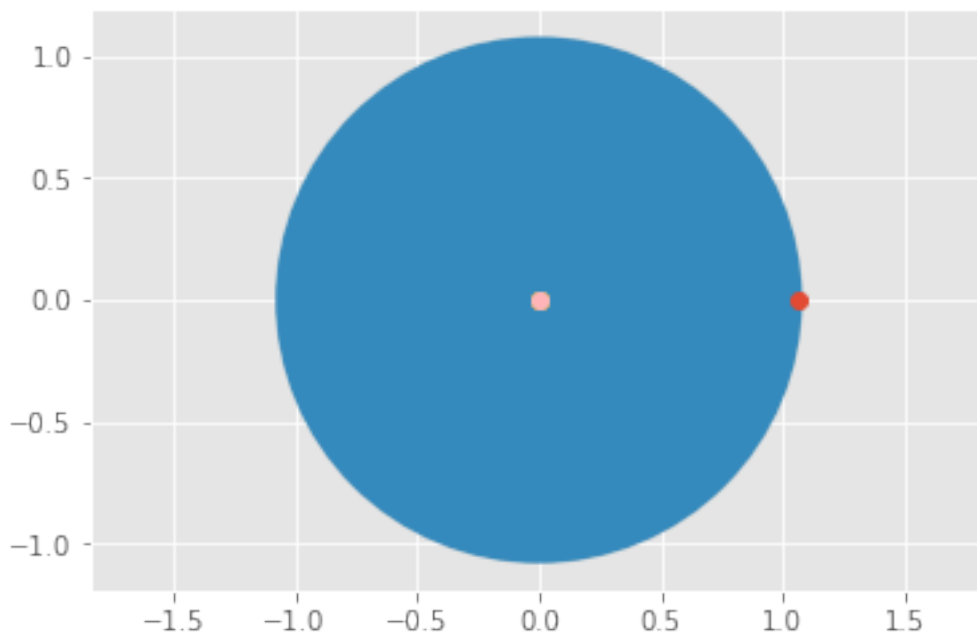
 9.18203593e-69+0.00000000e+00j -2.34050360e-70+0.00000000e+00j]
```




```
[ 1.02210653e+000+0.00000000e+00j  4.50452317e-004+0.00000000e+00j
 4.16556478e-007+0.00000000e+00j  5.62250074e-010+0.00000000e+00j
 9.01219447e-013+0.00000000e+00j  1.52026645e-015+0.00000000e+00j
 2.56331481e-018+0.00000000e+00j  4.45584063e-021+0.00000000e+00j
 9.19535925e-024+0.00000000e+00j  7.18418973e-026+0.00000000e+00j
 3.30055748e-028+3.87715017e-28j  3.30055748e-028-3.87715017e-28j
-7.44745228e-030+0.00000000e+00j  4.25106056e-030+0.00000000e+00j
 9.15290286e-031+0.00000000e+00j -1.63644117e-032+0.00000000e+00j
-1.08072617e-033+0.00000000e+00j  5.47723922e-034+0.00000000e+00j
-8.28087400e-037+0.00000000e+00j  6.39369966e-038+6.16843092e-38j
 6.39369966e-038-6.16843092e-38j  2.40163695e-041+0.00000000e+00j
 8.96838171e-042+3.90010534e-42j  8.96838171e-042-3.90010534e-42j
-2.37839625e-043+0.00000000e+00j  4.06121190e-045+0.00000000e+00j
 1.16123296e-046+0.00000000e+00j  1.87127519e-047+0.00000000e+00j
```

```
-4.97408031e-048+0.00000000e+00j -5.68751078e-051+0.00000000e+00j
 7.08384187e-053+0.00000000e+00j  1.60152113e-055+0.00000000e+00j
 6.73920438e-057+0.00000000e+00j -6.51991262e-059+0.00000000e+00j
-1.45631842e-060+0.00000000e+00j  5.20954517e-066+0.00000000e+00j
-6.85718359e-065+0.00000000e+00j  3.94770308e-069+0.00000000e+00j
-6.96880562e-071+0.00000000e+00j -1.09264567e-071+0.00000000e+00j
 5.82476101e-076+0.00000000e+00j -7.62647553e-076+0.00000000e+00j
-4.03306578e-078+0.00000000e+00j -3.85525132e-080+0.00000000e+00j
 6.14470327e-083+0.00000000e+00j -1.15168791e-082+0.00000000e+00j
-2.75284849e-086+0.00000000e+00j -2.72292707e-087+0.00000000e+00j
-4.05084878e-088+0.00000000e+00j  4.65472863e-091+0.00000000e+00j
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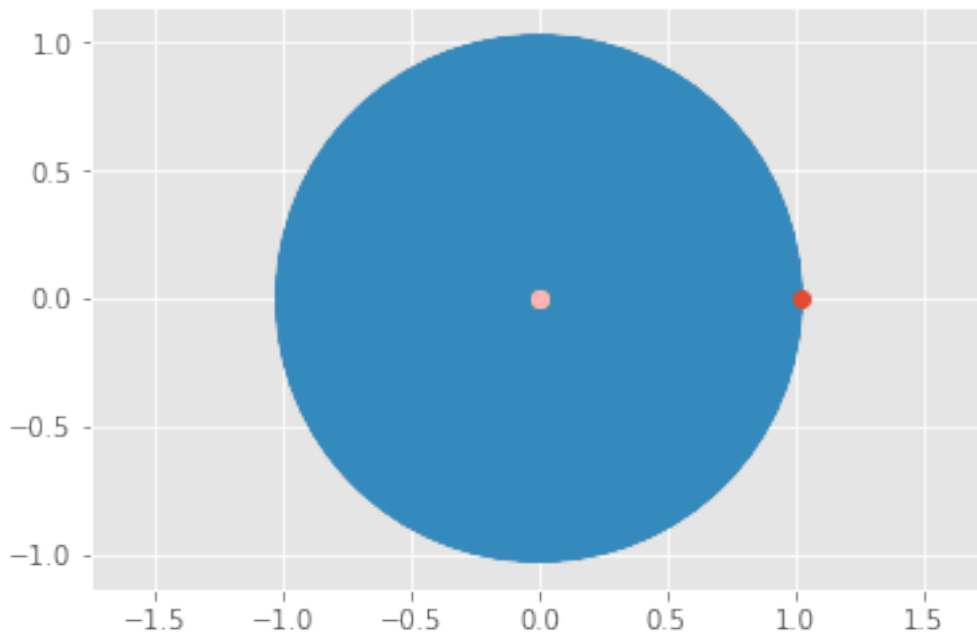
 3.25173970e-095+0.00000000e+00j  4.85833546e-097+0.00000000e+00j
-1.79418210e-099+0.00000000e+00j -4.15100671e-101+0.00000000e+00j]
```




```
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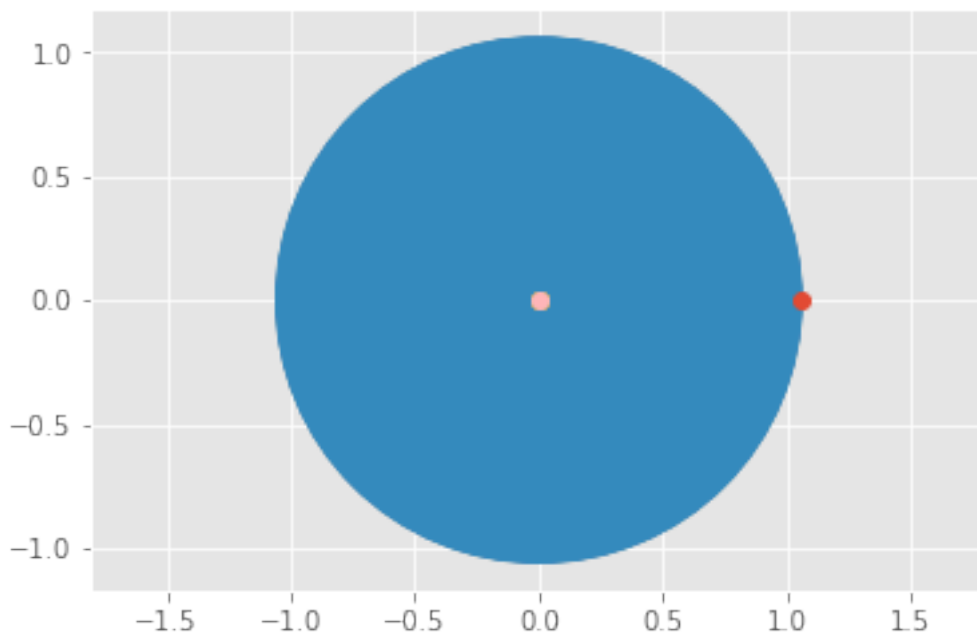
  1.42997744e-12+0.00000000e+00j  2.83475412e-15+0.00000000e+00j
  7.00110226e-18+0.00000000e+00j  1.91868125e-20+0.00000000e+00j
  4.23272590e-23+1.71056849e-23j  4.23272590e-23-1.71056849e-23j
  3.42555063e-24+0.00000000e+00j -7.18910297e-25+0.00000000e+00j
  2.55894192e-26+1.43788757e-25j  2.55894192e-26-1.43788757e-25j
 -1.87460320e-27+0.00000000e+00j -8.35683141e-30+2.34255745e-29j
 -8.35683141e-30-2.34255745e-29j -2.32927809e-30+0.00000000e+00j
  1.87706538e-30+0.00000000e+00j -2.59366933e-34+8.15024785e-34j
```

```
-2.59366933e-34-8.15024785e-34j  4.98568029e-35+0.00000000e+00j
 5.56935890e-37+0.00000000e+00j -1.37564994e-38+0.00000000e+00j
 1.08596314e-40+0.00000000e+00j -2.81124585e-41+0.00000000e+00j
-1.05906396e-42+0.00000000e+00j  3.09451440e-43+0.00000000e+00j
 1.04136498e-44+0.00000000e+00j  3.01856258e-46+0.00000000e+00j
-1.50798953e-46+0.00000000e+00j -7.52795051e-48+0.00000000e+00j
-9.38218886e-50+0.00000000e+00j  5.45718890e-50+0.00000000e+00j
 1.40465696e-53+0.00000000e+00j -5.25241396e-54+0.00000000e+00j
 3.41601216e-56+0.00000000e+00j -1.50197079e-57+0.00000000e+00j
-6.34492350e-59+0.00000000e+00j  1.84213531e-59+0.00000000e+00j
-7.93597070e-61+0.00000000e+00j  2.15490397e-62+0.00000000e+00j
-2.47289914e-62+0.00000000e+00j -6.21645689e-64+0.00000000e+00j
 1.58522285e-64+0.00000000e+00j  3.31358218e-65+0.00000000e+00j
 2.99602521e-68+6.14148418e-68j  2.99602521e-68-6.14148418e-68j
-1.94813639e-68+0.00000000e+00j -2.76639875e-70+0.00000000e+00j
-1.64801220e-72+0.00000000e+00j -1.57566372e-73+2.99175846e-74j
-1.57566372e-73-2.99175846e-74j  3.55387605e-76+0.00000000e+00j
-1.26251288e-75+0.00000000e+00j  5.60218199e-79+0.00000000e+00j]
```

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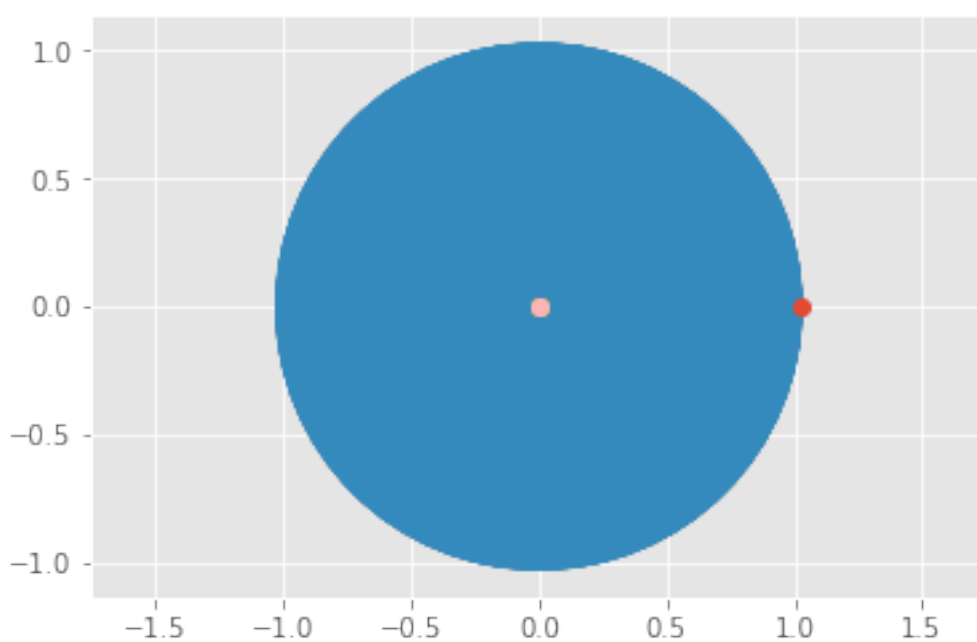
  9.78886225e-08+0.00000000e+00j  5.86439852e-11+0.00000000e+00j
  4.43437954e-14+0.00000000e+00j  3.96037598e-17+0.00000000e+00j
  4.10984169e-20+0.00000000e+00j  6.33883433e-23+0.00000000e+00j
  1.09207524e-24+0.00000000e+00j  1.07329716e-25+0.00000000e+00j
 -5.82478095e-27+0.00000000e+00j -4.70933641e-27+0.00000000e+00j
```

```
 1.03889373e-29+0.00000000e+00j -1.32298846e-32+4.35629792e-32j
-1.32298846e-32-4.35629792e-32j -5.91601217e-32+0.00000000e+00j
 3.37528253e-34+0.00000000e+00j -3.20572808e-36+0.00000000e+00j
 9.95454691e-39+1.44517172e-39j  9.95454691e-39-1.44517172e-39j
 3.21320673e-40+0.00000000e+00j -2.86962836e-41+0.00000000e+00j
 1.22254976e-42+0.00000000e+00j  3.09488533e-44+0.00000000e+00j
-4.51789831e-46+0.00000000e+00j -1.12479632e-47+0.00000000e+00j
 7.02031355e-48+0.00000000e+00j  4.79964783e-50+0.00000000e+00j
 2.85083369e-51+0.00000000e+00j  2.34983272e-53+0.00000000e+00j
 6.65333289e-55+0.00000000e+00j  1.56718854e-56+0.00000000e+00j
 2.32207879e-59+0.00000000e+00j -3.30826331e-60+0.00000000e+00j
 7.09612113e-61+0.00000000e+00j  3.18507439e-64+0.00000000e+00j
 2.60385897e-65+0.00000000e+00j  3.75934396e-66+0.00000000e+00j
 5.70773453e-69+0.00000000e+00j -8.32573054e-70+0.00000000e+00j
 4.31088328e-72+0.00000000e+00j -1.59467802e-73+0.00000000e+00j
 7.85509105e-77+0.00000000e+00j  2.67955431e-78+0.00000000e+00j
-5.86959333e-79+0.00000000e+00j  1.22772817e-81+0.00000000e+00j
 5.19894757e-83+0.00000000e+00j  8.30230628e-85+0.00000000e+00j
-1.40404761e-86+0.00000000e+00j  1.50310190e-87+0.00000000e+00j
-3.78220587e-89+0.00000000e+00j -6.86312244e-91+0.00000000e+00j
-1.33795859e-92+0.00000000e+00j -4.47579710e-95+0.00000000e+00j
 3.35712952e-95+0.00000000e+00j  1.58890444e-95+0.00000000e+00j]
```

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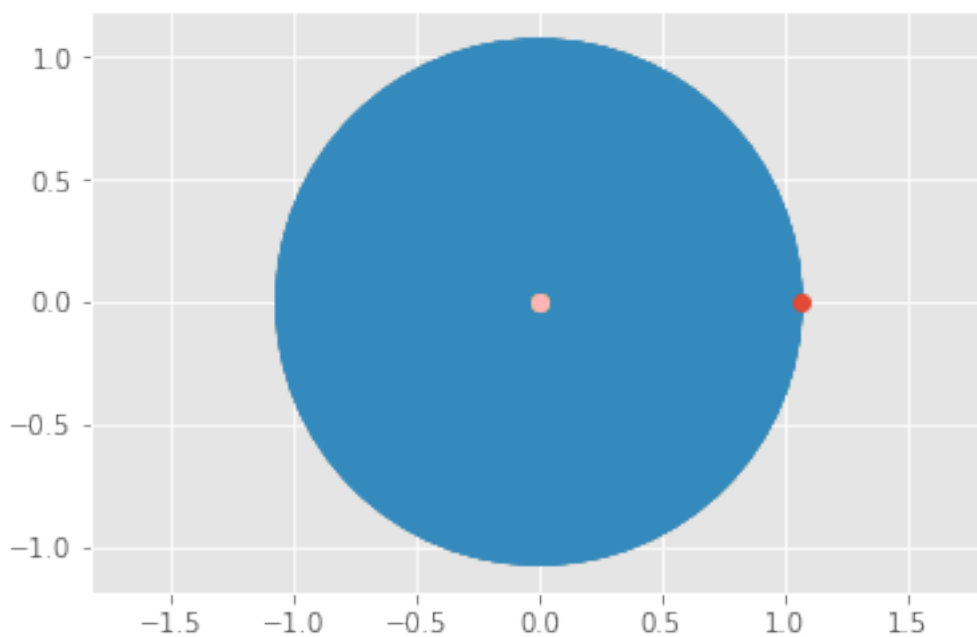
  1.49349604e-07+0.00000000e+00j  1.51647757e-10+0.00000000e+00j
```

```
 2.55042817e-13+0.00000000e+00j  4.41967651e-16+0.00000000e+00j
 5.77855605e-19+0.00000000e+00j -4.38123992e-21+0.00000000e+00j
 7.44771951e-23+0.00000000e+00j  5.68130346e-25+0.00000000e+00j
 6.19712298e-26+0.00000000e+00j  6.67896092e-27+0.00000000e+00j
-1.42440215e-27+0.00000000e+00j -4.22109188e-28+0.00000000e+00j
-1.20377041e-28+0.00000000e+00j  1.20060208e-28+0.00000000e+00j
 1.76035647e-29+2.73984474e-30j  1.76035647e-29-2.73984474e-30j
-1.21466889e-29+0.00000000e+00j  2.63144924e-30+0.00000000e+00j
-1.69032258e-31+0.00000000e+00j  2.73744871e-33+0.00000000e+00j
 1.31937173e-35+0.00000000e+00j -1.77442172e-35+0.00000000e+00j
-6.40415871e-36+0.00000000e+00j  6.82040201e-39+0.00000000e+00j
-1.22992481e-38+0.00000000e+00j  1.51847766e-43+1.84364093e-41j
 1.51847766e-43-1.84364093e-41j  2.36519936e-44+5.07289409e-44j
 2.36519936e-44-5.07289409e-44j -5.17957443e-45+0.00000000e+00j
 1.93512669e-47+0.00000000e+00j  5.93446581e-52+8.31491407e-50j
 5.93446581e-52-8.31491407e-50j  1.84911864e-52+0.00000000e+00j
-2.19245436e-52+0.00000000e+00j -6.56905807e-55+9.81493677e-55j
-6.56905807e-55-9.81493677e-55j  1.95080790e-57+0.00000000e+00j
 1.85608017e-58+0.00000000e+00j -4.69870402e-59+0.00000000e+00j
 2.70825880e-59+0.00000000e+00j -5.33125959e-61+0.00000000e+00j
 2.31095325e-63+0.00000000e+00j  1.68344223e-64+0.00000000e+00j
-5.89174382e-66+0.00000000e+00j  3.37938925e-66+1.94600144e-66j
 3.37938925e-66-1.94600144e-66j  1.98859468e-67+0.00000000e+00j
-3.79843076e-68+0.00000000e+00j  2.25861412e-68+0.00000000e+00j
-2.13735735e-71+0.00000000e+00j -2.48698359e-71+0.00000000e+00j
 7.81324055e-74+0.00000000e+00j -3.97625112e-74+0.00000000e+00j]
```



```
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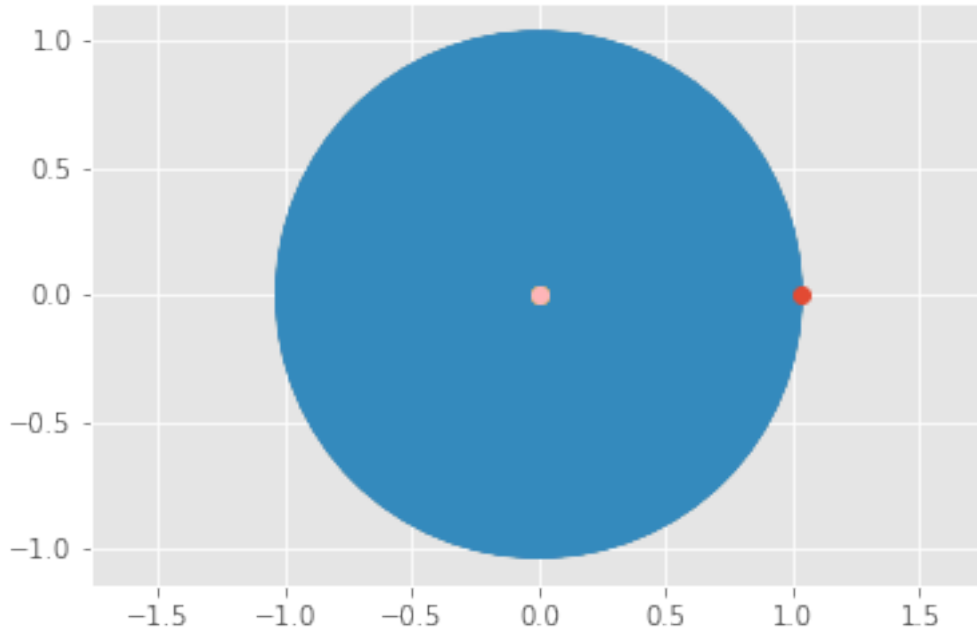
  1.30483148e-07+0.00000000e+00j  1.07552972e-10+0.00000000e+00j
  1.18635028e-13+0.00000000e+00j  1.63249304e-16+0.00000000e+00j
  2.44829632e-19+0.00000000e+00j  3.49644205e-22+0.00000000e+00j
 -1.24314873e-23+0.00000000e+00j -5.49218865e-25+0.00000000e+00j
 -7.49626677e-26+0.00000000e+00j  5.24226329e-26+0.00000000e+00j
 -5.32374360e-28+0.00000000e+00j  4.49607016e-29+0.00000000e+00j
  9.23745180e-32+0.00000000e+00j  2.06891622e-33+0.00000000e+00j
 -6.66055773e-35+5.66340806e-35j -6.66055773e-35-5.66340806e-35j
  1.50243159e-37+0.00000000e+00j  3.05829287e-38+0.00000000e+00j
  1.18734070e-40+0.00000000e+00j -4.07007844e-41+0.00000000e+00j
 -8.41907476e-43+0.00000000e+00j  2.97877023e-44+0.00000000e+00j
 -4.13326694e-46+0.00000000e+00j  8.13170440e-47+0.00000000e+00j
  2.01106593e-49+0.00000000e+00j  1.22497106e-50+0.00000000e+00j
 -4.64087772e-53+0.00000000e+00j  3.76464203e-53+0.00000000e+00j
 -5.27930305e-54+0.00000000e+00j -8.50550449e-56+2.79421686e-55j
 -8.50550449e-56-2.79421686e-55j  6.99268449e-59+0.00000000e+00j
 -3.48498947e-59+7.91504767e-60j -3.48498947e-59-7.91504767e-60j
 -2.58004729e-64+1.74504363e-62j -2.58004729e-64-1.74504363e-62j
 -7.08504191e-65+0.00000000e+00j  1.61691739e-65+0.00000000e+00j
  1.18061687e-67+0.00000000e+00j  3.76305577e-69+0.00000000e+00j
 -3.24885285e-70+0.00000000e+00j  9.98470768e-72+0.00000000e+00j
  2.18080648e-73+0.00000000e+00j  4.40295650e-76+1.65974111e-75j
  4.40295650e-76-1.65974111e-75j -1.46598549e-77+0.00000000e+00j
 -1.11681728e-77+0.00000000e+00j  6.35732278e-81+0.00000000e+00j
 -1.18374232e-80+0.00000000e+00j  4.59611105e-82+0.00000000e+00j
 -4.45809757e-85+3.73349896e-84j -4.45809757e-85-3.73349896e-84j
  1.28224734e-86+0.00000000e+00j -3.91384210e-87+0.00000000e+00j]
```




```
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  2.35127918e-008+0.00000000e+000j  8.95165879e-012+0.00000000e+000j
  4.78809985e-015+0.00000000e+000j  2.67680189e-018+0.00000000e+000j
  1.12983067e-021+0.00000000e+000j  3.96633690e-025+0.00000000e+000j
  1.01327713e-028+0.00000000e+000j -6.48582275e-030+0.00000000e+000j
  4.12252532e-030+0.00000000e+000j -7.41283384e-034+0.00000000e+000j
 -2.68109298e-033+0.00000000e+000j -1.25283727e-035+0.00000000e+000j
  5.65163631e-037+0.00000000e+000j -4.99583233e-040+0.00000000e+000j
 -3.18897930e-042+0.00000000e+000j -1.46463727e-043+0.00000000e+000j
 -1.10312832e-045+0.00000000e+000j -1.91918908e-048+0.00000000e+000j
 -1.69501562e-049+0.00000000e+000j  4.30337838e-050+0.00000000e+000j
 -5.80526869e-052+0.00000000e+000j  2.88925354e-054+0.00000000e+000j
  1.27790253e-056+0.00000000e+000j  2.79297185e-058+0.00000000e+000j
 -7.90147213e-061+0.00000000e+000j  2.17199586e-064+3.74335745e-063j
  2.17199586e-064-3.74335745e-063j  1.08560306e-065+0.00000000e+000j
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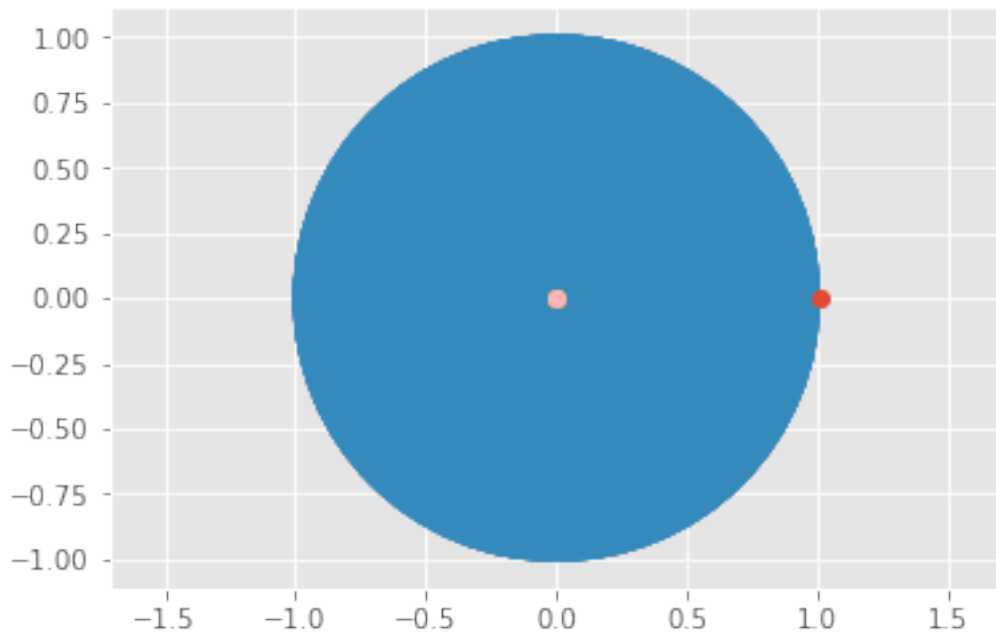
  2.67877172e-073+0.00000000e+000j  1.41577813e-075+0.00000000e+000j
 -2.88832235e-078+0.00000000e+000j -1.16096922e-080+0.00000000e+000j
 -1.08708577e-082+0.00000000e+000j  9.97542432e-085+0.00000000e+000j
 -1.78503388e-086+0.00000000e+000j  1.09404988e-088+0.00000000e+000j
 -1.75276864e-093+0.00000000e+000j -1.98279240e-092+0.00000000e+000j
 -8.89519378e-096+0.00000000e+000j -4.09290197e-098+0.00000000e+000j
 -2.05790516e-099+0.00000000e+000j -3.18894377e-102+0.00000000e+000j
  9.10741449e-105+0.00000000e+000j -7.42660337e-109+3.30895961e-108j
 -7.42660337e-109-3.30895961e-108j  1.29547028e-109+0.00000000e+000j
  2.01474306e-113+0.00000000e+000j -2.25664439e-115+0.00000000e+000j
```

```
  6.33782550e-118+0.00000000e+000j  1.38511641e-118+0.00000000e+000j
 -1.18079701e-123+0.00000000e+000j  2.04306089e-122+0.00000000e+000j]
```




```
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  1.69892725e-011+0.00000000e+000j  1.27368138e-016+0.00000000e+000j
  1.31585643e-021+0.00000000e+000j  1.93923335e-026+0.00000000e+000j
  3.82924397e-031+0.00000000e+000j  9.07508136e-036+0.00000000e+000j
  2.63440453e-040+0.00000000e+000j  2.07121787e-044+0.00000000e+000j
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 -1.17028243e-055+0.00000000e+000j  6.03635025e-058+0.00000000e+000j
  1.59495975e-060+0.00000000e+000j -2.22091652e-063+0.00000000e+000j
  2.95527594e-066+0.00000000e+000j  1.07261740e-068+0.00000000e+000j
 -6.23926845e-072+0.00000000e+000j  1.25801788e-075+0.00000000e+000j
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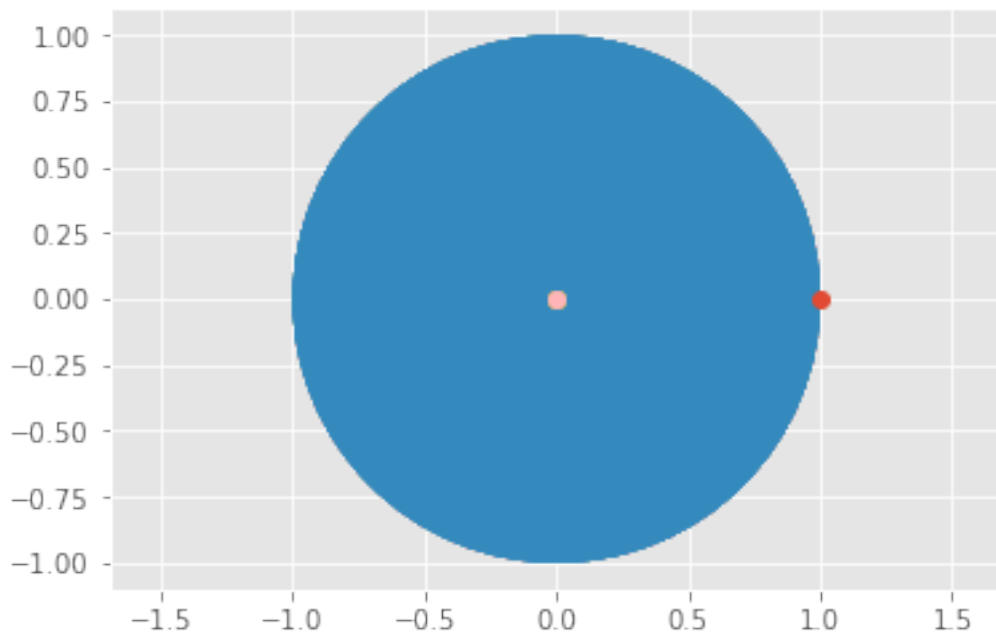
  8.78610484e-084+0.00000000e+000j  7.82390050e-086+0.00000000e+000j
 -2.49626630e-089+0.00000000e+000j -2.05053854e-092+0.00000000e+000j
  2.97560304e-095+0.00000000e+000j  9.84091578e-098+0.00000000e+000j
 -3.39394030e-101+0.00000000e+000j -1.54254320e-104+0.00000000e+000j
  1.31982775e-111+0.00000000e+000j  2.41593962e-109+0.00000000e+000j
  1.23801388e-116+0.00000000e+000j  1.30575781e-120+0.00000000e+000j
 -2.08394271e-125+0.00000000e+000j -4.27611414e-131+0.00000000e+000j
  1.13828748e-138+1.95366702e-138j  1.13828748e-138-1.95366702e-138j
  1.00652026e-143+0.00000000e+000j  8.04670806e-148+0.00000000e+000j
 -1.22117140e-150+0.00000000e+000j  1.68627132e-154+0.00000000e+000j
 -3.93863380e-157+0.00000000e+000j  1.83763956e-161+0.00000000e+000j
```

```
 8.62967422e-164+0.00000000e+000j -9.55285230e-169+0.00000000e+000j
-5.90578037e-172+0.00000000e+000j  2.23336672e-175+0.00000000e+000j
 8.97700780e-179+0.00000000e+000j -5.80499241e-183+0.00000000e+000j
-3.90821405e-188+0.00000000e+000j  8.62921278e-190+0.00000000e+000j
-8.41829649e-194+0.00000000e+000j -6.80593956e-198+0.00000000e+000j
 8.61220058e-202+0.00000000e+000j  2.76302307e-205+0.00000000e+000j]
```




```
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  3.24813468e-14+0.00000000e+00j  1.43274823e-17+0.00000000e+00j
  1.05648897e-20+0.00000000e+00j -5.23591922e-25+1.19700142e-24j
 -5.23591922e-25-1.19700142e-24j  2.86859804e-26+0.00000000e+00j
  1.36871108e-27+0.00000000e+00j  5.88702862e-30+7.54597533e-30j
  5.88702862e-30-7.54597533e-30j  2.76557652e-32+1.22304661e-31j
  2.76557652e-32-1.22304661e-31j  3.92098334e-33+0.00000000e+00j
 -1.39045150e-34+0.00000000e+00j  8.09885888e-36+0.00000000e+00j
 -4.34179998e-38+4.88906575e-38j -4.34179998e-38-4.88906575e-38j
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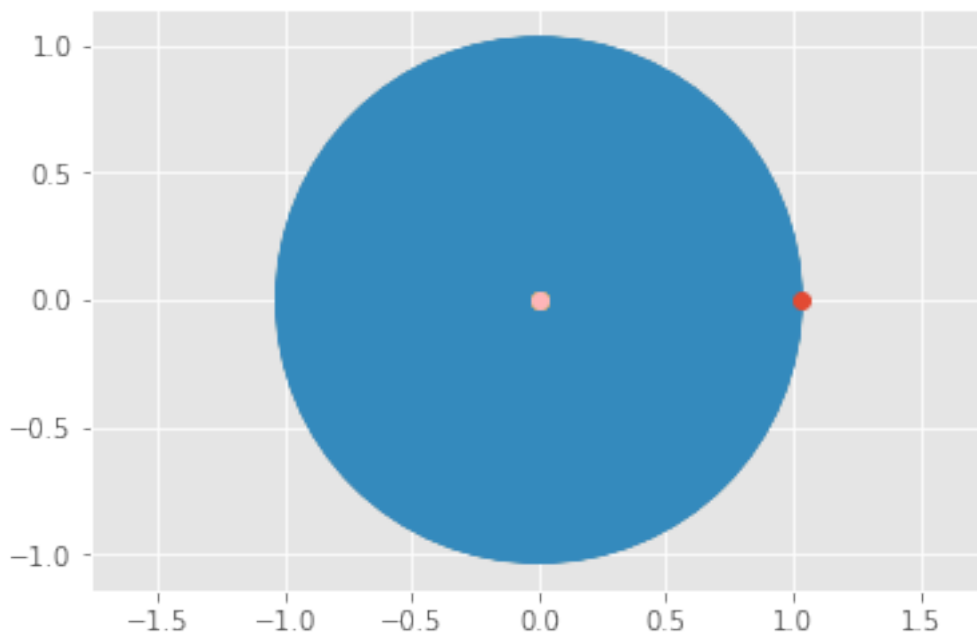
  1.65766404e-42+0.00000000e+00j  4.10809360e-43+0.00000000e+00j
 -1.37303899e-44+0.00000000e+00j  1.05495566e-46+0.00000000e+00j
  1.14223749e-46+0.00000000e+00j -4.50881248e-49+0.00000000e+00j
  1.62186173e-50+0.00000000e+00j -3.98189550e-51+0.00000000e+00j
  3.92495750e-53+0.00000000e+00j  3.86176821e-54+0.00000000e+00j
 -2.84542321e-55+0.00000000e+00j  1.85761580e-55+0.00000000e+00j
 -1.66444005e-56+0.00000000e+00j  8.81658625e-61+0.00000000e+00j
```

```
 4.17540559e-60+0.00000000e+00j -1.52002243e-62+0.00000000e+00j
-1.64881036e-64+0.00000000e+00j -5.46051374e-66+0.00000000e+00j
-1.11102945e-66+0.00000000e+00j  1.96877016e-68+0.00000000e+00j
 5.44512139e-70+0.00000000e+00j -1.87720513e-72+0.00000000e+00j
 2.47189391e-74+1.21730066e-74j  2.47189391e-74-1.21730066e-74j
-1.66214857e-75+0.00000000e+00j -5.73685668e-76+0.00000000e+00j
 4.51855915e-78+0.00000000e+00j -1.95456938e-79+0.00000000e+00j
-3.85244241e-81+0.00000000e+00j  2.85013319e-81+0.00000000e+00j
-2.92853059e-85+0.00000000e+00j  1.77033851e-84+0.00000000e+00j
 1.38802297e-86+0.00000000e+00j -1.93870378e-87+0.00000000e+00j]
```




```
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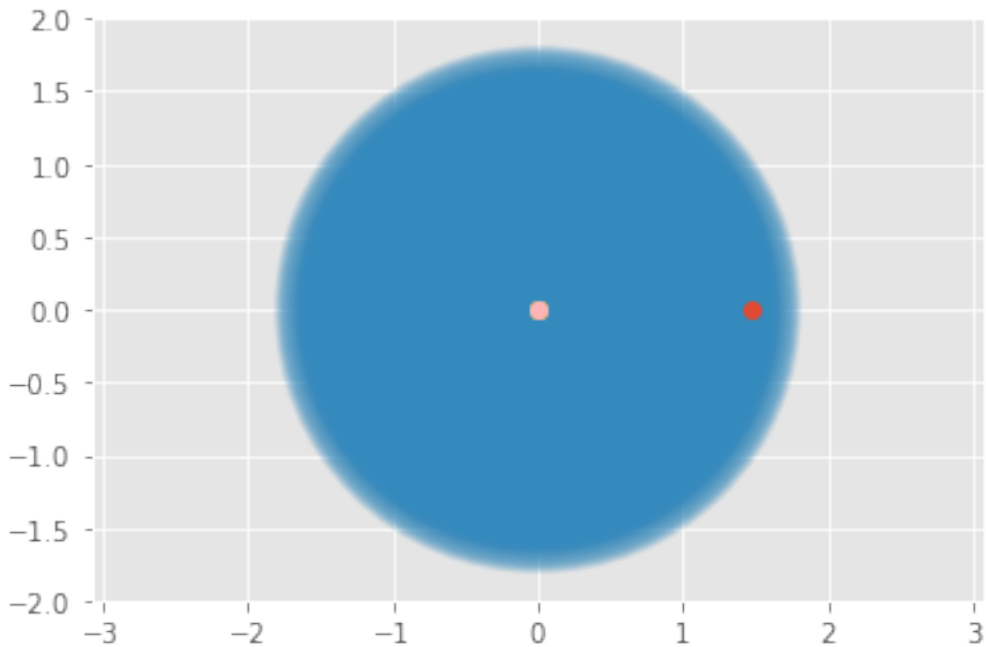
 1.42890990e-07+0.00000000e+00j  4.64304050e-09+0.00000000e+00j
 1.40506497e-10+0.00000000e+00j  4.62708514e-12+0.00000000e+00j
 1.97145928e-13+0.00000000e+00j  9.22649703e-15+0.00000000e+00j
 4.30519180e-16+0.00000000e+00j  2.40163051e-17+0.00000000e+00j
 4.16608179e-18+0.00000000e+00j  1.25002084e-18+0.00000000e+00j
-7.19054744e-19+0.00000000e+00j -1.82478854e-19+0.00000000e+00j
 8.10536595e-20+1.16038275e-20j  8.10536595e-20-1.16038275e-20j
-7.45381058e-20+0.00000000e+00j -2.73011431e-21+5.78574518e-20j
-2.73011431e-21-5.78574518e-20j -4.74852747e-20+0.00000000e+00j
 2.34849715e-20+0.00000000e+00j  8.13672213e-21+0.00000000e+00j
 4.80461644e-21+0.00000000e+00j -3.90985753e-21+9.01772764e-22j
-3.90985753e-21-9.01772764e-22j  9.24422655e-22+2.82464159e-22j
```

```
 9.24422655e-22-2.82464159e-22j  2.29804171e-23+2.64510104e-22j
 2.29804171e-23-2.64510104e-22j -1.86079477e-22+0.00000000e+00j
 6.25362125e-23+3.78958091e-23j  6.25362125e-23-3.78958091e-23j
-2.69837397e-23+0.00000000e+00j -9.74963616e-24+1.06995114e-23j
-9.74963616e-24-1.06995114e-23j  3.20583388e-24+0.00000000e+00j
-2.79885040e-26+1.29336753e-25j -2.79885040e-26-1.29336753e-25j
-1.99233393e-26+0.00000000e+00j  4.40823394e-27+0.00000000e+00j
-4.07054943e-27+1.49354936e-27j -4.07054943e-27-1.49354936e-27j
 9.03385900e-28+0.00000000e+00j -2.15956646e-28+0.00000000e+00j
-1.06733002e-28+1.32701406e-28j -1.06733002e-28-1.32701406e-28j
 1.01064971e-28+0.00000000e+00j  4.02603605e-29+0.00000000e+00j
 2.28547188e-29+0.00000000e+00j -1.15453569e-29+0.00000000e+00j
-1.86882211e-30+9.56118285e-31j -1.86882211e-30-9.56118285e-31j
 2.98905522e-30+0.00000000e+00j  7.38578778e-31+0.00000000e+00j]
```

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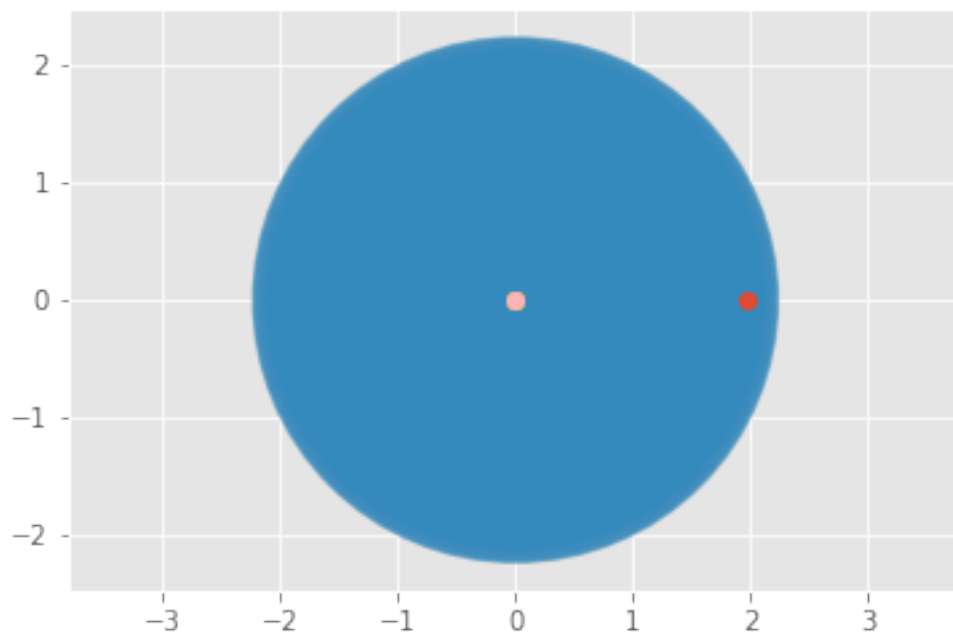
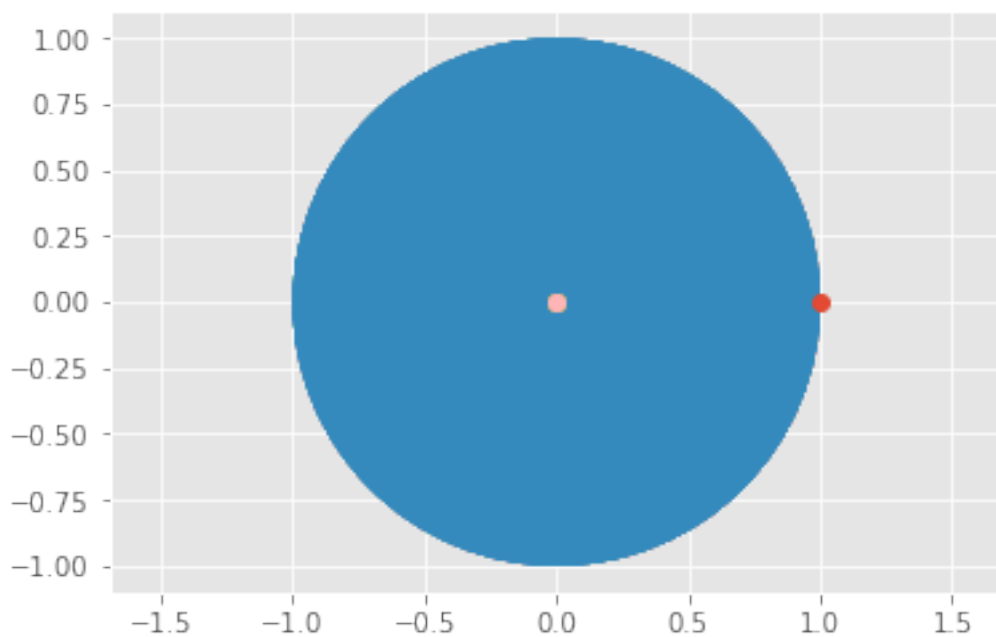
 4.04215432e-07+0.00000000e+00j  4.65862991e-09+0.00000000e+00j
 1.14655822e-10+0.00000000e+00j  6.48212366e-12+0.00000000e+00j
 4.68015632e-14+0.00000000e+00j -1.40824760e-14+0.00000000e+00j
 5.97198261e-16+0.00000000e+00j -1.53932552e-17+1.64009903e-17j
-1.53932552e-17-1.64009903e-17j  1.59458715e-17+0.00000000e+00j
 2.17691692e-19+1.04922545e-17j  2.17691692e-19-1.04922545e-17j
-1.45951187e-18+2.00177612e-18j -1.45951187e-18-2.00177612e-18j
-1.36440454e-19+5.68440162e-19j -1.36440454e-19-5.68440162e-19j
```

```
 -4.71332388e-19+0.00000000e+00j  1.99270187e-19+3.06601381e-19j
  1.99270187e-19-3.06601381e-19j  1.72459360e-19+0.00000000e+00j
  8.15601704e-20+0.00000000e+00j -5.38783930e-20+4.65259705e-20j
 -5.38783930e-20-4.65259705e-20j  3.62060172e-20+0.00000000e+00j
 -8.58490407e-21+1.60960925e-20j -8.58490407e-21-1.60960925e-20j
 -8.20136068e-22+7.70593243e-21j -8.20136068e-22-7.70593243e-21j
  6.65295833e-21+0.00000000e+00j -3.70657691e-21+0.00000000e+00j
  4.90339719e-21+0.00000000e+00j -5.36032808e-22+0.00000000e+00j
  3.52319290e-22+0.00000000e+00j  2.52971137e-22+0.00000000e+00j
 -1.42274521e-22+0.00000000e+00j  2.55505361e-23+6.99055817e-23j
  2.55505361e-23-6.99055817e-23j  7.00906699e-23+0.00000000e+00j
 -4.12561654e-23+0.00000000e+00j  1.66398178e-24+5.73085469e-24j
  1.66398178e-24-5.73085469e-24j -3.60374610e-24+0.00000000e+00j
 -1.17596036e-24+1.68112347e-24j -1.17596036e-24-1.68112347e-24j
 -2.04157478e-25+1.81593363e-25j -2.04157478e-25-1.81593363e-25j
  3.46267712e-26+2.33823410e-26j  3.46267712e-26-2.33823410e-26j
  1.08691007e-26+0.00000000e+00j -9.81731336e-27+0.00000000e+00j
 -4.00970191e-27+0.00000000e+00j  6.13889803e-29+0.00000000e+00j]
```



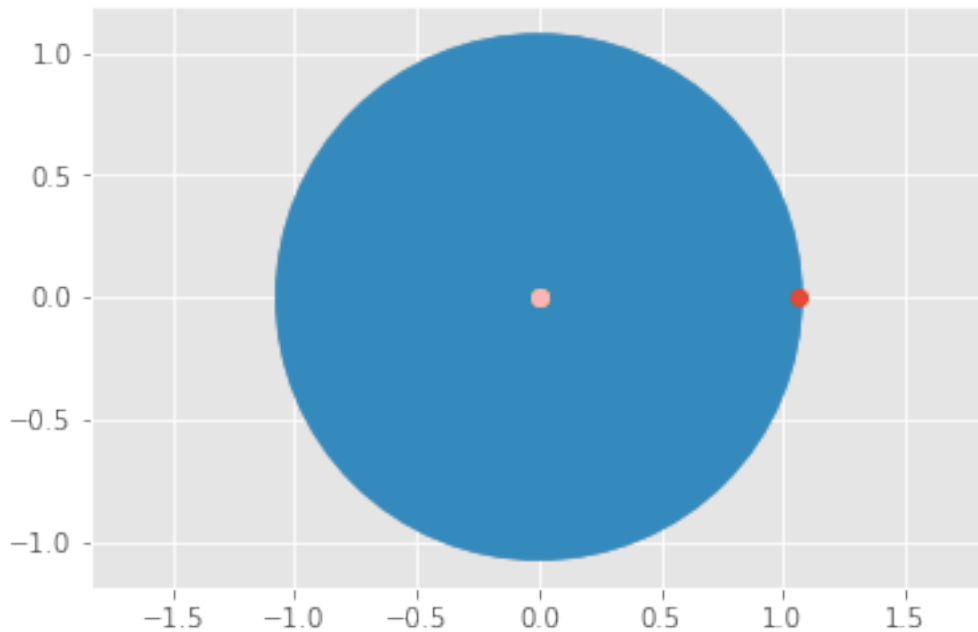

```
[ 1.00014481e+000+0.j  1.80334280e-006+0.j  9.18739317e-012+0.j
  8.02998553e-017+0.j  8.96172483e-022+0.j  1.05267345e-026+0.j
  1.25276119e-031+0.j  1.54429155e-036+0.j  1.94232291e-041+0.j
  2.51663885e-045+0.j  7.14797501e-049+0.j  2.05784735e-052+0.j
  4.78483428e-056+0.j  1.58078284e-060+0.j -1.74945830e-063+0.j
  1.09908835e-066+0.j -2.89647466e-071+0.j -2.65852043e-073+0.j
```

```
 1.02637654e-077+0.j -6.68630377e-080+0.j  4.32807601e-084+0.j
 4.42357402e-090+0.j -1.21243374e-090+0.j -7.15737232e-097+0.j
-1.40373947e-100+0.j  1.53057903e-108+0.j  1.20859309e-104+0.j
-4.77860783e-115+0.j -3.50295875e-113+0.j -4.46876260e-119+0.j
-2.29621686e-123+0.j -1.85434453e-124+0.j -8.31180363e-129+0.j
 2.18050335e-134+0.j  8.53766175e-137+0.j -7.08743929e-141+0.j
-7.47129283e-146+0.j  3.69729455e-148+0.j -3.80661968e-154+0.j
 3.19558367e-160+0.j -5.95168217e-159+0.j  8.52095265e-163+0.j
 4.44194535e-167+0.j  1.63782880e-168+0.j  5.11404256e-171+0.j
-3.68189920e-174+0.j  2.13306522e-178+0.j -3.58565261e-183+0.j
 1.53770839e-186+0.j  2.13219253e-188+0.j  5.07440409e-193+0.j
 5.23892200e-197+0.j  1.25483933e-199+0.j -1.09528967e-203+0.j
 6.30230127e-210+0.j -8.98775103e-208+0.j]
```




```
[ 1.05939015e+00+0.00000000e+00j  5.15361289e-04+0.00000000e+00j
  3.12668514e-07+0.00000000e+00j  1.83529162e-10+0.00000000e+00j
  2.51781351e-13+0.00000000e+00j  5.95785608e-16+0.00000000e+00j
  1.35640699e-18+0.00000000e+00j  3.16758275e-21+0.00000000e+00j
  4.75378201e-22+0.00000000e+00j -5.10380702e-24+0.00000000e+00j
  1.07143955e-24+0.00000000e+00j -9.08098993e-25+4.46876307e-25j
 -9.08098993e-25-4.46876307e-25j  1.06894238e-26+0.00000000e+00j
 -8.53887416e-27+3.78924332e-27j -8.53887416e-27-3.78924332e-27j
  1.67397126e-27+7.37088865e-27j  1.67397126e-27-7.37088865e-27j
  1.30342042e-27+0.00000000e+00j -7.41783211e-28+0.00000000e+00j
 -3.33415394e-30+0.00000000e+00j  5.38060925e-30+0.00000000e+00j
```

```
-1.30273538e-32+0.00000000e+00j  5.20236029e-33+0.00000000e+00j
-1.37950256e-33+0.00000000e+00j  1.46578357e-34+0.00000000e+00j
-3.82090695e-36+0.00000000e+00j -5.00770089e-38+1.85565445e-37j
-5.00770089e-38-1.85565445e-37j -3.04040223e-38+0.00000000e+00j
-2.04459863e-39+0.00000000e+00j -8.47170765e-41+0.00000000e+00j
 2.90220153e-41+0.00000000e+00j  6.65708997e-44+0.00000000e+00j
-1.68941605e-44+0.00000000e+00j  1.93886819e-48+0.00000000e+00j
-1.00715104e-47+0.00000000e+00j  8.43823727e-51+0.00000000e+00j
 1.29692941e-52+0.00000000e+00j  2.76074265e-54+0.00000000e+00j
-7.81162837e-55+3.34237562e-55j -7.81162837e-55-3.34237562e-55j
 2.89950137e-57+0.00000000e+00j  1.91714644e-57+0.00000000e+00j
 1.15423779e-59+0.00000000e+00j  1.29674915e-60+0.00000000e+00j
 7.80060228e-63+0.00000000e+00j  4.25755467e-64+0.00000000e+00j
-2.13375273e-64+0.00000000e+00j -1.25937200e-66+0.00000000e+00j
 3.43636571e-68+1.43062926e-67j  3.43636571e-68-1.43062926e-67j
 2.93181012e-70+0.00000000e+00j -1.41693685e-71+0.00000000e+00j
 4.31882224e-72+0.00000000e+00j  6.63333653e-75+0.00000000e+00j]
```




```
[ 1.07200460e+00+0.00000000e+00j  3.61451214e-04+0.00000000e+00j
  1.52002368e-07+0.00000000e+00j  9.19112207e-11+0.00000000e+00j
  8.17718474e-14+0.00000000e+00j  1.21843031e-16+0.00000000e+00j
  2.38644792e-19+0.00000000e+00j  3.98993860e-21+0.00000000e+00j
  7.50627157e-23+0.00000000e+00j  4.95911839e-23+0.00000000e+00j
  4.14824695e-25+3.03990142e-25j  4.14824695e-25-3.03990142e-25j
 -4.84580611e-26+0.00000000e+00j  2.33682511e-26+0.00000000e+00j
```

```
 2.14228462e-27+3.29170417e-27j  2.14228462e-27-3.29170417e-27j
 3.42119442e-28+0.00000000e+00j  3.42275246e-29+0.00000000e+00j
 2.01757307e-29+0.00000000e+00j -6.58248533e-32+0.00000000e+00j
 1.41142794e-31+0.00000000e+00j -3.93836239e-34+0.00000000e+00j
 7.76176961e-36+0.00000000e+00j -1.18760165e-36+0.00000000e+00j
 2.62874224e-37+0.00000000e+00j  5.83326262e-40+0.00000000e+00j
-4.61993239e-41+0.00000000e+00j  7.41457192e-42+0.00000000e+00j
-4.77790110e-43+0.00000000e+00j  5.61422237e-45+0.00000000e+00j
-1.36823956e-46+0.00000000e+00j  1.84056677e-47+0.00000000e+00j
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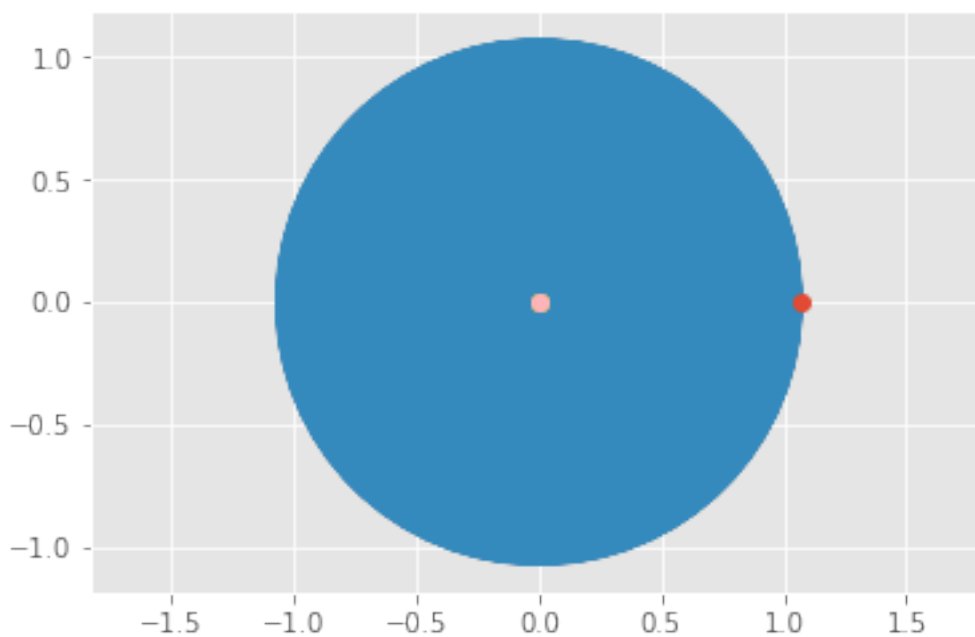
 7.65737409e-52+0.00000000e+00j  1.66186471e-53+0.00000000e+00j
-1.96048473e-54+0.00000000e+00j -4.28674731e-57+0.00000000e+00j
 1.67755905e-56+0.00000000e+00j  1.85376790e-59+7.12623719e-60j
 1.85376790e-59-7.12623719e-60j  2.07806258e-62+0.00000000e+00j
-1.25475939e-62+0.00000000e+00j  1.63473608e-63+0.00000000e+00j
 1.77049731e-64+0.00000000e+00j -1.67166594e-66+0.00000000e+00j
 9.18470986e-67+0.00000000e+00j -5.02455042e-68+0.00000000e+00j
-1.25954580e-69+0.00000000e+00j  3.33156166e-70+0.00000000e+00j
 6.40648564e-71+0.00000000e+00j  4.77957088e-73+1.21587069e-72j
 4.77957088e-73-1.21587069e-72j -4.87534336e-74+0.00000000e+00j
 2.98662475e-74+0.00000000e+00j  1.07008298e-76+0.00000000e+00j]
```




```
[ 1.00118224e+000+0.00000000e+000j  2.33183099e-005+0.00000000e+000j
  1.03676341e-009+0.00000000e+000j  6.71163884e-014+0.00000000e+000j
  6.07151430e-018+0.00000000e+000j  6.70784604e-022+0.00000000e+000j
```

```
 8.78137053e-026+0.00000000e+000j  1.23068828e-029+0.00000000e+000j
 1.38206344e-033+0.00000000e+000j -1.05633665e-037+0.00000000e+000j
 6.91170526e-040+0.00000000e+000j  1.60021517e-042+0.00000000e+000j
 3.39616178e-046+0.00000000e+000j -1.69987213e-045+0.00000000e+000j
 6.80107902e-050+0.00000000e+000j  1.26029818e-052+0.00000000e+000j
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 5.43636030e-061+0.00000000e+000j -2.53816973e-062+0.00000000e+000j
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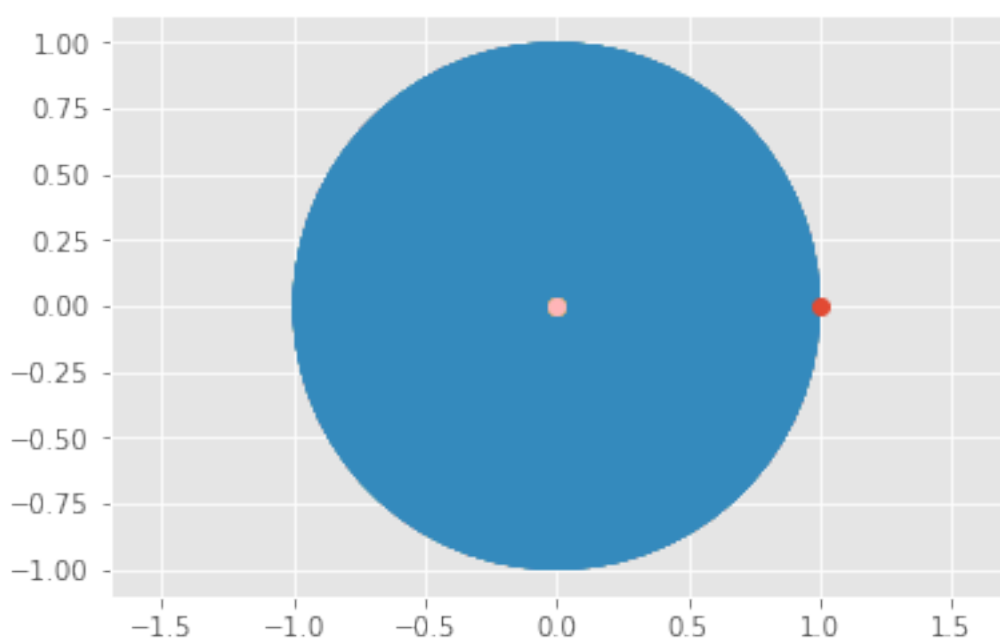
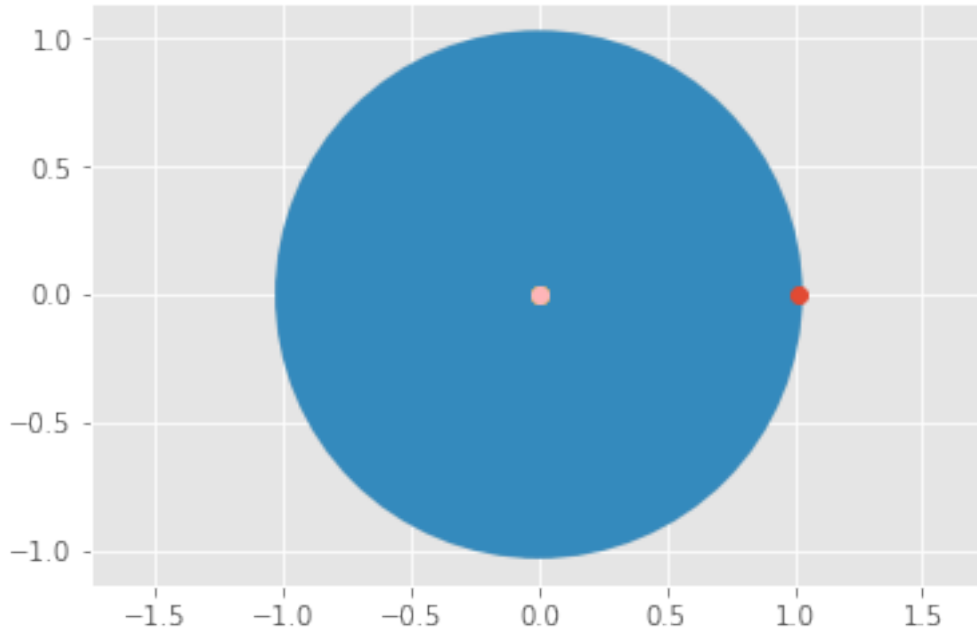
 1.73717645e-112+0.00000000e+000j  1.50567642e-114+0.00000000e+000j
 5.21524460e-117+0.00000000e+000j -3.91940762e-119+0.00000000e+000j
 3.37342979e-123+0.00000000e+000j -9.36265257e-128+0.00000000e+000j
-5.45642700e-127+0.00000000e+000j -3.03192255e-131+2.09850365e-131j
-3.03192255e-131-2.09850365e-131j -9.12007311e-133+0.00000000e+000j
 2.67709917e-135+0.00000000e+000j  1.39656737e-139+0.00000000e+000j
 1.18081206e-144+0.00000000e+000j  2.53746257e-147+0.00000000e+000j
-1.14172601e-150+2.03422119e-150j -1.14172601e-150-2.03422119e-150j
 7.64914236e-154+0.00000000e+000j -6.60280342e-157+0.00000000e+000j]
```

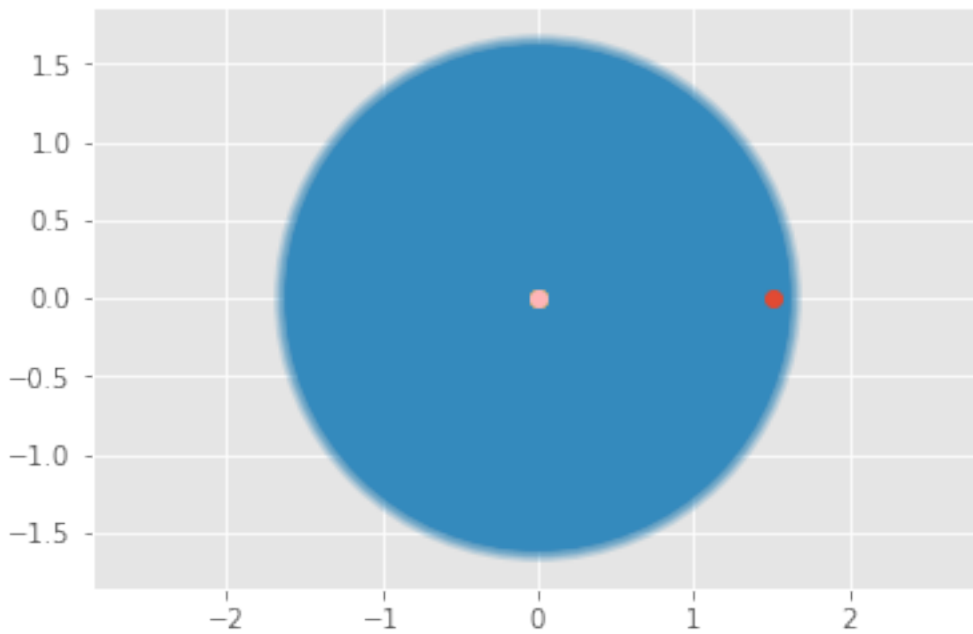


```
[ 1.01939199e+00+0.00000000e+00j -2.56317106e-05+0.00000000e+00j
  6.27708123e-09+0.00000000e+00j  3.26436700e-12+0.00000000e+00j
  1.45368070e-15+0.00000000e+00j  5.26052947e-19+0.00000000e+00j
  3.06318453e-22+0.00000000e+00j  8.79380017e-25+0.00000000e+00j
 -1.64971397e-25+0.00000000e+00j  9.82480953e-28+1.45725997e-26j
  9.82480953e-28-1.45725997e-26j -3.62008602e-27+0.00000000e+00j
  1.81847153e-27+0.00000000e+00j -1.39283765e-28+0.00000000e+00j
  2.51889189e-30+3.34123132e-30j  2.51889189e-30-3.34123132e-30j
  3.37346665e-31+0.00000000e+00j  4.36461533e-33+0.00000000e+00j
 -7.35433629e-34+0.00000000e+00j -1.04418906e-34+0.00000000e+00j
 -9.57293490e-37+0.00000000e+00j -8.85886545e-39+0.00000000e+00j
 -8.42674203e-42+2.30420639e-41j -8.42674203e-42-2.30420639e-41j
  6.27348194e-44+0.00000000e+00j -6.44248050e-45+0.00000000e+00j
  7.09635522e-48+0.00000000e+00j -2.28031632e-50+0.00000000e+00j
 -1.13778980e-51+0.00000000e+00j  3.08687283e-53+0.00000000e+00j
  4.81693776e-56+0.00000000e+00j  1.30526291e-58+2.04896161e-58j
  1.30526291e-58-2.04896161e-58j -1.12205210e-60+0.00000000e+00j
 -9.31940177e-62+0.00000000e+00j -6.43101755e-64+0.00000000e+00j
  1.68226586e-65+0.00000000e+00j -5.05760940e-66+0.00000000e+00j
 -2.83813386e-67+0.00000000e+00j  3.37065938e-69+0.00000000e+00j
  3.54307285e-70+0.00000000e+00j -1.31501657e-71+0.00000000e+00j
  1.51005057e-72+0.00000000e+00j  5.19589641e-73+0.00000000e+00j
 -1.32616734e-74+0.00000000e+00j  3.26451029e-76+0.00000000e+00j
  8.43282478e-77+0.00000000e+00j  2.76075155e-78+0.00000000e+00j
 -9.56780442e-79+0.00000000e+00j  2.32157563e-81+0.00000000e+00j
  2.45472108e-84+0.00000000e+00j  8.83338705e-88+0.00000000e+00j
  9.18049023e-90+0.00000000e+00j  1.03702931e-90+0.00000000e+00j
  2.29825078e-91+0.00000000e+00j -3.32595027e-92+0.00000000e+00j]
```




```
[ 1.51242059e+00+0.00000000e+00j  8.37051509e-03+0.00000000e+00j
  1.07613465e-04+0.00000000e+00j  1.79436849e-06+0.00000000e+00j
  3.79826265e-08+0.00000000e+00j  9.78643418e-10+0.00000000e+00j
  2.28533280e-11+0.00000000e+00j  4.92484679e-13+0.00000000e+00j
  9.31983549e-15+0.00000000e+00j  1.54374877e-16+0.00000000e+00j
  4.25615988e-18+0.00000000e+00j  1.84740167e-18+0.00000000e+00j
 -5.52298083e-19+2.75737682e-19j -5.52298083e-19-2.75737682e-19j
  5.90500399e-19+0.00000000e+00j  5.62775146e-20+7.78488124e-20j
  5.62775146e-20-7.78488124e-20j -3.50187193e-20+0.00000000e+00j
  1.78549127e-20+0.00000000e+00j -6.63273496e-21+0.00000000e+00j
  2.03663452e-21+0.00000000e+00j  2.89984369e-22+0.00000000e+00j
  8.77575970e-23+7.26576249e-23j  8.77575970e-23-7.26576249e-23j
 -2.38053671e-23+7.84648384e-24j -2.38053671e-23-7.84648384e-24j
  2.39647360e-23+0.00000000e+00j  2.08901720e-24+0.00000000e+00j
  4.13542215e-25+5.40950387e-25j  4.13542215e-25-5.40950387e-25j
  1.86141662e-25+0.00000000e+00j -9.60426087e-26+1.12616815e-25j
 -9.60426087e-26-1.12616815e-25j  1.84121356e-27+1.18041628e-26j
  1.84121356e-27-1.18041628e-26j  7.55000075e-27+0.00000000e+00j
 -1.12724704e-28+1.64587311e-27j -1.12724704e-28-1.64587311e-27j
 -5.12387994e-28+0.00000000e+00j -2.25574618e-29+5.83305881e-29j
 -2.25574618e-29-5.83305881e-29j  1.83394488e-29+0.00000000e+00j
 -8.57889284e-30+0.00000000e+00j  1.13580130e-30+3.24638088e-30j
  1.13580130e-30-3.24638088e-30j -1.27281866e-30+0.00000000e+00j
  7.27130703e-31+0.00000000e+00j -4.79420849e-31+0.00000000e+00j
  1.45431657e-31+0.00000000e+00j -2.89195106e-32+3.71163545e-32j
 -2.89195106e-32-3.71163545e-32j  3.16289560e-33+2.67532968e-32j
```

```
 3.16289560e-33-2.67532968e-32j -7.51940906e-33+0.00000000e+00j
 2.57578186e-35+1.97192312e-33j  2.57578186e-35-1.97192312e-33j]
```




```
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  4.03376077e-014+0.00000000e+00j  2.69886861e-017+0.00000000e+00j
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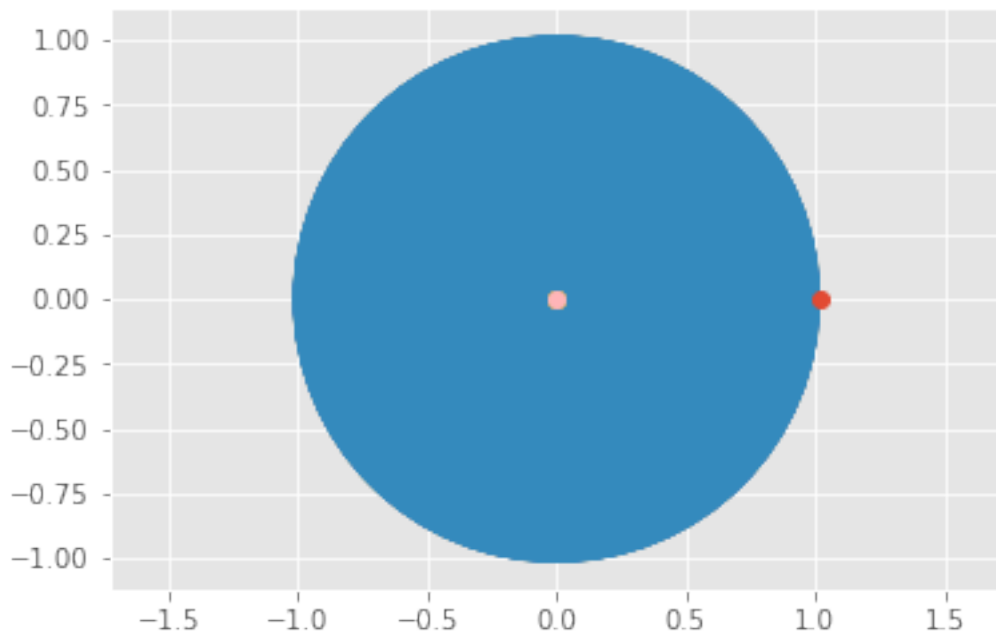
  1.17684751e-026+0.00000000e+00j  1.57468310e-028+0.00000000e+00j
 -1.14488066e-030+0.00000000e+00j  1.52152285e-031+0.00000000e+00j
  3.98817525e-032+0.00000000e+00j  7.86074705e-034+0.00000000e+00j
 -6.00459533e-035+0.00000000e+00j  3.99038957e-036+0.00000000e+00j
  2.94146428e-038+0.00000000e+00j -1.09606112e-040+0.00000000e+00j
 -7.80506495e-042+0.00000000e+00j  5.09442884e-043+0.00000000e+00j
 -2.66429617e-046+0.00000000e+00j  1.09488889e-045+0.00000000e+00j
  7.60317021e-050+0.00000000e+00j  2.89616899e-049+0.00000000e+00j
  1.10127697e-051+0.00000000e+00j  7.77135408e-054+0.00000000e+00j
 -9.74411132e-054+0.00000000e+00j  2.32018964e-056+0.00000000e+00j
  8.00124735e-057+0.00000000e+00j  1.40751359e-059+0.00000000e+00j
  5.78698721e-062+0.00000000e+00j -1.55828683e-063+0.00000000e+00j
  3.00987277e-066+0.00000000e+00j  3.81198996e-068+0.00000000e+00j
  1.65186605e-070+0.00000000e+00j -3.64721579e-073+0.00000000e+00j
 -6.23015009e-074+0.00000000e+00j  2.59140978e-075+0.00000000e+00j
 -4.37499695e-078+0.00000000e+00j  1.22983607e-079+0.00000000e+00j
 -1.20387161e-081+0.00000000e+00j  3.70957667e-084+0.00000000e+00j
 -1.88749914e-086+0.00000000e+00j  1.11110312e-088+0.00000000e+00j
```

```
 1.38881527e-091+3.09300696e-91j  1.38881527e-091-3.09300696e-91j
-1.21539322e-094+0.00000000e+00j  1.13838281e-095+0.00000000e+00j
-5.46551843e-097+0.00000000e+00j -2.32507951e-098+0.00000000e+00j
 3.36601051e-100+0.00000000e+00j -2.16384431e-101+0.00000000e+00j
 1.29568840e-101+0.00000000e+00j -5.62765037e-104+0.00000000e+00j
-4.32104957e-106+0.00000000e+00j  1.79482536e-108+0.00000000e+00j]
```

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  1.30199037e-016+0.j  1.70124213e-021+0.j  3.08191493e-026+0.j
  7.43764040e-031+0.j  2.22472982e-035+0.j  7.41066449e-040+0.j
  2.38906606e-044+0.j -4.24712673e-050+0.j  1.48338280e-051+0.j
  5.87093182e-053+0.j  2.62780694e-062+0.j  4.85483272e-058+0.j
 -1.39525174e-066+0.j  1.41076129e-066+0.j -5.34065957e-074+0.j
  9.18712390e-072+0.j  1.04275415e-078+0.j  8.86740449e-081+0.j
  1.26701493e-085+0.j -1.70174366e-088+0.j  5.63156464e-096+0.j
 -4.75987424e-094+0.j  1.88174787e-100+0.j -2.68480643e-104+0.j
 -7.79183762e-108+0.j  1.23063322e-113+0.j -2.00249031e-114+0.j
 -1.14676422e-118+0.j  6.31219193e-122+0.j -6.47062891e-129+0.j
  3.80499174e-127+0.j  5.81229863e-133+0.j  6.55009312e-137+0.j
  4.17885297e-140+0.j  1.84301513e-144+0.j  2.15222822e-147+0.j
 -1.33685049e-151+0.j -2.01474466e-155+0.j  3.33390459e-159+0.j
 -1.83529850e-163+0.j  6.66368571e-169+0.j -3.80909573e-169+0.j
  2.23042506e-174+0.j  3.40032077e-178+0.j -1.14667904e-181+0.j
  4.59954410e-185+0.j  5.87247531e-189+0.j -7.85110422e-193+0.j
  8.31457926e-196+0.j  2.84793258e-199+0.j  3.95099906e-205+0.j
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-9.01220463e-206+0.j -2.67307373e-211+0.j]
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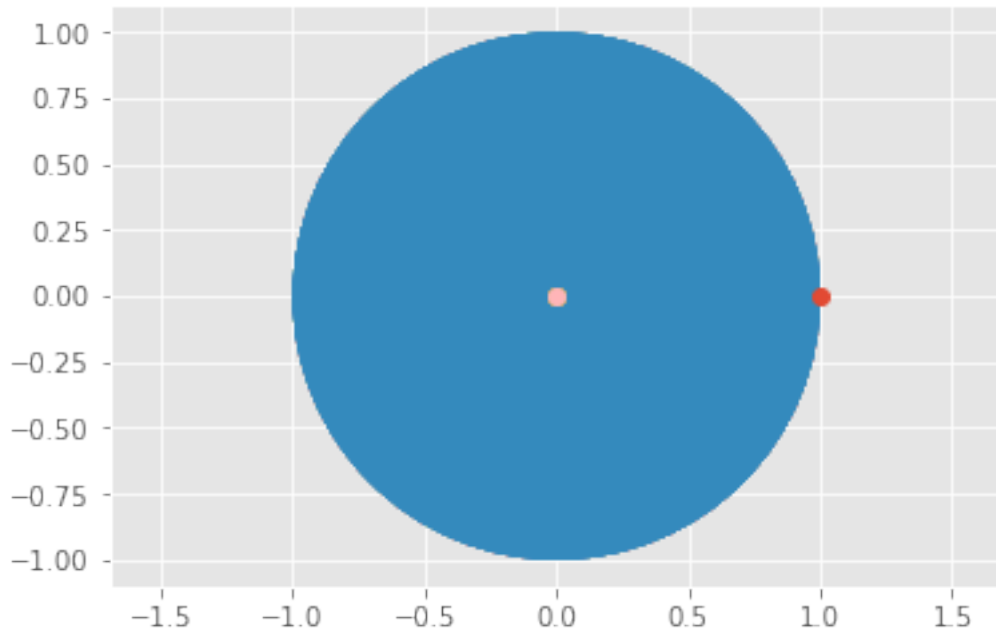


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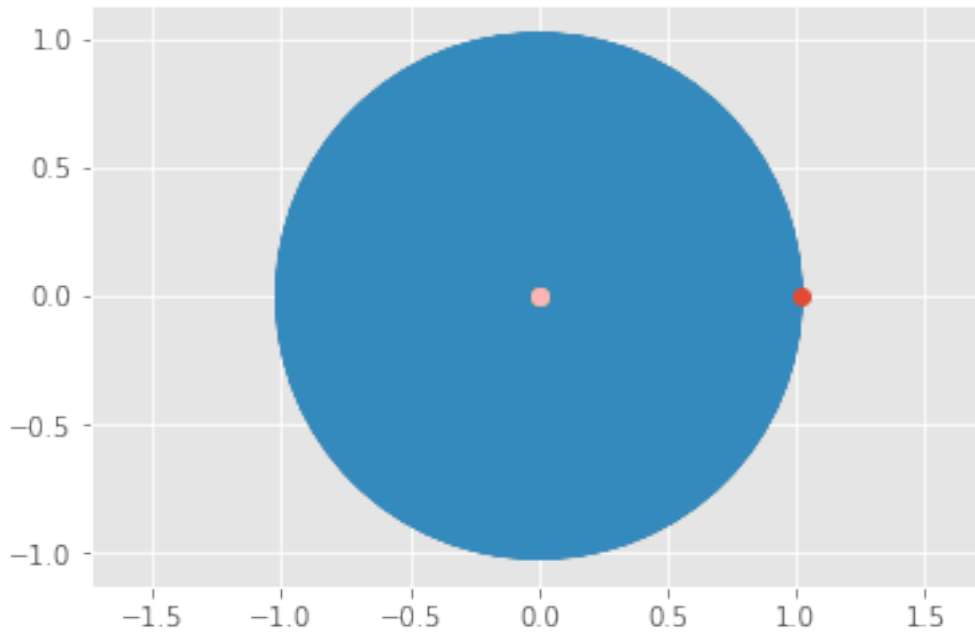
  1.65974053e-007+0.00000000e+00j  1.13596367e-010+0.00000000e+00j
  1.11358142e-013+0.00000000e+00j  1.39586541e-016+0.00000000e+00j
  2.08660386e-019+0.00000000e+00j  3.33004099e-022+0.00000000e+00j
  4.24496330e-025+0.00000000e+00j -1.80065052e-026+0.00000000e+00j
 -5.83394680e-028+0.00000000e+00j -6.96083588e-029+0.00000000e+00j
 -2.94329457e-030+0.00000000e+00j  1.50158638e-031+0.00000000e+00j
 -4.06787117e-033+0.00000000e+00j  3.73822315e-035+0.00000000e+00j
  9.43246520e-037+0.00000000e+00j -3.33372850e-038+0.00000000e+00j
  2.50725941e-039+0.00000000e+00j  1.70237144e-041+0.00000000e+00j
  3.76587538e-042+0.00000000e+00j  3.46448499e-044+0.00000000e+00j
  2.31105593e-046+0.00000000e+00j  4.99721204e-048+1.86142468e-48j
  4.99721204e-048-1.86142468e-48j -3.43951030e-049+0.00000000e+00j
  2.34879607e-050+0.00000000e+00j  1.02850815e-052+7.27045557e-53j
  1.02850815e-052-7.27045557e-53j -3.84960715e-055+0.00000000e+00j
 -9.90214091e-058+0.00000000e+00j -3.54643230e-059+0.00000000e+00j
 -1.52014838e-061+0.00000000e+00j -5.03103864e-066+2.43144523e-64j
 -5.03103864e-066-2.43144523e-64j -6.54851020e-067+0.00000000e+00j
 -4.34231281e-069+0.00000000e+00j -2.20114433e-071+0.00000000e+00j
  4.51070432e-073+0.00000000e+00j -7.30805385e-074+0.00000000e+00j
  3.68864184e-076+0.00000000e+00j  3.87121372e-078+0.00000000e+00j
 -1.72813859e-080+0.00000000e+00j  1.36067510e-081+0.00000000e+00j
 -6.60771258e-085+8.90254808e-85j -6.60771258e-085-8.90254808e-85j
```

```
-5.52550028e-087+0.00000000e+00j -2.07280848e-089+0.00000000e+00j
 2.04038672e-090+0.00000000e+00j -1.73476775e-093+0.00000000e+00j
 3.78586636e-094+0.00000000e+00j -1.51637539e-096+0.00000000e+00j
 7.51201552e-098+0.00000000e+00j  7.20297519e-100+0.00000000e+00j
-1.85749287e-101+0.00000000e+00j -2.28908712e-103+0.00000000e+00j]
```




```
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  1.20691965e-009+0.00000000e+000j  7.63220430e-014+0.00000000e+000j
  8.01714741e-018+0.00000000e+000j  1.04251551e-021+0.00000000e+000j
  5.89447316e-026+0.00000000e+000j  8.10036875e-031+0.00000000e+000j
 -1.46790100e-033+0.00000000e+000j -4.77260815e-036+0.00000000e+000j
 -5.41666497e-040+0.00000000e+000j  7.54476807e-039+0.00000000e+000j
  6.75555972e-041+0.00000000e+000j  6.05408491e-043+0.00000000e+000j
 -1.05287010e-044+0.00000000e+000j  1.56953253e-045+0.00000000e+000j
  4.18567776e-050+0.00000000e+000j  1.81750630e-053+0.00000000e+000j
  3.13283320e-056+0.00000000e+000j -1.71395502e-059+0.00000000e+000j
  6.02207774e-062+0.00000000e+000j  3.44093576e-064+0.00000000e+000j
 -3.63501041e-068+0.00000000e+000j -7.85013499e-070+0.00000000e+000j
  1.67016958e-071+0.00000000e+000j  2.14553236e-075+0.00000000e+000j
  1.58904813e-077+0.00000000e+000j  2.68412816e-078+0.00000000e+000j
  3.25840923e-081+0.00000000e+000j  5.07655904e-083+0.00000000e+000j
 -7.06200213e-084+0.00000000e+000j -8.10454193e-087+0.00000000e+000j
 -5.71545526e-089+0.00000000e+000j -3.21109167e-092+2.08460301e-092j
 -3.21109167e-092-2.08460301e-092j  1.90029518e-095+0.00000000e+000j
  1.07346130e-098+5.12253521e-099j  1.07346130e-098-5.12253521e-099j
```

```
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 1.18808958e-108+0.00000000e+000j -1.67031950e-110+0.00000000e+000j
-1.25241123e-112+0.00000000e+000j  1.52342474e-114+0.00000000e+000j
-1.15205393e-118+0.00000000e+000j  1.76983837e-117+0.00000000e+000j
-3.38159503e-121+0.00000000e+000j  1.13117009e-123+0.00000000e+000j
-1.10256687e-126+0.00000000e+000j  3.08245657e-128+3.05450322e-129j
 3.08245657e-128-3.05450322e-129j  1.53080602e-131+0.00000000e+000j
-2.07292065e-133+0.00000000e+000j -5.78578582e-135+0.00000000e+000j
-7.01619550e-137+0.00000000e+000j  3.51221132e-139+0.00000000e+000j]
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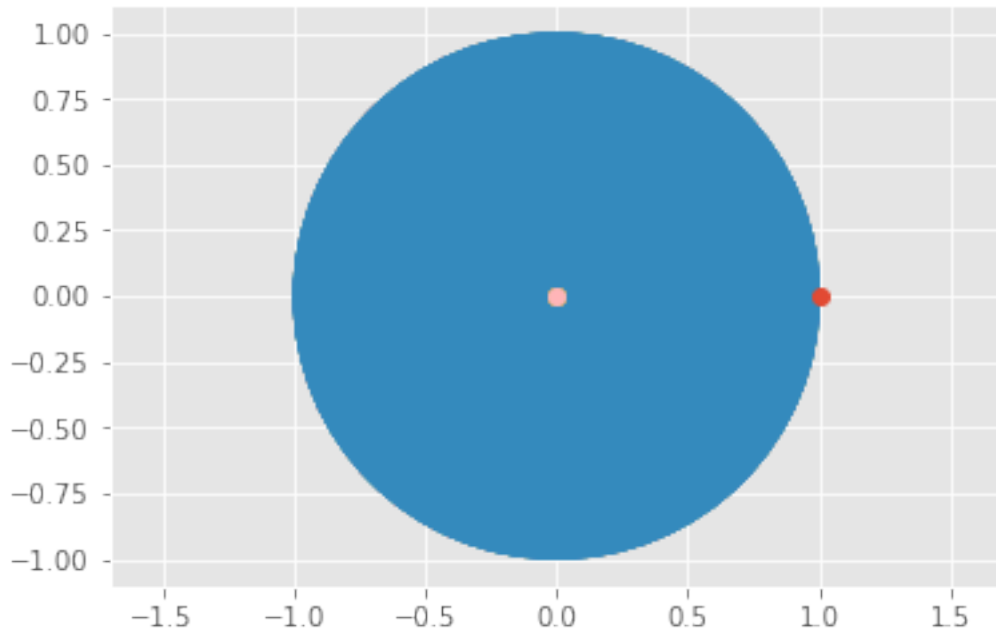


```
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  3.57251790e-008+0.00000000e+00j  1.57402843e-011+0.00000000e+00j
  1.03819563e-014+0.00000000e+00j  9.72614953e-018+0.00000000e+00j
  1.23857113e-020+0.00000000e+00j  1.90137590e-023+0.00000000e+00j
  2.73591798e-026+0.00000000e+00j -3.09496637e-028+0.00000000e+00j
  6.16427847e-030+0.00000000e+00j  2.33417266e-031+0.00000000e+00j
  5.93926240e-033+0.00000000e+00j  3.40693776e-036+2.97125450e-36j
  3.40693776e-036-2.97125450e-36j -5.29775392e-037+0.00000000e+00j
  2.33760647e-038+0.00000000e+00j -3.67527208e-041+0.00000000e+00j
 -5.24098024e-042+0.00000000e+00j -5.26093365e-044+0.00000000e+00j
 -1.86654271e-045+0.00000000e+00j  1.94662832e-047+0.00000000e+00j
  5.28797840e-050+0.00000000e+00j  1.32961938e-052+0.00000000e+00j
 -8.05298115e-054+0.00000000e+00j  1.44015519e-056+0.00000000e+00j
 -2.36950387e-060+3.47252464e-60j -2.36950387e-060-3.47252464e-60j
  1.02803399e-064+5.23255178e-64j  1.02803399e-064-5.23255178e-64j
```

```
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-5.63610736e-070-4.93726424e-70j  1.64056477e-072+0.00000000e+00j
-2.74384406e-074+0.00000000e+00j  5.99115171e-077+0.00000000e+00j
 5.56095132e-079+0.00000000e+00j -3.60369870e-081+0.00000000e+00j
 1.90866687e-082+0.00000000e+00j  3.89867017e-086+2.65881309e-85j
 3.89867017e-086-2.65881309e-85j -3.09890282e-088+0.00000000e+00j
 4.62734582e-089+0.00000000e+00j  3.48598528e-091+0.00000000e+00j
 1.82449146e-093+0.00000000e+00j -4.58607341e-096+0.00000000e+00j
 1.57440762e-097+0.00000000e+00j -9.59649918e-100+0.00000000e+00j
 3.07546407e-101+0.00000000e+00j  6.77513041e-104+0.00000000e+00j
-9.92427869e-105+0.00000000e+00j -3.37982025e-107+0.00000000e+00j
-6.14260609e-109+0.00000000e+00j  6.04320524e-111+0.00000000e+00j
 1.40002377e-115+0.00000000e+00j  1.74498326e-113+0.00000000e+00j]
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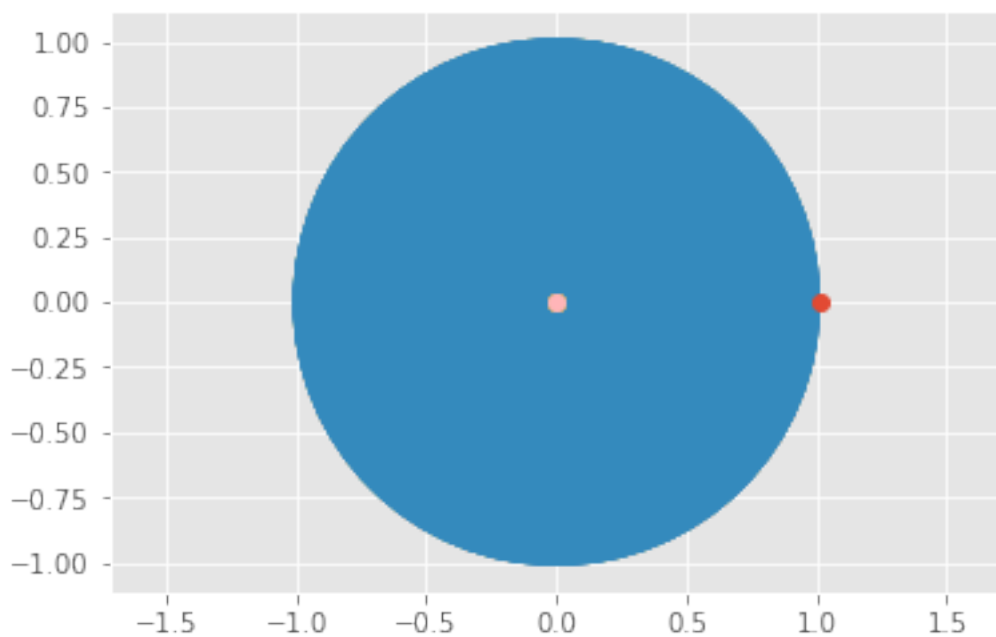


```
[ 1.00220101e+000+0.00000000e+00j -8.80226634e-006+0.00000000e+00j
  1.63586804e-010+0.00000000e+00j -4.49891863e-015+0.00000000e+00j
  1.66292309e-019+0.00000000e+00j -3.54951434e-023+0.00000000e+00j
  1.30120276e-026+0.00000000e+00j -2.00450861e-028+0.00000000e+00j
  1.19455025e-030+0.00000000e+00j -7.53792076e-033+0.00000000e+00j
 -1.62874701e-036+0.00000000e+00j  6.43362900e-038+0.00000000e+00j
  4.32205671e-042+0.00000000e+00j -6.45582975e-046+0.00000000e+00j
 -5.09842360e-047+0.00000000e+00j  4.64974154e-050+0.00000000e+00j
 -5.02599439e-055+0.00000000e+00j  8.07110698e-055+0.00000000e+00j
  4.96834272e-058+0.00000000e+00j  2.47121336e-061+0.00000000e+00j
 -1.12787087e-063+0.00000000e+00j  8.54774632e-067+0.00000000e+00j
```

```
 2.60522797e-068+0.00000000e+00j -2.15995513e-070+0.00000000e+00j
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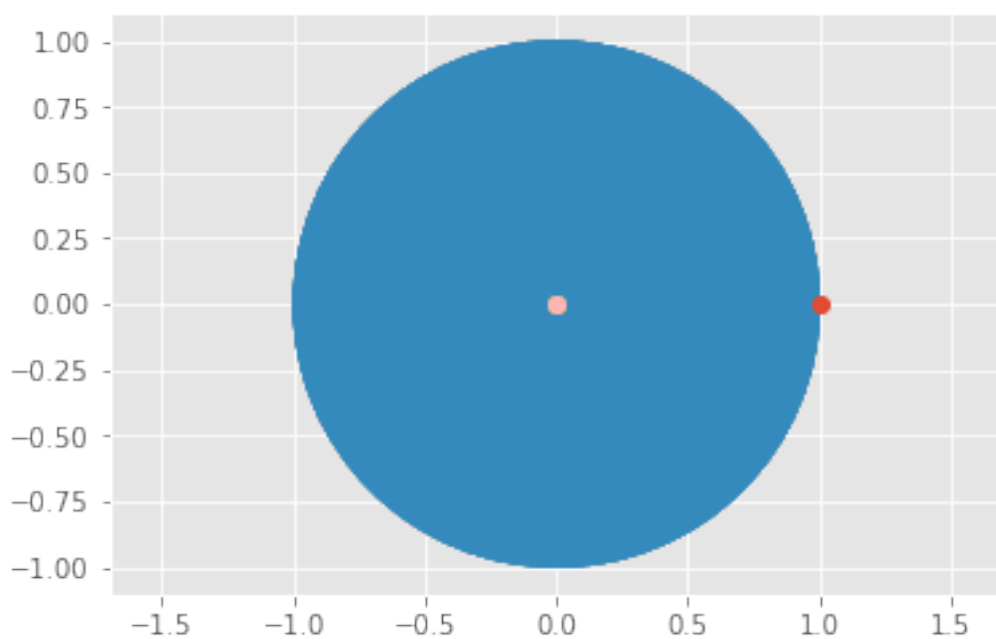
-2.24287887e-075+0.00000000e+00j  4.93229780e-079+0.00000000e+00j
 2.24062922e-082+0.00000000e+00j -1.15975062e-081+0.00000000e+00j
 3.47573628e-086+7.91200889e-86j  3.47573628e-086-7.91200889e-86j
-2.84901612e-090+0.00000000e+00j  1.69529723e-093+0.00000000e+00j
 1.52854745e-094+0.00000000e+00j  1.67957590e-096+0.00000000e+00j
-2.52182358e-102+0.00000000e+00j  1.27819377e-102+0.00000000e+00j
-1.75376327e-105+0.00000000e+00j -2.51649968e-109+0.00000000e+00j
 1.74056907e-110+0.00000000e+00j -6.47046411e-111+0.00000000e+00j
-5.33251029e-113+0.00000000e+00j -7.89413827e-120+0.00000000e+00j
 3.25029067e-118+0.00000000e+00j  1.25525896e-122+0.00000000e+00j
 2.04236694e-124+0.00000000e+00j  7.36849129e-127+0.00000000e+00j
-7.25156747e-132+0.00000000e+00j  3.30049635e-131+0.00000000e+00j
-2.85421115e-137+0.00000000e+00j  9.14577752e-136+0.00000000e+00j
 2.85805094e-139+0.00000000e+00j -5.44141186e-142+0.00000000e+00j
 1.16686818e-143+0.00000000e+00j  6.49501007e-147+0.00000000e+00j]
```




```
[ 1.01741476e+000+0.00000000e+00j  4.31933252e-004+0.00000000e+00j
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 5.38502951e-013+0.00000000e+00j  8.55536558e-016+0.00000000e+00j
 1.48360985e-018+0.00000000e+00j  2.74551802e-021+0.00000000e+00j
 5.69049261e-024+0.00000000e+00j  1.38854046e-026+0.00000000e+00j
 4.78245533e-029+0.00000000e+00j -6.04207460e-029+0.00000000e+00j
-2.98423495e-031+0.00000000e+00j  4.20474671e-031+0.00000000e+00j
```

```
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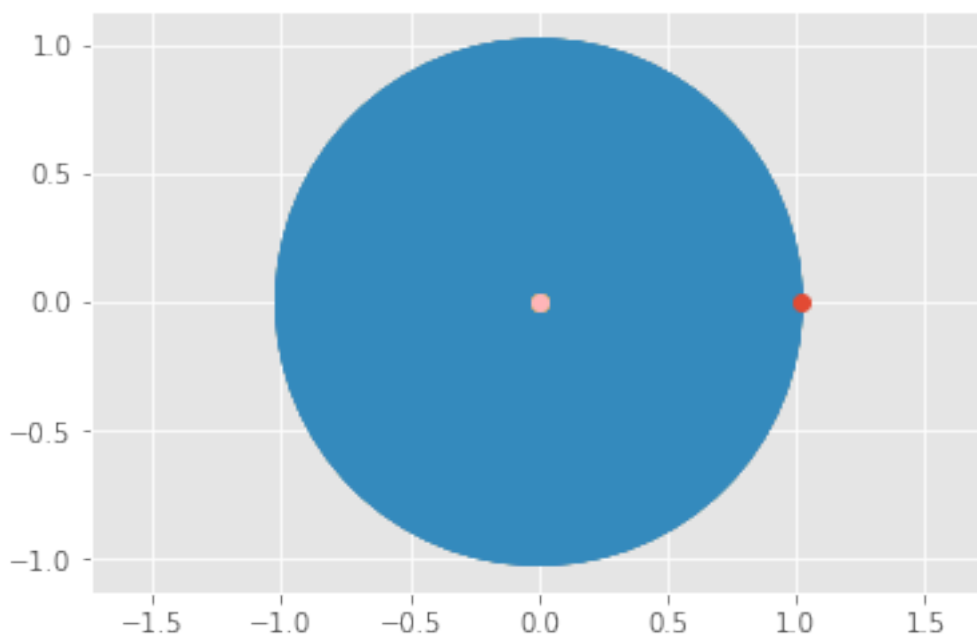
-1.59052142e-041+0.00000000e+00j  4.21450624e-042+0.00000000e+00j
-6.70865700e-044+0.00000000e+00j -2.68990785e-045+0.00000000e+00j
 3.49204619e-046+0.00000000e+00j  8.91717221e-049+0.00000000e+00j
-5.54649915e-050+0.00000000e+00j  3.25828727e-053+0.00000000e+00j
 3.40324189e-054+0.00000000e+00j -1.19186377e-055+0.00000000e+00j
-6.82128117e-058+0.00000000e+00j  7.19980758e-059+0.00000000e+00j
 3.81074072e-061+0.00000000e+00j -3.35059835e-063+0.00000000e+00j
-1.99832682e-065+0.00000000e+00j  7.96638948e-067+0.00000000e+00j
-5.81367238e-069+0.00000000e+00j -3.10590525e-071+0.00000000e+00j
 3.56911549e-071+0.00000000e+00j -1.09208028e-075+0.00000000e+00j
 3.33235894e-074+0.00000000e+00j -1.42820007e-078+0.00000000e+00j
 3.02660516e-080+0.00000000e+00j -2.03863050e-082+0.00000000e+00j
 6.91541873e-083+0.00000000e+00j  1.74798480e-084+0.00000000e+00j
 2.05239143e-086+0.00000000e+00j  6.70806113e-088+0.00000000e+00j
 7.48307902e-091+0.00000000e+00j -6.67933013e-092+0.00000000e+00j
-8.91333034e-094+0.00000000e+00j  2.02972520e-096+0.00000000e+00j
-1.96896781e-097+0.00000000e+00j -5.90461815e-099+0.00000000e+00j
-9.23188390e-104+0.00000000e+00j  7.83657971e-101+0.00000000e+00j]
```

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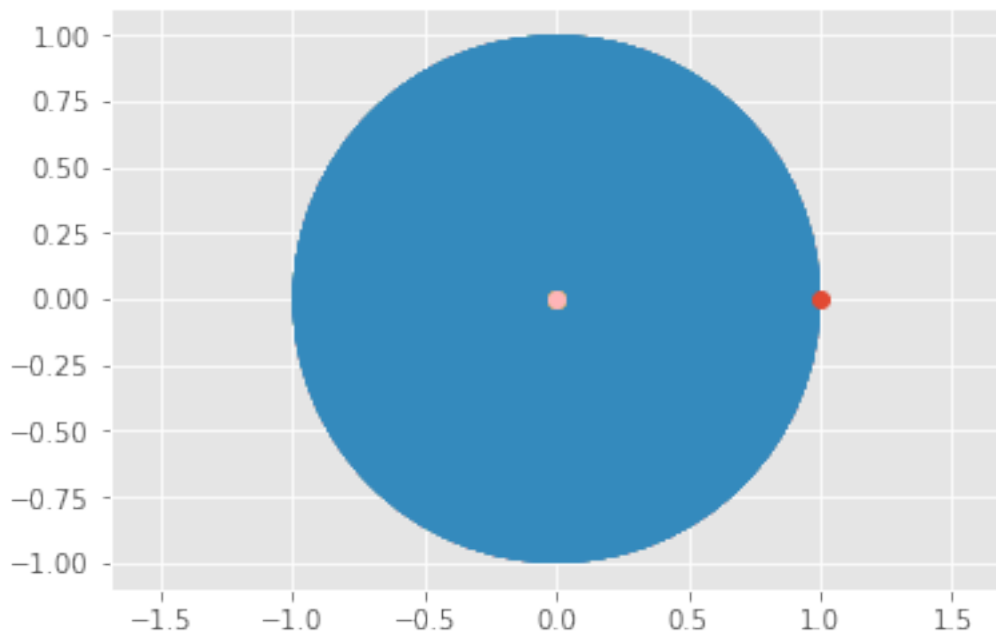
  2.32571576e-036+0.j  4.22459479e-041+0.j  3.59939883e-047+0.j
```

```
-2.95925590e-045+0.j -5.24011480e-049+0.j  1.72327802e-049+0.j
-3.94601475e-051+0.j  7.55378205e-054+0.j -8.29350125e-058+0.j
 1.34601750e-058+0.j  4.86067025e-063+0.j  1.96701040e-066+0.j
 1.46894335e-070+0.j  2.04322282e-075+0.j -1.50753866e-075+0.j
-3.77973274e-080+0.j  1.48701031e-084+0.j -3.24816226e-086+0.j
 6.75252190e-090+0.j -5.96856542e-094+0.j -2.17079324e-102+0.j
-3.20436189e-098+0.j -1.52927768e-106+0.j  1.94264104e-110+0.j
 4.61923517e-114+0.j  7.34383887e-122+0.j  3.70837136e-119+0.j
-3.40233272e-125+0.j -3.84171106e-128+0.j  8.86513084e-133+0.j
-1.05603944e-136+0.j -1.27217824e-141+0.j  2.58368671e-148+0.j
 8.17656758e-147+0.j  2.82184490e-157+0.j  9.52277012e-154+0.j
 1.87616968e-162+0.j  9.76242919e-164+0.j -6.15832954e-169+0.j
 3.06700869e-169+0.j  4.69329876e-174+0.j  1.88370446e-177+0.j
 4.27438109e-181+0.j  1.53867418e-183+0.j  8.58086743e-188+0.j
 2.43624024e-191+0.j  2.34499126e-200+0.j -3.63301526e-197+0.j
-7.54656081e-205+0.j  1.70278189e-208+0.j]
```




```
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  1.15259203e-008+0.00000000e+000j  2.44201532e-012+0.00000000e+000j
  4.13211741e-016+0.00000000e+000j  3.04843666e-020+0.00000000e+000j
  1.75424618e-026+0.00000000e+000j  2.34282096e-030+0.00000000e+000j
 -1.38304762e-032+0.00000000e+000j -2.30816142e-036+0.00000000e+000j
 -4.25701816e-038+0.00000000e+000j  5.88352229e-041+2.35592842e-040j
  5.88352229e-041-2.35592842e-040j -3.59457066e-043+0.00000000e+000j
  1.58877520e-042+0.00000000e+000j -2.66544646e-044+0.00000000e+000j
```

```
 1.45662681e-044+0.00000000e+000j -1.54205543e-047+0.00000000e+000j
-2.88646179e-049+0.00000000e+000j -5.02582499e-053+0.00000000e+000j
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-2.78817209e-057+0.00000000e+000j -5.14882689e-060+0.00000000e+000j
-2.37350995e-062+0.00000000e+000j -1.07257408e-064+0.00000000e+000j
-4.43366425e-069+0.00000000e+000j  1.09768665e-071+0.00000000e+000j
 3.10597004e-074+0.00000000e+000j -2.35539442e-078+0.00000000e+000j
 1.37434888e-080+2.01934910e-080j  1.37434888e-080-2.01934910e-080j
 1.15359487e-083+0.00000000e+000j -2.46164763e-087+0.00000000e+000j
-4.22788332e-089+0.00000000e+000j  1.42476242e-092+0.00000000e+000j
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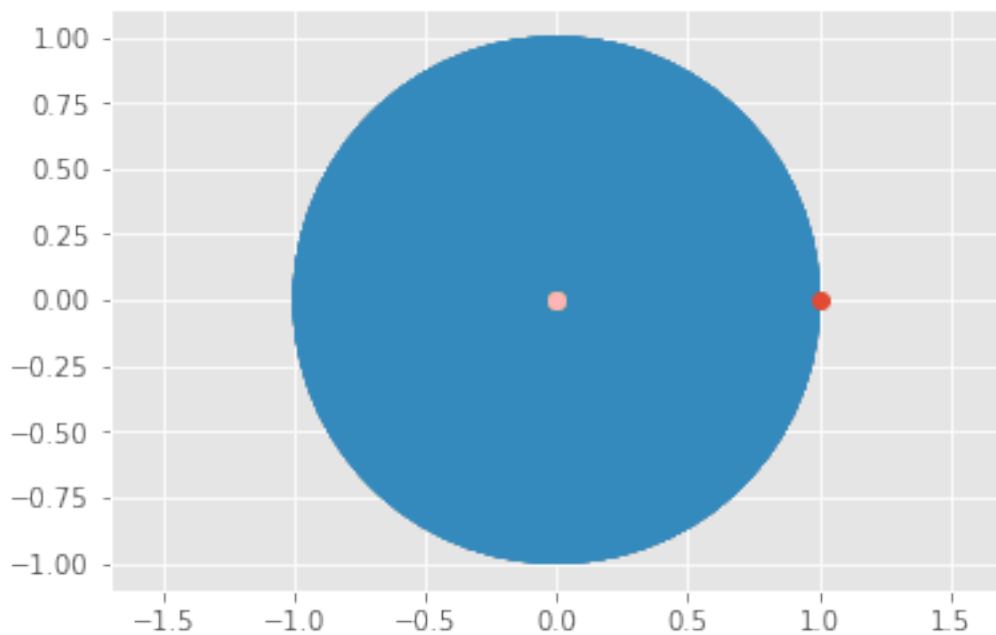
 8.01891273e-102+0.00000000e+000j  7.71682248e-101+0.00000000e+000j
 3.48867497e-104+0.00000000e+000j -3.51538430e-109+4.29359399e-109j
-3.51538430e-109-4.29359399e-109j  2.88054231e-112+0.00000000e+000j
-2.88453824e-114+0.00000000e+000j  1.06150318e-116+0.00000000e+000j
-7.79581704e-120+0.00000000e+000j  1.59249093e-121+0.00000000e+000j
 2.62220861e-124+0.00000000e+000j  2.82522207e-126+0.00000000e+000j
-4.45356246e-129+0.00000000e+000j -3.87170912e-131+0.00000000e+000j
-1.01970597e-133+0.00000000e+000j  7.44725835e-136+0.00000000e+000j
-1.22231161e-138+0.00000000e+000j  2.16657927e-141+0.00000000e+000j]
```




```
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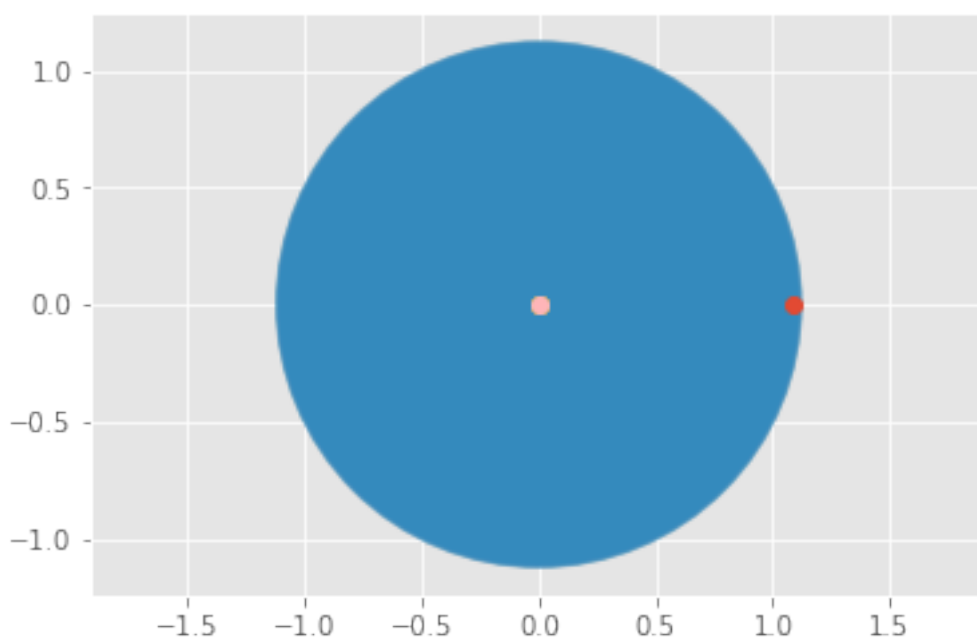
  3.55920752e-06+0.00000000e+00j  1.39059364e-08+0.00000000e+00j
  7.11571795e-11+0.00000000e+00j  4.12444494e-13+0.00000000e+00j
  2.90937575e-15+0.00000000e+00j  2.44775456e-17+0.00000000e+00j
```

```
 2.08596353e-19+0.00000000e+00j  3.81738085e-21+0.00000000e+00j
 3.57364824e-22+0.00000000e+00j -1.19360892e-22+0.00000000e+00j
-7.33754462e-24+5.91011114e-24j -7.33754462e-24-5.91011114e-24j
 1.24729689e-25+0.00000000e+00j -4.52995599e-27+1.11092855e-26j
-4.52995599e-27-1.11092855e-26j  6.56629261e-27+0.00000000e+00j
 7.35480000e-28+0.00000000e+00j -3.36751861e-30+0.00000000e+00j
 3.43196773e-31+0.00000000e+00j  3.92793734e-32+0.00000000e+00j
-1.02944916e-32+0.00000000e+00j  5.03417089e-33+0.00000000e+00j
-6.96309422e-34+0.00000000e+00j  2.46521453e-36+2.27489370e-36j
 2.46521453e-36-2.27489370e-36j -2.85665033e-37+0.00000000e+00j
 7.73299265e-39+0.00000000e+00j  1.31144868e-39+3.47418511e-39j
 1.31144868e-39-3.47418511e-39j  1.85928035e-40+0.00000000e+00j
-7.67986375e-41+0.00000000e+00j  6.14246395e-42+0.00000000e+00j
 4.90630382e-43+0.00000000e+00j  3.13302078e-44+0.00000000e+00j
 8.28342677e-46+0.00000000e+00j -7.47186619e-46+0.00000000e+00j
 9.04616149e-47+0.00000000e+00j -3.16565508e-48+0.00000000e+00j
 2.53660319e-48+0.00000000e+00j -2.28502453e-49+0.00000000e+00j
 8.43940287e-51+0.00000000e+00j  2.30988108e-52+0.00000000e+00j
-3.61343484e-53+0.00000000e+00j  2.44733029e-53+0.00000000e+00j
 3.08210439e-55+0.00000000e+00j  2.74371376e-56+0.00000000e+00j
-1.87561232e-57+0.00000000e+00j  1.51793786e-57+0.00000000e+00j
 1.20812759e-58+0.00000000e+00j  1.95227831e-60+0.00000000e+00j
-1.35981779e-60+0.00000000e+00j -2.03249560e-61+0.00000000e+00j
 5.60491438e-64+0.00000000e+00j  1.01230815e-63+0.00000000e+00j]
```



```
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  3.88392163e-14+0.00000000e+00j  4.19470435e-17+0.00000000e+00j
  5.85076671e-20+0.00000000e+00j  9.84223848e-23+0.00000000e+00j
  7.93499534e-26+7.81831443e-26j  7.93499534e-26-7.81831443e-26j
 -5.65195543e-28+3.61851230e-27j -5.65195543e-28-3.61851230e-27j
  6.26151799e-28+0.00000000e+00j  3.61775194e-29+0.00000000e+00j
 -1.55443559e-31+0.00000000e+00j -1.12904267e-32+0.00000000e+00j
  6.11691979e-34+0.00000000e+00j  7.10386062e-36+0.00000000e+00j
 -3.49886547e-37+0.00000000e+00j -1.60347162e-38+0.00000000e+00j
  2.20388244e-40+0.00000000e+00j -2.27390785e-41+0.00000000e+00j
 -4.31819786e-43+0.00000000e+00j  6.51467744e-46+1.87973342e-45j
  6.51467744e-46-1.87973342e-45j  9.13349349e-48+0.00000000e+00j
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  4.25284396e-53+0.00000000e+00j  5.14727332e-57+7.44749673e-57j
  5.14727332e-57-7.44749673e-57j  5.10043370e-60+0.00000000e+00j
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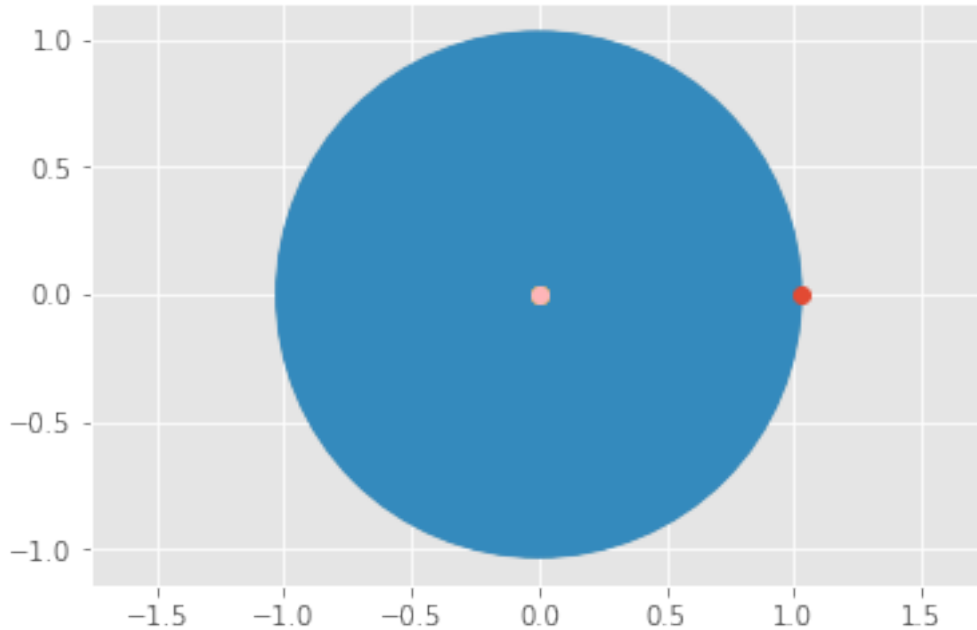
  5.21364907e-64+0.00000000e+00j -1.04406508e-64+0.00000000e+00j
  2.79470999e-65+0.00000000e+00j -2.65846398e-69+0.00000000e+00j
  6.36055076e-68+0.00000000e+00j  2.93343146e-71+0.00000000e+00j
  5.69813600e-73+0.00000000e+00j  5.80507641e-75+1.28843161e-74j
  5.80507641e-75-1.28843161e-74j  3.46251796e-77+5.03140891e-77j
  3.46251796e-77-5.03140891e-77j -1.09764172e-78+0.00000000e+00j
  3.63826207e-80+0.00000000e+00j -3.44947301e-82+7.43220373e-82j
 -3.44947301e-82-7.43220373e-82j -1.52243341e-83+0.00000000e+00j
  5.99245671e-85+1.92972232e-84j  5.99245671e-85-1.92972232e-84j
 -4.10389491e-88+0.00000000e+00j -1.75631073e-86+0.00000000e+00j]
```

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  4.51799319e-09+0.00000000e+00j  5.73249433e-11+0.00000000e+00j
  1.02304249e-12+0.00000000e+00j  2.50950901e-14+0.00000000e+00j
  6.42674123e-16+0.00000000e+00j  2.00865401e-17+0.00000000e+00j
  9.78948593e-19+0.00000000e+00j  1.82828353e-19+0.00000000e+00j
  2.87033024e-20+0.00000000e+00j -3.73556315e-22+1.10622975e-20j
 -3.73556315e-22-1.10622975e-20j  4.11455408e-21+0.00000000e+00j
  5.70338266e-22+4.47074768e-22j  5.70338266e-22-4.47074768e-22j
 -6.07602573e-22+0.00000000e+00j  2.70935857e-23+2.01829042e-22j
  2.70935857e-23-2.01829042e-22j  5.33409954e-23+0.00000000e+00j
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  1.15579417e-23+0.00000000e+00j -2.73315263e-24+2.21298760e-24j
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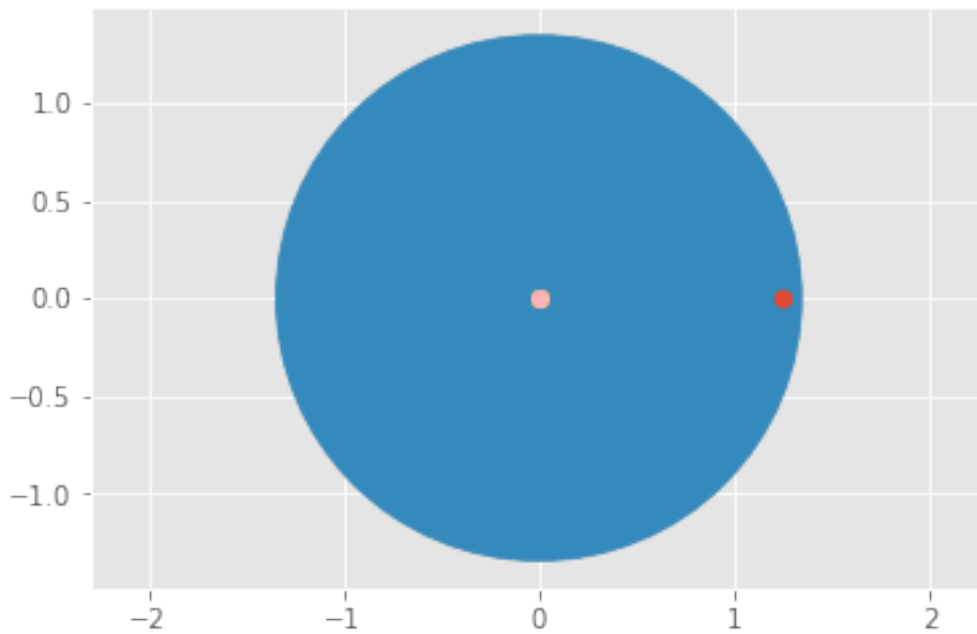
 -1.96232879e-28+0.00000000e+00j -7.16609876e-30+0.00000000e+00j
 -1.05662191e-30+0.00000000e+00j -9.82352051e-32+0.00000000e+00j
  2.90721368e-33+6.02488017e-33j  2.90721368e-33-6.02488017e-33j
 -1.33785489e-33+0.00000000e+00j  1.36410885e-34+0.00000000e+00j
  9.16935199e-36+0.00000000e+00j -9.84629138e-36+1.07369230e-36j
 -9.84629138e-36-1.07369230e-36j  1.19221354e-38+4.63206257e-38j
  1.19221354e-38-4.63206257e-38j  4.50875532e-39+0.00000000e+00j
  1.04380108e-40+9.86280376e-41j  1.04380108e-40-9.86280376e-41j
  3.89318370e-42+6.44135108e-42j  3.89318370e-42-6.44135108e-42j
 -5.22986744e-43+0.00000000e+00j -8.03478240e-44+7.85109613e-44j
```

```
 -8.03478240e-44-7.85109613e-44j -2.30802081e-45+0.00000000e+00j
  4.93937834e-45+0.00000000e+00j  1.31426187e-46+0.00000000e+00j]
```




```
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  5.62874527e-15+0.00000000e+00j -1.12212533e-17+0.00000000e+00j
 -2.24325845e-20+0.00000000e+00j -1.15787563e-22+7.84449273e-23j
 -1.15787563e-22-7.84449273e-23j  8.97798755e-23+0.00000000e+00j
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  4.85106943e-25+0.00000000e+00j -3.45682129e-26+0.00000000e+00j
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 -3.23126975e-29+0.00000000e+00j -1.15794837e-30+0.00000000e+00j
 -1.18409113e-31+0.00000000e+00j  1.12903660e-31+0.00000000e+00j
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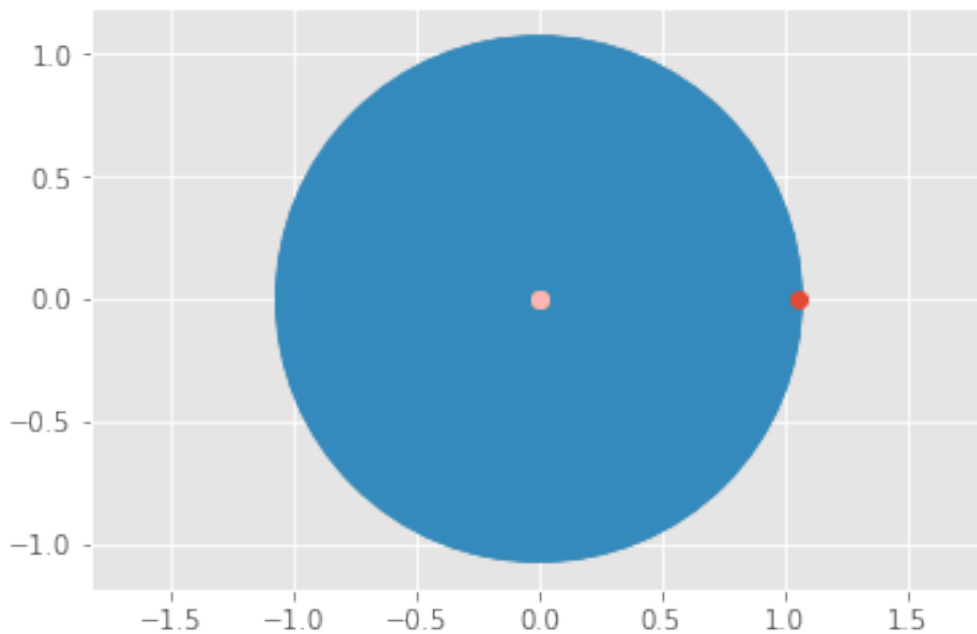
 -2.26324864e-37+4.69199623e-37j -2.26324864e-37-4.69199623e-37j
 -2.71612116e-39+0.00000000e+00j  9.94147950e-41+0.00000000e+00j
  1.41319039e-41+0.00000000e+00j  1.14521757e-42+0.00000000e+00j
  7.48026974e-45+0.00000000e+00j -3.01696735e-46+0.00000000e+00j
  1.89418139e-47+0.00000000e+00j  6.91154912e-49+0.00000000e+00j
 -2.16455149e-49+0.00000000e+00j  9.56004817e-52+5.73746413e-51j
  9.56004817e-52-5.73746413e-51j -3.14852703e-52+1.32726327e-51j
 -3.14852703e-52-1.32726327e-51j  1.50398153e-52+0.00000000e+00j
  1.21965013e-53+0.00000000e+00j  7.43770295e-55+0.00000000e+00j
  9.11395328e-57+0.00000000e+00j  2.05426280e-58+0.00000000e+00j
```

```
-6.58357327e-59+0.00000000e+00j  6.07479646e-61+0.00000000e+00j
 2.31138526e-62+0.00000000e+00j  4.92209875e-64+0.00000000e+00j
 1.71555796e-65+0.00000000e+00j -2.64093047e-66+0.00000000e+00j
 1.20077024e-66+0.00000000e+00j -1.57726174e-68+0.00000000e+00j
 1.46198085e-70+0.00000000e+00j -2.23571684e-71+0.00000000e+00j
-1.68021075e-74+0.00000000e+00j -5.81228585e-73+0.00000000e+00j]
```




```
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  1.03739637e+00+0.00000000e+00j -5.51666856e-01+0.00000000e+00j
  8.40377897e-02+0.00000000e+00j  6.00704055e-03+0.00000000e+00j
  9.40632720e-04+0.00000000e+00j  7.23211622e-05+0.00000000e+00j
  6.11403498e-06+0.00000000e+00j  8.80370865e-07+0.00000000e+00j
  5.86954870e-08+0.00000000e+00j  2.26330095e-09+0.00000000e+00j
  2.67937480e-11+0.00000000e+00j -1.92224546e-11+0.00000000e+00j
  2.17369557e-13+0.00000000e+00j  6.46633061e-14+0.00000000e+00j
 -9.40375198e-15+0.00000000e+00j -2.76011641e-15+0.00000000e+00j
 -1.74673820e-15+7.94990617e-16j -1.74673820e-15-7.94990617e-16j
  7.82004886e-16+1.38145171e-15j  7.82004886e-16-1.38145171e-15j
  1.65985213e-15+0.00000000e+00j -5.12952397e-17+1.18486917e-15j
 -5.12952397e-17-1.18486917e-15j -3.37599912e-16+1.09334883e-15j
 -3.37599912e-16-1.09334883e-15j  7.60015644e-16+8.63251051e-16j
  7.60015644e-16-8.63251051e-16j  1.02450128e-15+4.38184060e-16j
  1.02450128e-15-4.38184060e-16j -7.67573308e-16+4.31072353e-16j
 -7.67573308e-16-4.31072353e-16j  9.05386544e-16+0.00000000e+00j
 -5.98028763e-16+1.67260833e-16j -5.98028763e-16-1.67260833e-16j
```

```
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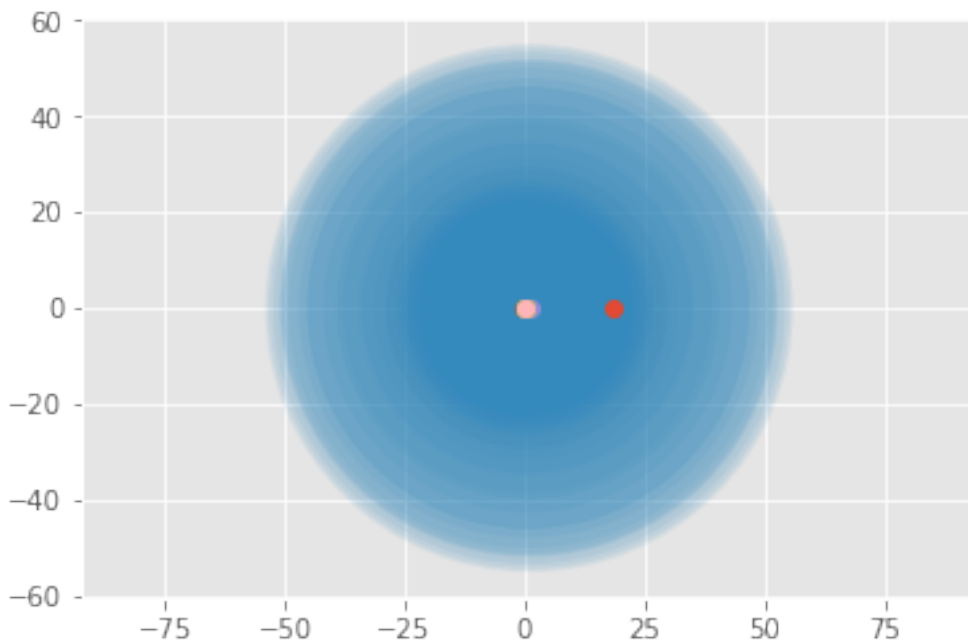
 4.80531778e-16+2.73623499e-16j  4.80531778e-16-2.73623499e-16j
 1.50393737e-16+5.15452425e-16j  1.50393737e-16-5.15452425e-16j
 4.41291617e-16+0.00000000e+00j -1.39734842e-16+4.55341585e-16j
-1.39734842e-16-4.55341585e-16j  1.85343514e-16+3.56778275e-16j
 1.85343514e-16-3.56778275e-16j -3.08599860e-16+0.00000000e+00j
-7.46219683e-17+2.36332790e-16j -7.46219683e-17-2.36332790e-16j
-1.56245359e-16+1.35515297e-16j -1.56245359e-16-1.35515297e-16j
 1.41730896e-16+0.00000000e+00j  1.91317549e-17+0.00000000e+00j
-6.48002934e-18+8.02223595e-17j -6.48002934e-18-8.02223595e-17j]
```

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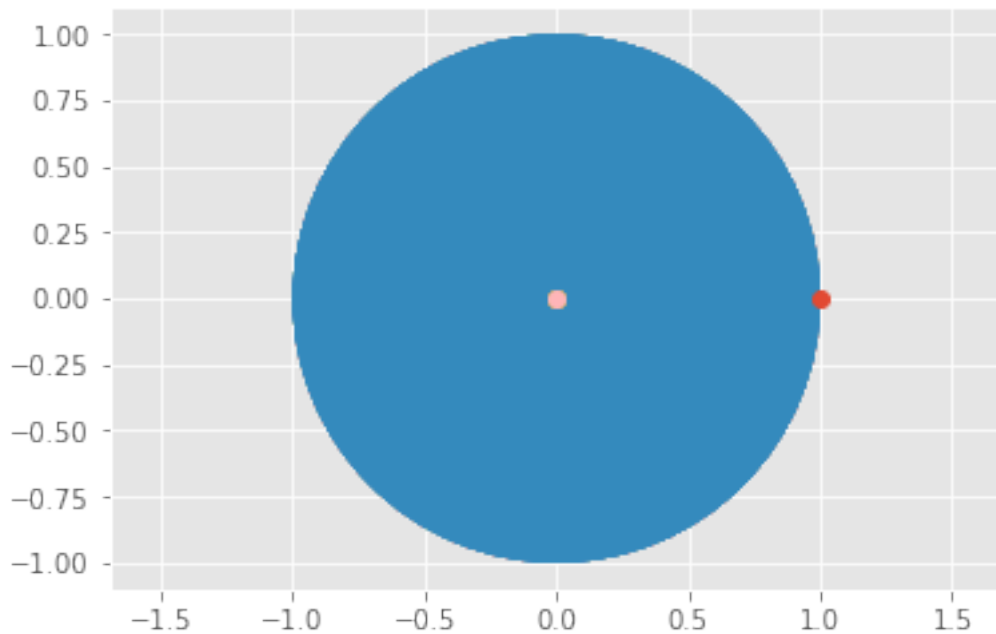
  1.45877606e-021+0.00000000e+00j  1.94309640e-026+0.00000000e+00j
  3.00363833e-031+0.00000000e+00j  5.17022346e-036+0.00000000e+00j
  8.45806841e-041+0.00000000e+00j  1.27859829e-045+0.00000000e+00j
  7.09261011e-050+0.00000000e+00j  4.37817624e-055+0.00000000e+00j
 -6.29317916e-058+0.00000000e+00j -1.13868679e-061+0.00000000e+00j
  1.46698131e-065+0.00000000e+00j -2.98936360e-069+0.00000000e+00j
  1.16926856e-073+0.00000000e+00j -2.13711970e-076+2.00620355e-76j
 -2.13711970e-076-2.00620355e-76j -1.30667188e-081+0.00000000e+00j
 -5.55116453e-085+0.00000000e+00j  1.22638613e-088+0.00000000e+00j
  1.00032399e-092+0.00000000e+00j -4.35655688e-097+0.00000000e+00j
  6.42643945e-099+0.00000000e+00j  4.09134333e-104+0.00000000e+00j
 -1.41925932e-106+0.00000000e+00j  1.13690973e-110+0.00000000e+00j
```

```
 5.56200147e-118+0.00000000e+00j  2.15077237e-114+0.00000000e+00j
-1.39042874e-121+0.00000000e+00j  2.61952917e-124+0.00000000e+00j
-9.72549481e-131+0.00000000e+00j -6.09938598e-129+0.00000000e+00j
-1.22936910e-137+0.00000000e+00j  3.32839299e-136+0.00000000e+00j
-4.96299980e-142+0.00000000e+00j  7.89293247e-147+0.00000000e+00j
 3.59017189e-150+0.00000000e+00j -4.45827722e-154+0.00000000e+00j
-1.21178666e-157+0.00000000e+00j  3.86264494e-161+0.00000000e+00j
 1.04337669e-166+0.00000000e+00j -2.32598100e-166+0.00000000e+00j
-5.26164503e-173+0.00000000e+00j  5.30743387e-176+0.00000000e+00j
 5.33896575e-179+0.00000000e+00j  4.77779888e-183+0.00000000e+00j
 1.98712880e-186+0.00000000e+00j  3.48016195e-190+0.00000000e+00j
-2.31268423e-193+0.00000000e+00j  7.33675173e-198+0.00000000e+00j
 2.90020693e-201+0.00000000e+00j  2.63445158e-206+0.00000000e+00j
-1.20012117e-208+0.00000000e+00j  4.42055882e-212+0.00000000e+00j]
```




```
[ 1.00010354e+00+0.00000000e+00j  2.65068591e-06+0.00000000e+00j
  1.49183918e-11+0.00000000e+00j  1.29325610e-16+0.00000000e+00j
  1.45877606e-21+0.00000000e+00j  1.94309633e-26+0.00000000e+00j
  3.00363010e-31+0.00000000e+00j  5.16924399e-36+0.00000000e+00j
  1.80260548e-39+0.00000000e+00j  8.37515282e-41+0.00000000e+00j
  2.12985786e-43+0.00000000e+00j  9.83577888e-46+0.00000000e+00j
  3.17925732e-46+0.00000000e+00j  4.36368511e-49+0.00000000e+00j
  4.65222539e-50+0.00000000e+00j  1.12458691e-52+1.11340618e-52j
  1.12458691e-52-1.11340618e-52j  1.62074927e-53+0.00000000e+00j
  2.44232085e-55+9.12478974e-55j  2.44232085e-55-9.12478974e-55j
```

```
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 -8.45355966e-56+1.53044891e-56j -8.45355966e-56-1.53044891e-56j
 -2.13031837e-57+8.86234506e-56j -2.13031837e-57-8.86234506e-56j
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  3.97704518e-56+3.28995768e-56j  3.97704518e-56-3.28995768e-56j
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  8.70616928e-57+0.00000000e+00j  2.47517725e-57+0.00000000e+00j
 -1.47104116e-57+0.00000000e+00j -6.34788469e-58+0.00000000e+00j
 -1.17651172e-59+1.67847460e-59j -1.17651172e-59-1.67847460e-59j
 -1.30246602e-59+0.00000000e+00j -8.22774505e-62+0.00000000e+00j
 -2.83396252e-63+0.00000000e+00j  4.82231349e-65+1.01917800e-64j
  4.82231349e-65-1.01917800e-64j  7.95564465e-66+0.00000000e+00j
 -3.64710024e-66+0.00000000e+00j  1.14199987e-67+0.00000000e+00j
  3.53090487e-69+0.00000000e+00j -3.49020115e-69+0.00000000e+00j
  1.31716464e-70+0.00000000e+00j -6.76663047e-72+0.00000000e+00j
 -3.82726674e-73+0.00000000e+00j  4.35143677e-74+0.00000000e+00j
  1.00563486e-75+0.00000000e+00j -1.13561531e-77+0.00000000e+00j]
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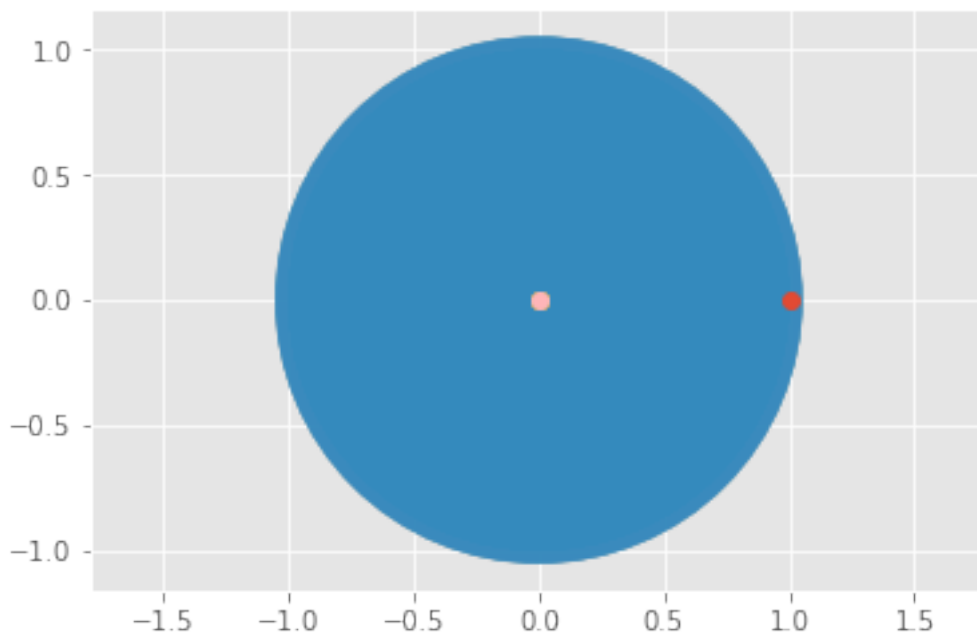


```
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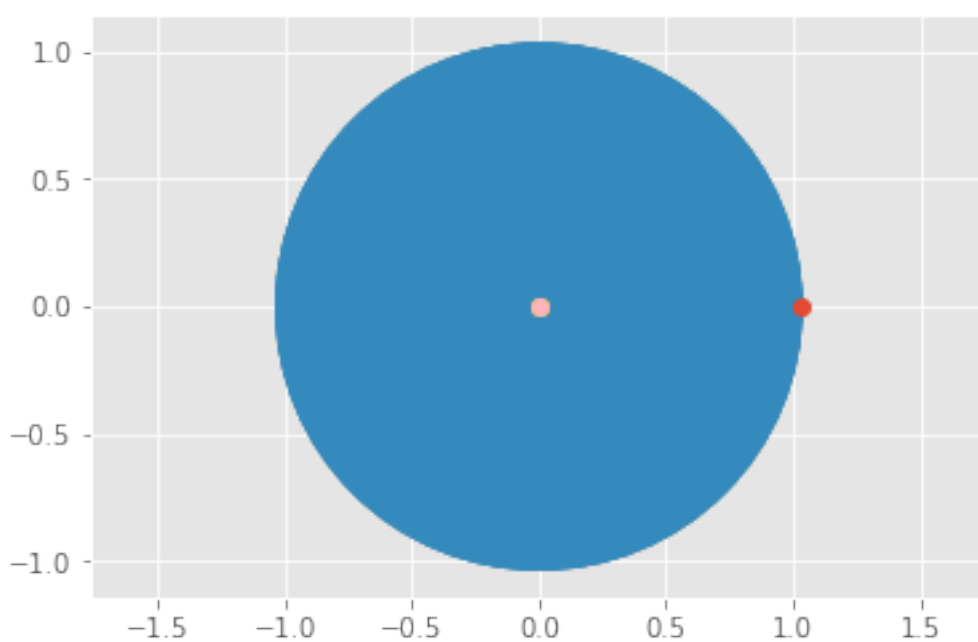
  4.08126149e-07+0.00000000e+00j  3.18459742e-10+0.00000000e+00j
  3.64651896e-13+0.00000000e+00j  4.90759718e-16+0.00000000e+00j
  7.07359368e-19+0.00000000e+00j  1.03870382e-21+0.00000000e+00j
  4.74438198e-25+0.00000000e+00j -6.62510171e-25+0.00000000e+00j
  3.63991230e-27+0.00000000e+00j -4.52844348e-28+0.00000000e+00j
```

```
 1.57717570e-28+0.00000000e+00j  7.93650934e-29+0.00000000e+00j
-1.30927418e-31+0.00000000e+00j -1.99490591e-33+9.38791304e-33j
-1.99490591e-33-9.38791304e-33j -6.93291645e-34+0.00000000e+00j
 6.81586074e-36+0.00000000e+00j -1.33134843e-37+3.17638781e-37j
-1.33134843e-37-3.17638781e-37j  9.87139302e-40+0.00000000e+00j
-1.21971709e-39+0.00000000e+00j  1.69506818e-41+0.00000000e+00j
-1.95125043e-42+0.00000000e+00j  4.75856066e-44+0.00000000e+00j
-7.50615879e-46+0.00000000e+00j  1.68632039e-47+0.00000000e+00j
 2.42746119e-49+0.00000000e+00j -1.76062708e-50+0.00000000e+00j
 1.50430191e-52+0.00000000e+00j  1.43729824e-55+0.00000000e+00j
 1.28959670e-56+0.00000000e+00j -7.20295171e-57+0.00000000e+00j
 3.74607106e-58+0.00000000e+00j  2.65137999e-60+0.00000000e+00j
 3.89392717e-62+0.00000000e+00j  1.55378737e-62+0.00000000e+00j
 3.12890635e-65+0.00000000e+00j  2.98046980e-67+0.00000000e+00j
 2.35884472e-68+0.00000000e+00j -6.89578315e-70+0.00000000e+00j
-8.80149903e-72+0.00000000e+00j  2.51700859e-73+0.00000000e+00j
-7.43569611e-75+0.00000000e+00j  1.74046915e-76+0.00000000e+00j
-1.09564234e-78+0.00000000e+00j -2.90832446e-79+0.00000000e+00j
-2.61648829e-81+0.00000000e+00j  1.21456471e-82+0.00000000e+00j
 1.36602643e-84+0.00000000e+00j  6.01075631e-87+4.27916595e-86j
 6.01075631e-87-4.27916595e-86j  1.59051941e-89+0.00000000e+00j
 7.27656409e-90+0.00000000e+00j  5.30904952e-91+0.00000000e+00j]
```




```
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  8.89977375e-009+0.00000000e+000j  3.28303312e-012+0.00000000e+000j
```

```
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 2.36426133e-026+0.00000000e+000j -9.12575120e-029+0.00000000e+000j
-2.27672471e-030+0.00000000e+000j  1.41182927e-031+0.00000000e+000j
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 1.60159005e-042+0.00000000e+000j  5.51750908e-046+1.20612991e-045j
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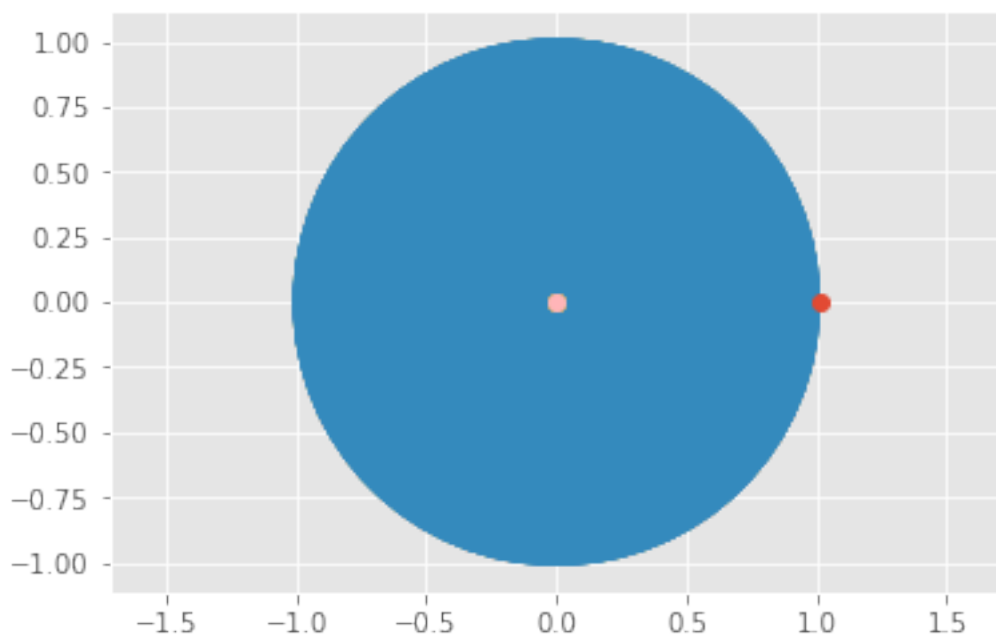
-1.64101466e-066+0.00000000e+000j  1.12688684e-067+0.00000000e+000j
-3.95180401e-071+0.00000000e+000j  1.24603168e-073+4.19372495e-073j
 1.24603168e-073-4.19372495e-073j -1.02661098e-076+0.00000000e+000j
 2.80141188e-078+0.00000000e+000j  5.86937583e-082+0.00000000e+000j
-2.58749891e-081+0.00000000e+000j  4.26525415e-083+0.00000000e+000j
 6.19151473e-089+0.00000000e+000j  2.32764093e-086+0.00000000e+000j
 3.51035817e-091+7.55004439e-091j  3.51035817e-091-7.55004439e-091j
 1.17397761e-094+0.00000000e+000j  1.29370769e-096+1.16158323e-096j
 1.29370769e-096-1.16158323e-096j -4.48419773e-098+0.00000000e+000j
 5.88198101e-100+0.00000000e+000j -2.20226158e-102+0.00000000e+000j
 6.89570713e-104+0.00000000e+000j -1.82517155e-106+0.00000000e+000j
 1.91764119e-108+1.32828288e-108j  1.91764119e-108-1.32828288e-108j
-6.30188543e-112+0.00000000e+000j  2.36258053e-114+0.00000000e+000j]
```



```
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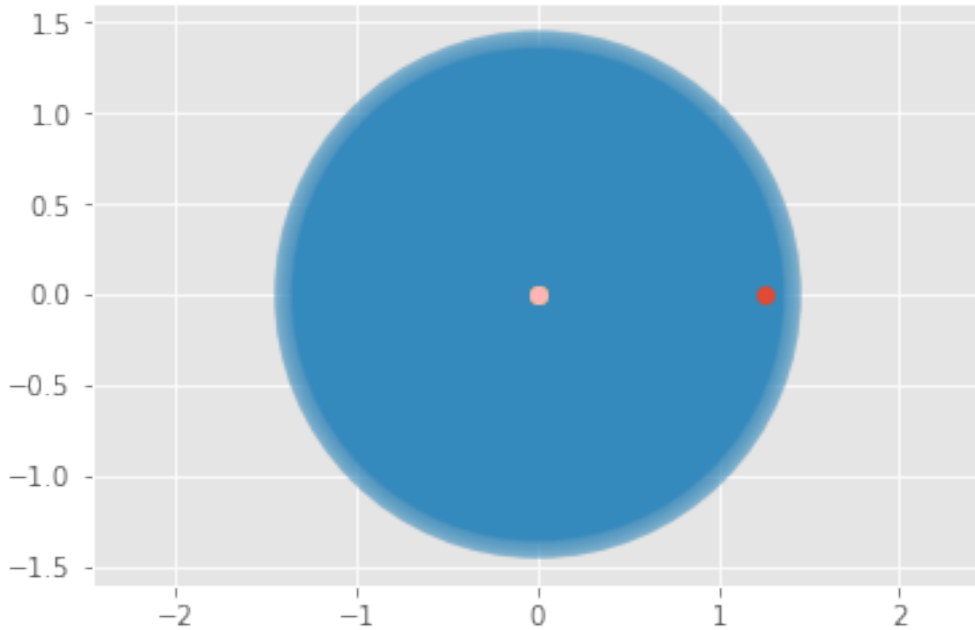
  2.52156848e-05+0.00000000e+00j  3.00736664e-07+0.00000000e+00j
  5.06364866e-09+0.00000000e+00j  1.00516376e-10+0.00000000e+00j
  2.57911075e-12+0.00000000e+00j  8.41500194e-14+0.00000000e+00j
  2.74688967e-15+0.00000000e+00j  8.70801048e-17+0.00000000e+00j
  3.04514407e-18+0.00000000e+00j  6.80173630e-19+0.00000000e+00j
  7.19510913e-20+0.00000000e+00j -7.58692188e-21+3.58131422e-20j
 -7.58692188e-21-3.58131422e-20j  7.63127959e-21+1.57069121e-20j
  7.63127959e-21-1.57069121e-20j  1.21497900e-20+0.00000000e+00j
 -5.09065883e-21+1.09588603e-21j -5.09065883e-21-1.09588603e-21j
  1.05716933e-21+0.00000000e+00j -2.23862881e-22+0.00000000e+00j
  1.11834560e-22+0.00000000e+00j -1.05614800e-23+0.00000000e+00j
  4.76315585e-24+0.00000000e+00j -2.53498228e-24+0.00000000e+00j
  2.57833946e-25+0.00000000e+00j  1.40074077e-25+0.00000000e+00j
  3.19310762e-26+0.00000000e+00j -4.91538801e-27+0.00000000e+00j
 -1.18607118e-27+0.00000000e+00j  5.49034061e-28+0.00000000e+00j
  2.06162811e-28+0.00000000e+00j -1.40970160e-30+3.30853853e-29j
 -1.40970160e-30-3.30853853e-29j -1.17472333e-29+0.00000000e+00j
  3.17004147e-30+0.00000000e+00j -1.50042126e-30+8.81328694e-31j
 -1.50042126e-30-8.81328694e-31j  2.21672024e-31+0.00000000e+00j
  7.63536694e-32+0.00000000e+00j -8.38855039e-33+0.00000000e+00j
  1.71003869e-34+2.97350124e-33j  1.71003869e-34-2.97350124e-33j
  2.59781211e-34+0.00000000e+00j -2.18580844e-34+0.00000000e+00j
  8.05073890e-35+0.00000000e+00j -5.31479773e-35+0.00000000e+00j
 -1.70792887e-36+0.00000000e+00j -6.88884813e-37+0.00000000e+00j
  3.92275346e-37+0.00000000e+00j  6.90968604e-38+0.00000000e+00j
 -1.13783847e-38+0.00000000e+00j  1.10462310e-38+0.00000000e+00j
 -1.74552031e-40+0.00000000e+00j  4.55906009e-39+0.00000000e+00j]
```




```
[ 1.05301995e+00+0.00000000e+00j  2.82658199e-04+0.00000000e+00j
  1.92745409e-07+0.00000000e+00j  2.56648821e-10+0.00000000e+00j
  6.14833357e-13+0.00000000e+00j  1.68120901e-15+0.00000000e+00j
  4.94092176e-18+0.00000000e+00j  1.67659020e-20+0.00000000e+00j
 -5.73253652e-22+0.00000000e+00j  9.38652778e-24+5.79567328e-24j
  9.38652778e-24-5.79567328e-24j -9.01825119e-25+0.00000000e+00j
  6.91878890e-26+0.00000000e+00j -3.02143630e-28+0.00000000e+00j
  5.22572496e-29+0.00000000e+00j  5.09192605e-30+0.00000000e+00j
  4.35466999e-31+0.00000000e+00j  6.64314239e-33+0.00000000e+00j
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  2.79937532e-37+0.00000000e+00j -1.54260533e-36+0.00000000e+00j
  1.04622823e-38+0.00000000e+00j  2.85490436e-40+0.00000000e+00j
  5.10286241e-41+0.00000000e+00j  6.40975979e-42+0.00000000e+00j
 -3.34986824e-44+2.34281933e-43j -3.34986824e-44-2.34281933e-43j
  2.52018966e-45+0.00000000e+00j -1.70460359e-45+0.00000000e+00j
 -1.14532349e-45+0.00000000e+00j -3.88399378e-48+0.00000000e+00j
  1.43977324e-49+0.00000000e+00j -1.26168036e-49+0.00000000e+00j
  3.02781390e-51+0.00000000e+00j  2.63100883e-52+0.00000000e+00j
 -8.78473100e-54+0.00000000e+00j -4.88155068e-56+0.00000000e+00j
  5.31488629e-58+0.00000000e+00j  1.52474153e-58+0.00000000e+00j
  1.69209036e-59+0.00000000e+00j  7.45169618e-61+0.00000000e+00j
 -4.73113414e-61+0.00000000e+00j  1.63119337e-62+0.00000000e+00j
  9.00645954e-65+0.00000000e+00j -7.81050220e-65+0.00000000e+00j
 -6.04439337e-66+2.55375981e-66j -6.04439337e-66-2.55375981e-66j
 -4.90259504e-68+7.16088823e-68j -4.90259504e-68-7.16088823e-68j
 -8.98605888e-69+0.00000000e+00j  5.41069651e-70+0.00000000e+00j
```

```
-1.45001714e-70+0.00000000e+00j  3.22282366e-73+8.04300101e-73j
 3.22282366e-73-8.04300101e-73j  9.51458361e-74+0.00000000e+00j]
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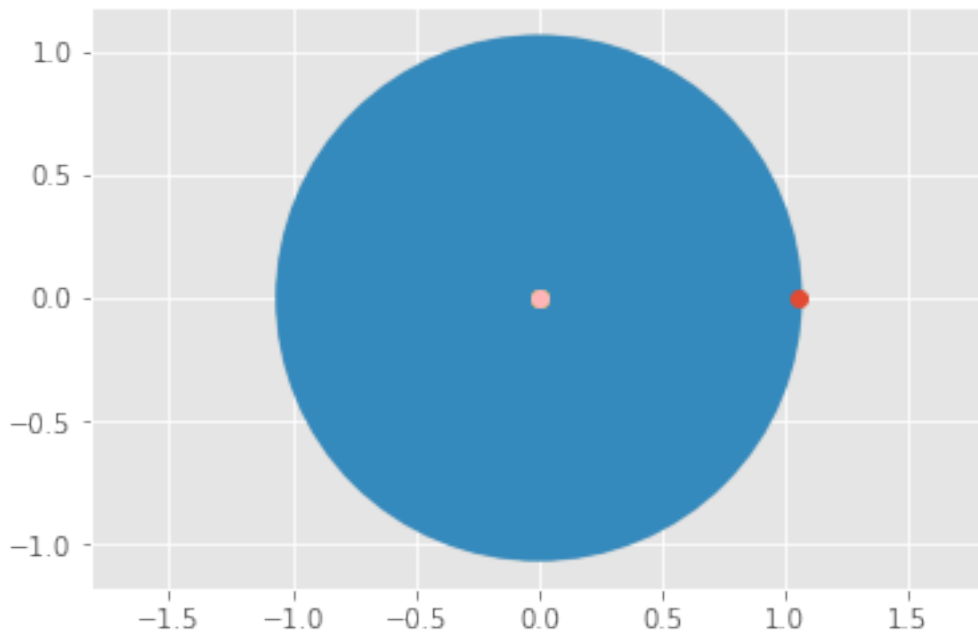


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  7.91827163e-12+0.00000000e+00j  1.43314865e-14+0.00000000e+00j
  3.52820186e-17+0.00000000e+00j  9.65170448e-20+0.00000000e+00j
  2.50029472e-22+0.00000000e+00j -2.67284841e-24+0.00000000e+00j
  2.17639775e-25+0.00000000e+00j  4.67529341e-27+0.00000000e+00j
  3.55549210e-28+0.00000000e+00j -5.53804288e-29+0.00000000e+00j
  1.76868512e-29+0.00000000e+00j  2.67432229e-32+0.00000000e+00j
 -7.10374911e-34+0.00000000e+00j  5.39737799e-35+0.00000000e+00j
  2.08083443e-37+0.00000000e+00j  9.37463224e-39+0.00000000e+00j
 -1.71356764e-39+0.00000000e+00j -2.54133366e-40+1.46767612e-40j
 -2.54133366e-40-1.46767612e-40j  6.91196351e-43+1.29317005e-42j
  6.91196351e-43-1.29317005e-42j -8.52485116e-45+0.00000000e+00j
  9.09214920e-46+0.00000000e+00j -2.68542738e-46+0.00000000e+00j
  1.03942114e-47+0.00000000e+00j -1.21214812e-48+0.00000000e+00j
 -1.69165964e-49+0.00000000e+00j  1.63442514e-50+0.00000000e+00j
 -2.38933556e-51+0.00000000e+00j  5.27639916e-52+0.00000000e+00j
  9.61136318e-54+0.00000000e+00j  3.66885468e-55+0.00000000e+00j
 -7.96432814e-57+0.00000000e+00j  9.07055124e-59+3.05839146e-59j
  9.07055124e-59-3.05839146e-59j  5.29909776e-62+1.14992242e-61j
  5.29909776e-62-1.14992242e-61j  1.17761986e-63+2.97267351e-63j
  1.17761986e-63-2.97267351e-63j -1.41185862e-66+0.00000000e+00j
```

```
-6.52116762e-66+0.00000000e+00j -1.19807955e-67+0.00000000e+00j
 6.89798315e-69+0.00000000e+00j -2.36875704e-69+0.00000000e+00j
 2.31049023e-70+0.00000000e+00j  2.05836375e-71+0.00000000e+00j
 2.65331937e-73+3.56710275e-73j  2.65331937e-73-3.56710275e-73j
 5.33452024e-75+5.03548115e-75j  5.33452024e-75-5.03548115e-75j
-9.25083511e-76+0.00000000e+00j -5.52209333e-78+0.00000000e+00j]
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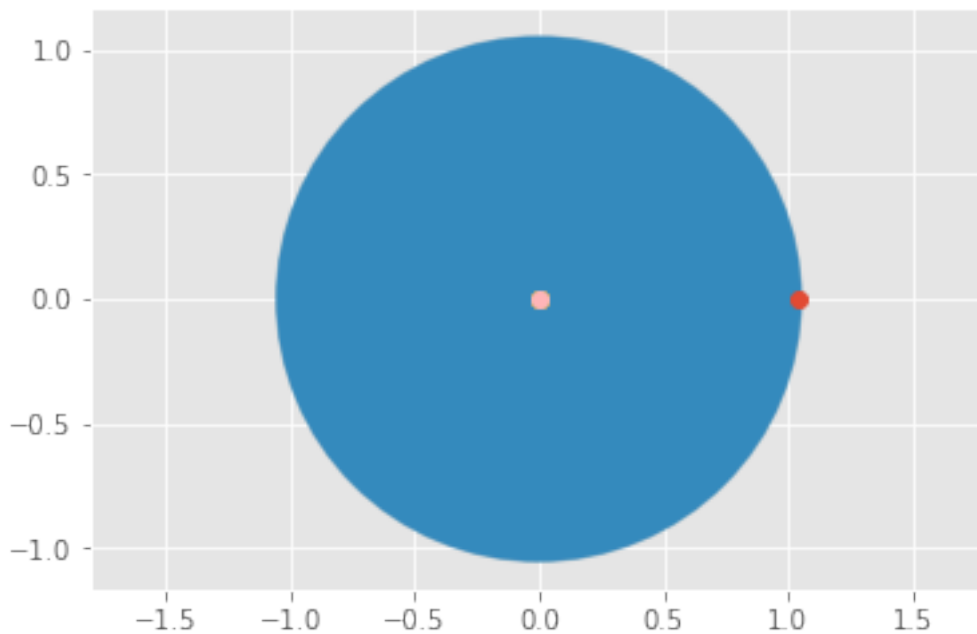


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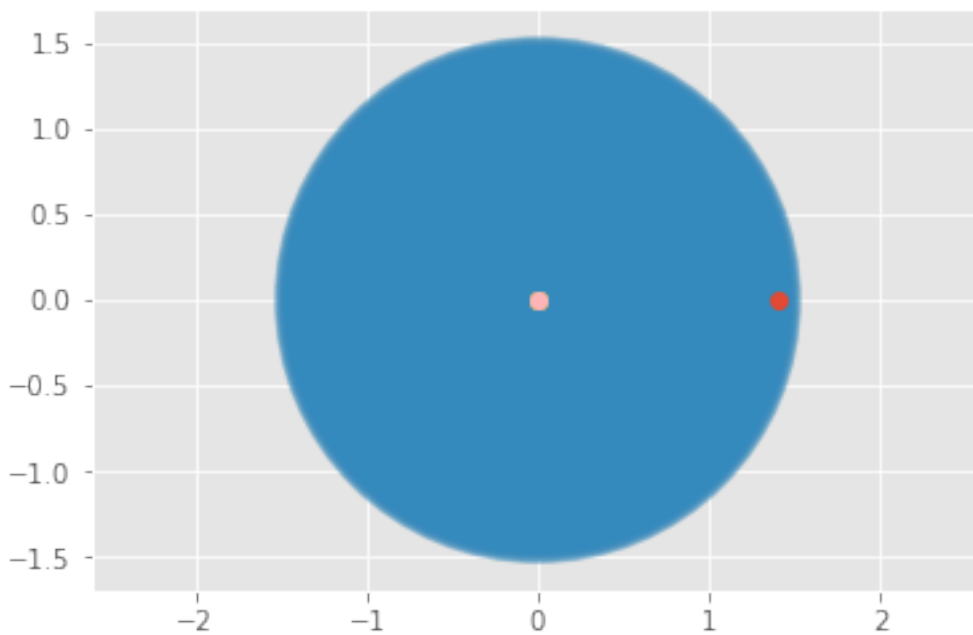
  1.37064508e-04+0.00000000e+00j  2.30856464e-06+0.00000000e+00j
  4.73800856e-08+0.00000000e+00j  1.16147571e-09+0.00000000e+00j
  3.00871082e-11+0.00000000e+00j  8.04065597e-13+0.00000000e+00j
  2.23515291e-14+0.00000000e+00j  6.39621028e-16+0.00000000e+00j
  1.94299624e-17+0.00000000e+00j  3.98346942e-18+0.00000000e+00j
  3.07080078e-19+0.00000000e+00j  1.63074552e-19+1.30528181e-19j
  1.63074552e-19-1.30528181e-19j  1.82942338e-20+3.34638721e-20j
  1.82942338e-20-3.34638721e-20j -9.54681291e-21+1.28977950e-20j
 -9.54681291e-21-1.28977950e-20j  3.44647443e-21+0.00000000e+00j
  1.72705244e-21+0.00000000e+00j  4.11958253e-24+2.22410426e-22j
  4.11958253e-24-2.22410426e-22j  2.03049483e-22+0.00000000e+00j
 -2.06264824e-23+8.57356453e-23j -2.06264824e-23-8.57356453e-23j
  7.64113888e-24+0.00000000e+00j -2.54914610e-24+0.00000000e+00j
 -7.28278738e-25+5.18739459e-25j -7.28278738e-25-5.18739459e-25j
  3.60496701e-26+0.00000000e+00j -1.16767868e-26+3.58933955e-26j
 -1.16767868e-26-3.58933955e-26j  1.05818774e-27+7.93973866e-28j
  1.05818774e-27-7.93973866e-28j -4.98323154e-28+0.00000000e+00j
```

```
 3.44332966e-28+0.00000000e+00j -2.18078041e-29+0.00000000e+00j
 1.93163424e-29+0.00000000e+00j -8.78310685e-31+0.00000000e+00j
 9.50273753e-32+2.01626596e-31j  9.50273753e-32-2.01626596e-31j
-7.69482010e-33+0.00000000e+00j -5.67507192e-33+0.00000000e+00j
 3.61437835e-34+0.00000000e+00j -1.23076423e-34+7.06106245e-35j
-1.23076423e-34-7.06106245e-35j  6.37415768e-35+0.00000000e+00j
 3.61271619e-36+0.00000000e+00j -2.30298135e-37+0.00000000e+00j
 2.14160305e-38+6.55172818e-38j  2.14160305e-38-6.55172818e-38j
-2.43528942e-38+0.00000000e+00j -8.84255577e-40+0.00000000e+00j
 5.94858178e-40+0.00000000e+00j  6.06107071e-41+0.00000000e+00j]
```

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 2.24876322e-08+0.00000000e+00j  2.69251726e-10+0.00000000e+00j
 2.44783588e-12+0.00000000e+00j  1.82695440e-14+0.00000000e+00j
 1.80733075e-16+0.00000000e+00j  5.86882085e-18+3.15175618e-18j
 5.86882085e-18-3.15175618e-18j -1.84200180e-19+2.19826966e-18j
-1.84200180e-19-2.19826966e-18j -4.59715652e-19+0.00000000e+00j
-3.76246649e-19+0.00000000e+00j  5.55367301e-20+4.23857425e-20j
 5.55367301e-20-4.23857425e-20j  3.42479275e-20+7.63166799e-21j
 3.42479275e-20-7.63166799e-21j  7.35535328e-21+0.00000000e+00j
-3.62536612e-21+1.50487269e-21j -3.62536612e-21-1.50487269e-21j
-7.74371270e-22+0.00000000e+00j -1.63994008e-22+2.23807083e-22j
-1.63994008e-22-2.23807083e-22j  2.58540946e-22+0.00000000e+00j
-6.20227970e-23+0.00000000e+00j  6.20840238e-23+0.00000000e+00j
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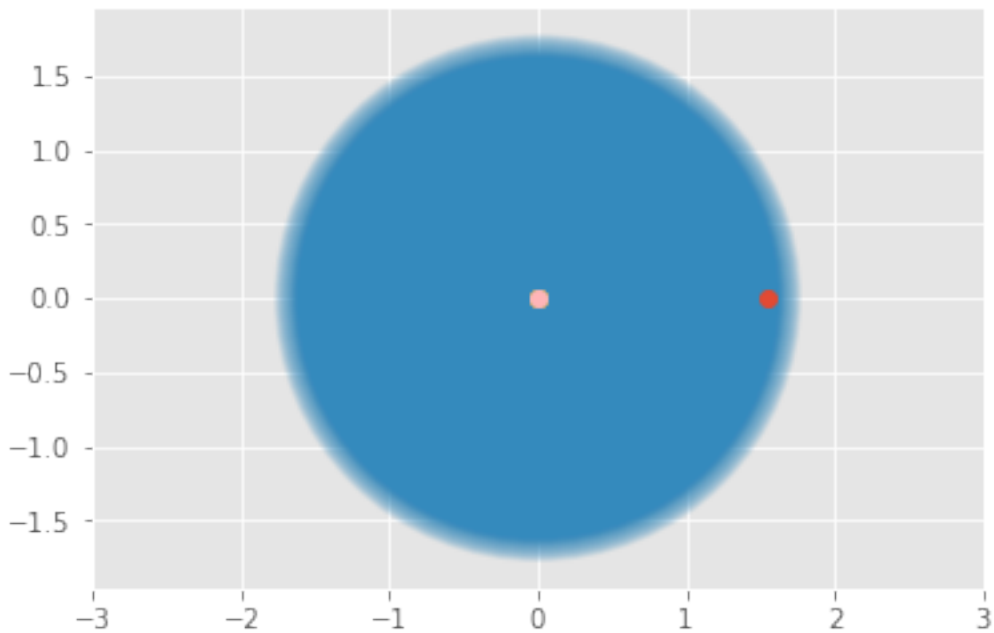
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 8.08766544e-24+4.13862145e-23j  8.08766544e-24-4.13862145e-23j
 3.50913933e-23+0.00000000e+00j -1.85414531e-23+2.53391797e-23j
-1.85414531e-23-2.53391797e-23j  1.58485900e-23+0.00000000e+00j
-9.89820004e-24+0.00000000e+00j  5.02676357e-24+0.00000000e+00j
-2.10465830e-24+0.00000000e+00j  2.68387948e-25+8.15644095e-25j
 2.68387948e-25-8.15644095e-25j  6.68399367e-25+0.00000000e+00j
-2.96679819e-26+0.00000000e+00j  1.09002588e-26+0.00000000e+00j
-9.44471794e-27+0.00000000e+00j  2.98387054e-27+1.51546750e-27j
 2.98387054e-27-1.51546750e-27j -1.09611711e-27+4.13842891e-28j
-1.09611711e-27-4.13842891e-28j  9.63814819e-28+0.00000000e+00j
-8.75487354e-29+0.00000000e+00j  6.79226595e-29+5.98122854e-29j
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-6.53991602e-30+4.13245929e-29j -6.53991602e-30-4.13245929e-29j
 2.96173593e-29+0.00000000e+00j  2.37175055e-31+0.00000000e+00j]
```